



XXII Escola de Geometria Diferencial

Teresina, Piauí – 2026

Ancient solutions to free boundary curve shortening flow

Theodora Bourni

Nathan Burns

Coleção IMPA Lecture Notes

impa



Instituto de
Matemática
Pura e Aplicada

Ancient solutions to free boundary curve shortening flow

Ancient solutions to free boundary curve shortening flow – 1ª edição
Julho de 2026

Copyright © 2026 Theodora Bourni e Nathan Burns.

Direitos reservados por Associação Instituto Nacional de Matemática Pura e Aplicada – IMPA. Não é permitida a reprodução total ou parcial desta obra, por quaisquer meios, sem a prévia autorização por escrito da instituição.

Publicado no Brasil / Published in Brazil

ISBN 978-85-244-0559-4 (impresso), 978-85-244-0562-4 (e-book)

MSC (2020) Primário: 53E20

Coleção IMPA Lecture Notes

Produção **Editora Pi** **Capa** Roniê Bezerra da Silva e Comunicação IMPA

Realização da Editora do IMPA

IMPA

Estrada Dona Castorina, 110

Jardim Botânico

22460-320 Rio de Janeiro RJ

www.impa.br

editora@impa.br

B775a Bourni, Theodora.

Ancient solutions to free boundary curve shortening flow
/ Theodora Bourni, Nathan Burns. 1. ed. Rio de Janeiro:
IMPA, 2026.

106 p.: il. (Coleção IMPA Lecture Notes)

“Apresentado na XXII Escola de Geometria Diferencial”

ISBN 978-85-244-0559-4 (impresso)

ISBN 978-85-244-0562-4 (e-book)

1. Geometria diferencial. 2. Fluxo de encurvamento de
curvas. I. Burns, Nathan. II. Escola de Geometria Diferencial
(22. : 2026 : Teresina, PI). III. Título. IV. Série.

CDD: 516.36

Carolina Celano Lima / CRB-7: 2438



XXII Escola de Geometria Diferencial

Teresina, Piauí – 2026

Ancient solutions to free boundary curve shortening flow

Theodora Bourni

Nathan Burns

Coleção IMPA Lecture Notes

impa



Instituto de
Matemática
Pura e Aplicada

Preface

The study of geometric evolution equations has become a central theme in modern differential geometry and geometric analysis. Among these, curvature driven flows provide a powerful framework for understanding the geometry and topology of submanifolds through their deformation over time. A particularly fundamental example is the curve shortening flow, the gradient flow of length for planar curves, in which a family of immersions evolves in the direction of its curvature vector. Despite its simple formulation, this flow exhibits rich behavior and serves as a model for the highly important higher-dimensional mean curvature flow, which has numerous applications.

The classical theory of curve shortening flow is now well developed. Foundational results show that embedded planar curves evolve toward canonical geometries: convex curves shrink smoothly to round points in finite time, while arbitrary embedded curves become convex before collapsing. These results not only reveal the regularizing nature of the flow but also establish a paradigm for the analysis of singularity formation. In particular, the study of blow-up limits near singularities has led to a classification of possible asymptotic profiles and has motivated the investigation of special solutions such as self-similar shrinkers and ancient solutions.

Ancient solutions, that are solutions defined on backward-infinite time intervals, play a distinguished role in this theory. They arise naturally as singularity models and exhibit strong rigidity and symmetry properties. Their classification has been a major achievement in the boundary-less setting and continues to inform our understanding of singularity formation and stability in geometric flows.

In contrast, the theory becomes significantly more subtle when boundary conditions are introduced. The free boundary curve shortening flow considers curves whose endpoints are constrained to lie on a fixed support

curve, which they meet orthogonally, while otherwise evolving by curvature. This natural Neumann-type problem introduces new analytical and geometric challenges, as the evolution is influenced not only by intrinsic curvature effects but also by interaction with the boundary.

Early work on free boundary problems for curvature flows established foundational results on existence, regularity, and qualitative behavior. In convex domains, solutions with suitable initial data remain well behaved and exhibit convergence phenomena analogous to the closed-curve case. However, the presence of the boundary leads to new behaviors, including convergence to boundary points, formation of half-shrinking profiles, and the emergence of critical stationary configurations.

Recent progress has significantly advanced the understanding of long-time behavior and singularity formation in the free boundary setting. In particular, it has been shown that, depending on the geometry and initial conditions, solutions may either converge in infinite time to a stationary chord or contract in finite time to a boundary point, with rescaled limits given by canonical self-similar solutions such as the shrinking semicircle. These results can be viewed as free boundary analogues of the classical theorems for closed curves, but their proofs require new ideas to handle the lack of symmetry and the evolving interaction with the support curve.

A central theme of current interest is the quantitative analysis of these asymptotics. Beyond qualitative convergence, one seeks precise rates, stability properties, and a detailed description of the spectrum of the linearized flow near self-similar solutions. Such results are essential for understanding uniqueness and rigidity phenomena, as well as for connecting forward evolution with backward-in-time behavior.

Closely related to this analysis is the classification of ancient solutions. In the free boundary setting, this theory is still in its infancy compared to the closed case. The presence of the boundary restricts possible symmetries while introducing new geometric constraints, leading to a delicate interplay between curvature, topology, and boundary geometry. Recent work has begun to classify ancient solutions under certain geometric conditions. For instance, a complete classification is now known for convex ancient free boundary curve shortening flows in compact convex domains.

This book is devoted to a systematic study of the free boundary curve shortening flow, with an emphasis on three interconnected themes: existence theory, stability and asymptotics, and ancient solutions. We begin

by establishing short- and long-time existence results under natural geometric assumptions, highlighting the role of boundary conditions in the analytical framework. These are classical results that provide the foundation for the subject. We then turn to more recent developments and present a detailed analysis of the forward evolution, focusing on stability properties and convergence to self-similar solutions, including quantitative rates where available. Finally, we study the backward-in-time theory, presenting classification results and structural properties of ancient solutions.

Throughout, our goal is to present a coherent picture of how boundary conditions influence curvature-driven evolution and to develop techniques that bridge the gap between the classical theory of closed curves and the emerging theory of free boundary flows. The interplay between analysis, geometry, and topology in this setting continues to present challenging problems and rich structures, many of which remain to be fully understood.

April 2026.

Theodora Bourni, Nathan Burns

Acknowledgment.

The authors were supported by the NSF grant DMS-2405007.

Contents

1	The Free Boundary Curve Shortening Flow	1
1.1	Short-Time Existence	4
1.2	Evolution Equations	7
1.3	Maximum Principles	11
1.3.1	The Avoidance Principle	16
1.4	Long-Time Existence - Curvature Blow-Up	18
1.4.1	The Interior Estimate	24
1.4.2	The Boundary Estimate	26
2	Forwards in Time Convergence and Stability	31
2.1	Forwards in Time Convergence	31
2.1.1	The Intrinsic Blow-Up	33
2.1.2	A Brief Comment on the General Case	35
2.2	Forwards in time stability	36
2.2.1	Geometric Functionals and Their Evolution Equations	39
2.2.2	Dynamic stability	46
2.2.3	L^2 decay	51
3	Ancient solutions	59
3.1	Introduction	59
3.2	Classification in the case of S^1	63
3.2.1	Existence	63
3.2.2	Asymptotics for the Height of a Constructed Solution	70
3.2.3	Uniqueness	74
3.3	The General Case; A Second Proof	79
3.3.1	Graphs Using a Local Parametrization	81
3.3.2	Graphical Curve Shortening Flow	84

3.3.3	Constructing an Ancient Solution	86
3.3.4	Uniqueness	90
	Bibliography	97

1

The Free Boundary Curve Shortening Flow

We consider Ω a closed domain in $\mathbb{R}^2$, with a smooth and convex boundary $\Sigma := \partial\Omega$. We say that a one-parameter family of immersed curves $\gamma : I \times [0, T) \rightarrow \Omega$, where I is a one dimensional manifold, is a solution to the *free boundary curve shortening flow* in Ω if $\gamma(\partial I \times [0, T)) \subset \Sigma$ and

$$\begin{cases} \frac{\partial \gamma}{\partial t} = -\kappa \nu & \text{in } I \times [0, T) \\ \langle \nu_\Sigma \circ \gamma, \nu \rangle = 0 & \text{on } \partial I \times [0, T). \end{cases} \quad (1.1)$$

where $\kappa = \kappa(x, t)$ is the curvature of the curve $\gamma(x, t)$, with respect to the unit normal $\nu = \nu(x, t)$, and ν_Σ is the outward unit normal vector field to Σ . Our sign convention is such that a convex curve has positive curvature with respect to the outward pointing unit normal. We will denote by $\Gamma_t \subset \mathbb{R}^2$ the image of γ at a time t , and sometimes refer to the family $\{\Gamma_t\}_{t \in [0, T)}$ as a solution to the free boundary curve shortening flow.

Intuitively, equation (1.1) describes a family of curves moving by their curvature, and whose end-points are tethered orthogonally to the prescribed curve Σ (see Figure 1.1). Quantities related to the curve Σ will be denoted by a subscript Σ , à la ν_Σ . Finally, unless otherwise specified, all curves and evolving families of curves considered in this book are assumed to be embedded.

Example 1.1 (Shrinking semicircles). Let $\Omega = \{(x, y) \in \mathbb{R}^2 \mid y \geq 0\}$ and, for some $R > 0$, define the semi-circles

$$S(\theta, t) = \sqrt{R^2 - 2t} (\cos \theta, \sin \theta), \quad \text{for } (\theta, t) \in [0, \pi] \times [0, \frac{R^2}{2}).$$

Then one can see that the curvature at any point, with respect to the outward pointing normal $\nu(\theta, t) = (\cos \theta, \sin \theta)$, is given by $\kappa(\theta, t) = \frac{1}{\sqrt{R^2 - 2t}}$.

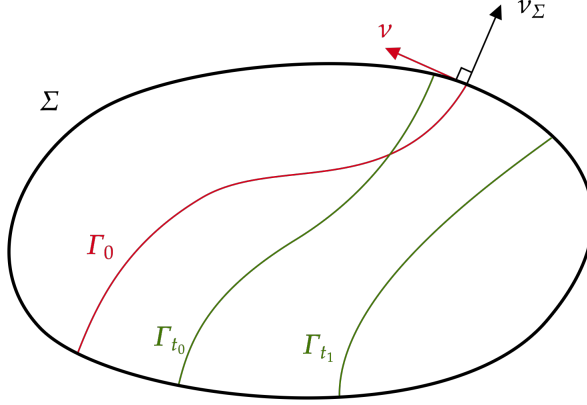


Figure 1.1: Three time-slices of a free boundary curve shortening flow.

Therefore,

$$\frac{\partial S}{\partial t}(\theta, t) = -\frac{1}{\sqrt{R^2 - 2t}}(\cos \theta, \sin \theta) = -\kappa(\theta, t)\nu(\theta, t).$$

Since at $\theta = 0$ and $\theta = \pi$, we have $\nu = e_1$ and $\nu = -e_1$ respectively, and also $\nu_\Sigma = e_2$, we have $\langle \nu_\Sigma \circ \gamma, \nu \rangle = 0$ at the boundary $\{0, \pi\} \times [0, \frac{R^2}{2})$. Hence this family of semicircles is a solution to the free boundary curve shortening flow in Ω .

The previous example is a demonstration of a more general phenomenon, if $\{\Gamma_t\}_{t \in [0, T)}$ is a solution to the periodic curve shortening flow which is symmetric about a given line, then the two component families which lie on either side of the line of symmetry form solutions to the free boundary curve shortening flow.

Example 1.2. Let $y : I \times [0, T) \rightarrow \mathbb{R}$ be a one-parameter family of real-valued functions defined over a fixed interval $I \subset \mathbb{R}$. Denote by

$$\tilde{\Gamma}_t := \{(x, y(x, t)) \mid x \in I\}$$

the graphs of the functions $y(x, t)$. Then one may check (see for instance (Andrews *et al.*, 2020, Lemma 2.7)) that the curves $\{\tilde{\Gamma}_t\}_{t \in [0, T)}$ move with respect to the curve shortening flow if and only if

$$\frac{\partial y}{\partial t}(x, t) = \frac{y''(x, t)}{1 + y'(x, t)^2}.$$

If we further assume that the curves $\{\tilde{\Gamma}_t\}_{t \in [0, T]}$ intersect the boundary of a convex domain $\Sigma = \partial\Omega$ at a point $p_0 = (x_0, y(x_0, t))$, then this intersection will be orthogonal if

$$y'(x_0, t_0) = \tan \theta_0, \quad (1.2)$$

where θ_0 is the angle between the unit normal to Σ and the x -axis. If each intersection point satisfies (1.2), then

$$\{\Gamma_t = \tilde{\Gamma}_t \cap \Omega\}_{t \in [0, T]},$$

is a (possibly disconnected) free boundary curve shortening flow in Ω .

With the previous example in mind, we can give a non-trivial example of the free boundary curve shortening flow. The authors would like to thank Mat Langford for bringing the following example to our attention.

Example 1.3 (The Dueling Grim-Reapers). Let Σ be the curve in $\mathbb{R}^2$ given by the graph of the function

$$u(x) = -\log(\cos x),$$

where $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, which is known as the *grim reaper curve*, and let Ω be the convex region bounded by Σ . Then for any $x \in (-\frac{\pi}{2}, \frac{\pi}{2})$, an easy computation reveals that the downward pointing unit normal vector to Σ at $(x, u(x))$ is given by $\nu_\Sigma = (\sin x, -\cos x)$, and note that the angle between ν_Σ and the x -axis is given by $x - \frac{\pi}{2}$. Now consider the family of curves $\{\tilde{\Gamma}_t\}_{t \in (-\infty, \infty)}$ given by graphs of the real-valued functions

$$y(x, t) = t - \log(\cos(x - \frac{\pi}{2})),$$

where $x \in (0, \pi)$. One may check that $y(x, t)$ is a solution to

$$\frac{\partial y}{\partial t}(x, t) = \frac{y''(x, t)}{1 + y'(x, t)^2}$$

for any $x \in (0, \pi)$. Therefore, the family $\{\tilde{\Gamma}_t\}_{t \in (-\infty, \infty)}$ is a solution to curve shortening flow that moves by translation. This translating solution is known as the *grim reaper*.

Furthermore, we have

$$y'(x, t) = \tan(x - \frac{\pi}{2}),$$

and so (1.2) is automatically satisfied at any point $p \in \tilde{\Gamma}_t \cap \Sigma$. Therefore, as per Example 1.2, the family of curves given by $\Gamma_t = \tilde{\Gamma}_t \cap \Omega$ satisfy free boundary curve shortening flow in Ω . This solution is visualized in Figure 1.2.

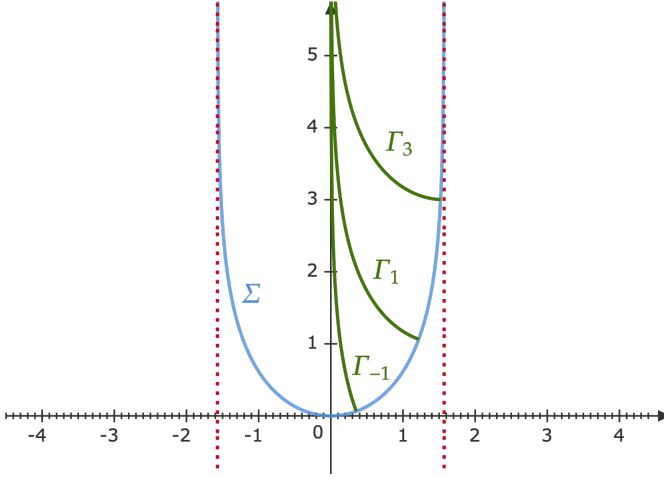


Figure 1.2: The Duelling Grim-Reapers

1.1 Short-Time Existence

The first step in developing the theory of free boundary curve shortening flow is to establish short-time existence for sufficiently smooth initial data. For this purpose, we prove the following short-time existence theorem.

Theorem 1.4. *Let Σ be a smooth curve in $\mathbb{R}^2$ enclosing a convex region Ω and let $\gamma_0 : I \rightarrow \mathbb{R}^2$ be a smooth, compact embedding satisfying*

$$\begin{cases} \gamma_0(I) \subset \bar{\Omega} \\ \gamma_0(\partial I) \subset \Sigma, \end{cases} \quad (1.3)$$

with free boundary on Σ . Then for any $\alpha \in (0, 1)$, equation (1.1) has a unique solution

$$\gamma(x, t) \in C^\infty(I \times (0, \delta]) \cap C^{2+\alpha, \frac{2+\alpha}{2}}(I \times [0, \delta]),$$

for a small constant $\delta > 0$ and with initial condition $\gamma(x, 0) = \gamma_0(x)$.

We follow the argument of Stahl (Stahl, 1996b, Section 2) and introduce coordinates that allow us to seek solutions represented as graphs over the initial curve. We note that the theorem remains valid for immersions; however, for the purposes of this book, we restrict our attention to embeddings.

Definition 1.5. Let $\Gamma_0 := \gamma_0(I)$ be a smooth embedding inside Ω with free boundary on Σ . Define a smooth vector field ξ inside an ε -tubular neighborhood of Γ_0 , $\mathcal{U}_\varepsilon$, such that

$$\xi|_{\Gamma_0} = \nu, \quad \xi|_{\Sigma \cap \mathcal{U}_\varepsilon} \in T\Sigma, \quad \|\xi\| = 1.$$

We consider $\Phi : I \times (-\varepsilon, \varepsilon) \rightarrow \Omega$, the flow of the vector field ξ starting at Γ_0 . The integral curves $s \mapsto \Phi(x, s)$ are normal to Γ_0 and, for $x \in \partial I$, lie in Σ . Therefore, given $p = \Phi(x, s) \in \mathcal{U}_\varepsilon$, we can define the ‘height’ coordinate

$$y(p) = s \iff p = \Phi(x, s) \text{ for some } s \in (-\varepsilon, \varepsilon).$$

Using x as the I -coordinate, any point $p \in \mathcal{U}_\varepsilon$ can be expressed as

$$(x, y) \in I \times (-\varepsilon, \varepsilon).$$

We refer to these coordinates as *generalized Gaussian coordinates centered at Γ_0* . Note that in the definition we identify I with its image under γ_0 .

Proof of Theorem 1.4. Let $\gamma_0 : I \rightarrow \mathbb{R}^2$ be a compact smooth embedding satisfying (1.3). Wishing to emulate Example 1.2, in generalized Gaussian coordinates centered at Γ_0 , as in Definition 1.5, consider the following scalar initial-boundary-value problem:

$$\begin{cases} \frac{\partial w}{\partial t}(x, t) = -\frac{\kappa_w}{\langle \nu_w, \xi \rangle}(x, t), & (x, t) \in I \times [0, \delta] \\ w(x, 0) = 0, & x \in I \\ w_x(x, t) = -\langle \Phi_x(x, w(x, t)), \xi \rangle & x \in \partial I \times [0, \delta] \end{cases} \quad (1.4)$$

for some small $\delta > 0$, and where κ_w denotes the curvature with respect to ν_w , a choice of unit normal field to

$$\hat{\Gamma}_t := \{(x, w(x)) \mid x \in I\}.$$

Then, (1.4) is a parabolic PDE with a *transversal* boundary condition. Therefore, applying standard PDE theory (see (Ladyzhenskaya; Solonnikov;

(Ural'tseva, 1968, Chapter IV) and (Hamilton, 1975)), we obtain a unique $C^{2+\alpha, \frac{2+\alpha}{2}}(I \times [0, \delta])$ -solution for any $\alpha \in (0, 1)$.

Next, we note that (1.4) is equivalent to (1.1), up to a tangential diffeomorphism. To see this, we argue as in the interior case, following (Ecker; Huisken, 1991, Section 1). Define

$$\tilde{F}(x, t) := \Phi(x, w(x, t)), \quad (x, t) \in I \times [0, \delta].$$

By construction, $\tilde{F}$ satisfies

$$\left\langle \frac{\partial \tilde{F}}{\partial t}, \nu_w \right\rangle = -\kappa_w.$$

Furthermore, at every boundary point $x \in \partial I$,

$$\left\langle \frac{\partial \tilde{F}}{\partial x}, \xi \right\rangle = \langle \Phi_x, \xi \rangle + w_x = 0.$$

Thus, $\tilde{F}$ satisfies the normal component of the curve shortening flow equation, together with the free boundary condition.

We now remove the tangential component of the velocity by reparametrising the curve. Let

$$\varphi: I \times [0, \delta] \longrightarrow I$$

be a solution of the system of ordinary differential equations

$$\begin{cases} \frac{\partial \tilde{F}}{\partial x}(\varphi(x, t), t) \frac{\partial \varphi}{\partial t}(x, t) = - \left(\frac{\partial \tilde{F}}{\partial t}(\varphi(x, t), t) \right)^T, \\ \varphi(x, 0) = x, \end{cases}$$

where the superscript T denotes the tangential component to the curve. As in the boundaryless case, the reparametrised map

$$F(x, t) := \tilde{F}(\varphi(x, t), t)$$

then satisfies the curve shortening flow equation.

It remains to verify that F satisfies the free boundary condition. Since $\tilde{F}$ does satisfy the free boundary condition (as noted above), it suffices enough to show that

$$\varphi(x, t) \in \partial I \quad \text{for } x \in \partial I,$$

and $t \in [0, \delta]$. To this end, observe that, at a boundary point,

$$\left(\frac{\partial \tilde{F}}{\partial t} \right)^T = 0.$$

Consequently, the ODE for φ , together with the initial condition $\varphi(x, 0) = x$, gives

$$\frac{\partial \varphi}{\partial t}(x, t) = 0 \quad \text{for } x \in \partial I.$$

Thus, the reparametrisation fixes the boundary points and, in particular, preserves the free boundary condition. Hence F is a solution of the free boundary curve shortening flow, showing that (1.4) and (1.1) are equivalent up to a tangential diffeomorphism. ■

Remark 1.6. We remark here, as it will be used later, that one can write the boundary condition of the graphical free boundary curve shortening flow, (1.4), as

$$\nu_{\Sigma}^1 w_x = \nu_{\Sigma}^2,$$

where $\nu^{\Sigma} = (\nu_{\Sigma}^1, \nu_{\Sigma}^2)$ is written in generalized Gaussian coordinates, that is

$$\nu^{\Sigma} = \nu_{\Sigma}^1 \Phi_x + \xi \nu_{\Sigma}^2.$$

This can be verified, by dotting the above expression by ξ and using the boundary condition

$$w_x = -\langle \Phi_x, \xi \rangle.$$

1.2 Evolution Equations

We now turn our attention to the evolution of various geometric quantities defined on a curve evolving via the free boundary curve shortening flow.

Lemma 1.7 ((Gage; Hamilton, 1986), (Andrews et al., 2020)). *Let $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$ be a solution to the (free boundary) curve shortening flow. Then*

$$\frac{\partial}{\partial t} \frac{\partial}{\partial s} = \frac{\partial}{\partial s} \frac{\partial}{\partial t} + \kappa^2 \frac{\partial}{\partial s},$$

where s is the arc-length parameter.

Proof. $\frac{\partial}{\partial t} \frac{\partial}{\partial s} = \frac{\partial}{\partial t} \left(\frac{1}{|\gamma'|} \frac{\partial}{\partial x} \right) = \kappa^2 \frac{1}{|\gamma'|} \frac{\partial}{\partial x} + \frac{1}{|\gamma'|} \frac{\partial}{\partial x} \frac{\partial}{\partial t} = \kappa^2 \frac{\partial}{\partial s} + \frac{\partial}{\partial s} \frac{\partial}{\partial t}. \quad \blacksquare$

This commutator formula for the arc-length and time derivatives will greatly simplify the computations in the next proposition.

Proposition 1.8 (The Evolution Equations). *Let $\gamma : I \times [0, T)$ be a solution to the (free boundary) curve shortening flow. Then*

1. $\partial_t \tau = -\kappa_s \nu$, where τ denotes the unit tangent vector.
2. $\partial_t \nu = \kappa_s \tau$.
3. $\partial_t \kappa = \kappa_{ss} + \kappa^3$.
4. $\partial_t ds = -\kappa^2 ds$.
5. $\partial_t \theta = \kappa_s$, where θ is either the turning angle or the normal angle at a point (that is the angle between τ and the x -axis or ν and the x -axis respectively).

Proof.

1. This is a simple application of the commutator formula for the arc-length and time derivatives:

$$\partial_t \tau = \gamma_{st} = \gamma_{ts} + \kappa^2 \tau = (-\kappa \nu)_s + \kappa^2 \tau = -\kappa_s \nu - \kappa^2 \tau + \kappa^2 \tau = -\kappa_s \nu.$$

2. Differentiating the equation $\langle \nu, \nu \rangle = 1$ with respect to time implies $\langle \partial_t \nu, \nu \rangle = 0$, which tells us that $\partial_t \nu$ lies in the tangential direction, i.e., $\partial_t \nu = \langle \partial_t \nu, \tau \rangle \tau$. To compute $\langle \partial_t \nu, \tau \rangle$, we differentiate the equation $\langle \nu, \tau \rangle = 0$ with respect to time and use $\partial_t \tau = -\kappa_s \nu$:

$$\langle \partial_t \nu, \tau \rangle = -\langle \nu, \partial_t \tau \rangle = \kappa_s.$$

3. The curvature is given by $\kappa = \langle \nu_s, \tau \rangle$, and so

$$\begin{aligned} \partial_t \kappa &= \langle \nu_{st}, \tau \rangle + \langle \nu_s, \tau_t \rangle \\ &= \langle \nu_{ts} + \kappa^2 \nu_s, \tau \rangle - \kappa_s \langle \nu_s, \nu \rangle \\ &= \langle (\kappa_s \tau)_s, \tau \rangle + \kappa^3 \langle \tau, \tau \rangle - \kappa_s \kappa \langle \tau, \nu \rangle \\ &= \kappa_{ss} \langle \tau, \tau \rangle - \kappa_s \kappa \langle \nu, \tau \rangle + \kappa^3 \langle \tau, \tau \rangle - \kappa_s \kappa \langle \tau, \nu \rangle \\ &= \kappa_{ss} + \kappa^3. \end{aligned}$$

4. First we compute:

$$\begin{aligned}
 \partial_t |\gamma'| &= \partial_t \langle \gamma', \gamma' \rangle^{1/2} \\
 &= \frac{1}{|\gamma'|} \langle -(\kappa \nu)', \gamma' \rangle \\
 &= -\langle \kappa_s \nu, \gamma' \rangle - \kappa^2 \langle \tau, \gamma' \rangle \\
 &= -\kappa^2 |\gamma'|.
 \end{aligned}$$

Writing $ds = |\gamma'| dx$, we have $\partial_t ds = -\kappa^2 |\gamma'| dx = -\kappa^2 ds$.

5. Since $\tau = (\cos \theta, \sin \theta)$, taking a time derivative, yields

$$-\kappa_s \nu = -\theta_t (\sin \theta, -\cos \theta),$$

which of course means $\partial_t \theta = \kappa_s$. The case where $\nu = (\cos \theta, \sin \theta)$ instead is similar. ■

Remark 1.9. Proposition 1.8 implies that the length functional defined by

$$L(\gamma_t) = \int_{\Gamma_t} ds$$

is a non-increasing function of time, justifying the name-sake of the flow.

Proposition 1.8 also provides us with many of the *bread and butter* tools we shall use going forward. Before going any further, we shall present a technical lemma, which shows us how to take time derivatives at the boundary.

Lemma 1.10. *Let $f : \Sigma \rightarrow \mathbb{R}$ be a function defined on Σ , and let $\hat{f} : I \times [0, T) \rightarrow \mathbb{R}$ be a function defined on a free boundary curve shortening flow with*

$$\hat{f}(x, t) = f(\gamma(x, t)), \quad \forall x \in \partial I.$$

Then

$$\partial_t \hat{f} = -\kappa \langle \nabla_\Sigma f \circ \gamma, \nu \rangle$$

at $\partial I \times [0, T)$.

Proof. $\partial_t \hat{f} = \partial_t (f \circ \gamma) = \langle \nabla_\Sigma f \circ \gamma, \partial_t \gamma \rangle = -\kappa \langle \nabla_\Sigma f \circ \gamma, \nu \rangle$. ■

We may use this result to demonstrate a very useful curvature condition at the boundary.

Lemma 1.11. *Let $\{\Gamma_t\}_{t \in [0, T)}$ be a free boundary curve shortening flow in Ω . Then*

$$\kappa_s = \langle \nabla \kappa, \nu_\Sigma \circ \gamma \rangle = \kappa \kappa_\Sigma \circ \gamma$$

at the boundary points, where κ_Σ denotes the curvature of Σ .

Proof. Differentiating the boundary condition

$$\langle \nu, \nu_\Sigma \circ \gamma \rangle = 0$$

with respect to time and using Lemma 1.10 yields

$$0 = \partial_t \langle \nu, \nu_\Sigma \circ \gamma \rangle = \langle \nabla \kappa, \nu_\Sigma \circ \gamma \rangle - \kappa \kappa_\Sigma \circ \gamma,$$

and thus the result. ■

We finish this section by computing the evolution of the enclosed area.

Lemma 1.12 (Evolution of the enclosed area). *Let $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$ be a free boundary curve shortening flow in Ω . Let $A(t)$ denote the area of the region bounded by Γ_t and Σ . Then at any time for which $A(t) < \infty$, we have*

$$\frac{d}{dt} A(t) = - \int_{\Gamma_t} \kappa ds,$$

where the curvature κ is with respect to the unit normal that points outside the enclosed region.

Remark 1.13. Of course when Ω is bounded, there are two such regions and the formula works for both. Note that the curvature switches sign depending on the region.

Remark 1.14. The integral that describes the evolution of the enclosed area is the total turning angle of the curve γ , and so, if γ is an embedding at time t , we must have

$$0 \geq \frac{d}{dt} A(t) \geq -2\pi.$$

Proof of Lemma 1.12. By the divergence theorem, the area enclosed by the curves Γ_t and Σ is given by integrating the support function, $\sigma := \langle \gamma, \nu \rangle$, along the boundary. Therefore,

$$2A(t) = \int_{\Gamma_t} \sigma ds + \int_{\Sigma_t} \sigma_\Sigma ds_\Sigma,$$

where Σ_t denotes the portion of the boundary of the enclosed (by Γ_t and Σ) region that lies on Σ and σ_Σ denotes the support function of Σ . Taking the time derivative of the first term, we compute

$$\begin{aligned} \partial_t \left(\int_{\Gamma_t} \sigma ds \right) &= \int_{\Gamma_t} (\partial_t \sigma - \kappa^2 \sigma) ds \\ &= \int_{\Gamma_t} (-\kappa + \kappa_s \langle \gamma, \tau \rangle - \kappa^2 \sigma) ds \\ &= \int_{\Gamma_t} -2\kappa ds + \sum_{\partial \Gamma_t} \kappa \langle \gamma, \nu_\Sigma \rangle. \end{aligned}$$

Note that along Σ we have $\sigma_\Sigma = \pm \langle \gamma_\Sigma, \tau \rangle$, which is positive if τ points out of Ω and negative if τ points into Ω . The only contribution to the time derivative comes from the motion of the endpoints of Σ_t . Thus,

$$\partial_t \left(\int_{\Sigma_t} \sigma_\Sigma ds_\Sigma \right) = - \sum_{\partial \Gamma_t} \kappa \langle \gamma, \nu_\Sigma \rangle.$$

Combining the two expressions, the boundary terms cancel, yielding the desired result. $\blacksquare$

1.3 Maximum Principles

In this section, we will slowly build up the machinery that will be used repeatedly in our study of the free boundary curve shortening flow. Given a family of functions $u : I \times [0, T) \rightarrow \mathbb{R}$ defined on a free boundary curve shortening flow, we shall consider linear parabolic operators of the form

$$Lu = u_t - \Delta u - \langle \nabla u, b \rangle - cu,$$

where, $b \in \mathbb{R}^2$ and $c \in \mathbb{R}$ (we will often impose a sign condition on c). We remark that, by a family of functions defined on the flow, we mean functions

u , that are define via an ambient function $U : \Omega \rightarrow \mathbb{R}$ as $u(x, t) = U(\gamma(x, t))$. Moreover, the Laplacian and gradient in the definition of L are those on the curves. We begin with the following version of the weak maximum principle.

Lemma 1.15 (The Weak Maximum Principle). *Let $\{\Gamma_t\}_{t \in [0, T]}$ be a free boundary curve shortening flow inside Ω , and suppose that a family of functions $u : I \times [0, T) \rightarrow \mathbb{R}$ satisfies*

$$Lu \leq 0,$$

where $c \equiv 0$. Then, for any $(x, t) \in I \times [0, T)$,

$$u(x, t) \leq \sup_{(I \times \{t=0\}) \cup (\partial I \times [0, T))} u.$$

Proof. First we assume that $Lu < 0$. If the conclusion is not true, then, for some $\delta > 0$, there is an interior point x_0 of I and a first time $t_0 > 0$ for which

$$u(x_0, t_0) - \delta \geq \sup_{(I \times \{t=0\}) \cup (\partial I \times [0, T))} u.$$

Then at (x_0, t_0) we must have $\partial_t u \geq 0$, $\nabla u = 0$ and $\Delta u \leq 0$, and therefore

$$0 \leq \partial_t u - \Delta u - \langle \nabla u, b \rangle = Lu < 0,$$

which is a contradiction. To extend this to the case $Lu \leq 0$, we define the function $u_\varepsilon = u + \varepsilon e^{-t}$, for which we note that

$$Lu_\varepsilon \leq -\varepsilon e^{-t} < 0.$$

The estimate can be thus applied to u_ε and for any point $(x, t) \in I \times (0, T)$ we obtain

$$\begin{aligned} u(x, t) &\leq u_\varepsilon(x, t) \leq \sup_{(I \times \{t=0\}) \cup (\partial I \times [0, T))} (u + \varepsilon e^{-t}) \\ &\leq \sup_{(I \times \{t=0\}) \cup (\partial I \times [0, T))} u + \varepsilon. \end{aligned}$$

Taking $\varepsilon \rightarrow 0$ yields the result. ■

For most purposes the above result will be sufficient, however, in the case that $c \leq 0$, we have the following result.

Lemma 1.16. *Let $\{\Gamma_t\}_{t \in [0, T)}$ be a free boundary curve shortening flow in Ω and suppose that a family of functions $u : I \times [0, T) \rightarrow \mathbb{R}$ satisfies*

$$Lu \leq 0,$$

where $c \leq 0$. Then for any $(x, t) \in I \times [0, T)$

$$u(x, t) \leq \max \left\{ \sup_{(I \times \{t=0\}) \cup (\partial I \times [0, T))} u, 0 \right\}.$$

Proof. Define the operator $\tilde{L} := L + c$ and note that it satisfies $\tilde{L}u \leq Lu \leq 0$ in the subdomain of $I \times [0, T)$ where $u \geq 0$. Therefore, the result follows by Lemma 1.15 after replacing L by $\tilde{L}$ in the hypotheses. ■

The next result is useful to deal with potential boundary maxima.

Lemma 1.17 (Hopf–Boundary Point Lemma). *Let $\{\Gamma_t\}_{t \in [0, T)}$ be a free boundary curve shortening flow in Ω , and suppose that a family of functions $u : I \times [0, T) \rightarrow \mathbb{R}$ satisfies*

$$Lu \leq 0$$

where $c \leq 0$ and $u \geq 0$ or $c \equiv 0$ and no sign condition on u is imposed. Furthermore, suppose that there is a point $(x_0, t_0) \in \partial I \times [0, T)$, for which:

$$u(x_0, t_0) > u(x, t)$$

for any (x, t) with $t \leq t_0$ and $(x, t) \neq (x_0, t_0)$. Then

$$\langle \nabla u, \nu_\Sigma \circ \gamma \rangle > 0.$$

at the point (x_0, t_0) .

Proof. Let B be a ball of radius r inside Ω which touches Σ at $\gamma(x_0, t_0)$, and small enough so that $\Gamma_{t_0} \cap B$ consists of one connected component. Choose the origin to be the centre of the ball B , and let $J \subset I$ be such that $\gamma(x, t_0) \in \partial B$ for any $x \in \partial J$. Let $v : J \times [0, T) \rightarrow \mathbb{R}$, be defined by

$$v(x, t) = e^{-\alpha|\gamma(x, t_0)|^2} - e^{-\alpha r^2},$$

where $\alpha > 0$ is a constant to be determined. Note that $v \geq 0$ for any $x \in J$ and $v = 0$ at any $x \in \partial J$. Defining $\tilde{L} = L + c$, we may compute

$$\tilde{L}v \leq \left(2\alpha - 2\alpha\kappa \langle \gamma, \tau \rangle + 2\alpha \langle \gamma, \tau \rangle \langle b, \tau \rangle - 4\alpha^2 \langle \gamma, \tau \rangle^2 \right) e^{-\alpha|\gamma(x, t_0)|^2}.$$

Fix a small constant ρ and let $B_{r-\rho}$ be a ball of a radius $r - \rho$ centred at the origin. Let $K \subset J$ be such that

1. $x_0 \in \partial K$,
2. $\gamma(\partial K, t_0)$ contains the point on $\partial B_{r-\rho}$ closest to $\gamma(x_0, t_0)$,
3. $\gamma(K, t_0)$ has a single connected component.

Then, for ρ sufficiently small, we can find a constant $\alpha := \alpha(\rho)$ and a small constant δ , such that $\tilde{L}v \leq 0$ for any $x \in K$ and $t \in [t_0 - \delta, t_0]$ (because we can ensure that $\langle \gamma, \tau \rangle^2 > 0$ in $K \times [t_0 - \delta, t_0]$). By continuity, we may find a positive constant ε , such that $u(x, t) + \varepsilon v(x, t) \leq u(x_0, t_0)$ in $\partial K \times [t_0 - \delta, t_0]$ and in $K \times \{t_0 - \delta\}$. Since $\tilde{L}(u + \varepsilon v) \leq 0$, for $(x, t) \in K \times [t_0 - \delta, t_0]$, by the weak maximum principle, we conclude that $u(x, t_0) + \varepsilon v(x, t_0) \leq u(x_0, t_0)$ for all $x \in K$. Taking the outward normal derivative therefore implies that

$$\langle \nabla u, \nu_\Sigma \circ \gamma \rangle \geq -\varepsilon \langle \nabla v, \nu_\Sigma \circ \gamma \rangle > 0. \quad \blacksquare$$

We remark that this proof does not use the strong maximum principle (see conversation and Lemma 1.18 below). It is more common, and perhaps more straightforward, to prove this result using the strong maximum principle (see, for instance, (Lieberman, 1996, Chapter II)).

The maximum principles above will be sufficient for most practical applications. However, a stronger result can be obtained: an interior maximum cannot be attained by non-constant data. This result is known as the *strong maximum principle*, which we state below.

Lemma 1.18 (Strong Maximum Principle). *Let $\{\Gamma_t\}_{t \in [0, T]}$ be a free boundary curve shortening flow inside Ω and suppose that a family of functions $u : I \times [0, T) \rightarrow \mathbb{R}$ satisfies*

$$Lu \leq 0,$$

where $c \leq 0$. Suppose that there is some interior point $(x_0, t_0) \in I \times (0, T)$ for which

$$u(x_0, t_0) = \sup_{\bar{I} \times [0, T)} u,$$

then u is a constant function.

The proof of the strong maximum principle can be found for instance in (Lieberman, 1996, Chapter II).

A first application of the maximum principle that we show is the preservation of convexity of a free boundary curve shortening flow.

Lemma 1.19. *Let $\{\Gamma_t\}_{t \in [0, T]}$ be a free boundary curve shortening flow inside Ω . If Γ_0 is strictly convex (up to and including its boundary), then so are Γ_t for all $t \in (0, T)$.*

Proof. Our assumption implies that $\kappa(x, 0) > 0$, for all $x \in \bar{I}$. By Proposition 1.8 and Lemma 1.11, we know that κ satisfies

$$\begin{cases} (\partial_t - \Delta)\kappa = \kappa^3 = \kappa^2\kappa & \text{in } I, \\ \langle \nabla \kappa, \nu_\Sigma \rangle = \kappa\kappa_\Sigma & \text{on } \partial I. \end{cases} \quad (1.5)$$

By the weak maximum principle (applied to $-\kappa$), we have that, for all $(x, t) \in I \times [0, T)$

$$\kappa(x, t) \geq \min_{(I \times \{t=0\}) \cup (\partial I \times [0, T))} \kappa.$$

Next we claim that the right hand side of the preceding inequality is strictly positive. If not, there exists $x \in \partial I$, and a first time t , such that $\kappa(x, t) = 0$. By the boundary condition in (1.5), at that point we must have

$$\langle \nabla \kappa, \nu_\Sigma \rangle = 0,$$

which is ruled out by the Hopf-boundary point lemma 1.17. ■

We remark that one can relax the hypothesis in Lemma 1.19 to convex (instead of strictly convex) by invoking the use of the strong maximum principle.

Finally, as it will be often useful later, we state the following Sturm-type Theorem due to Angenent.

Theorem 1.20 ((Angenent, 1988)). *Assume that $u : I \times [0, T] \rightarrow \mathbb{R}$ is a bounded solution of*

$$u_t = a(x, t)u_{xx} + b(x, t)u_x + c(x, t)u,$$

where

$$a + a^{-1} + a_t + a_x + a_{xx} + b + b_t + b_x + c \leq C.$$

Also assume that for any $x_0 \in \partial I$, we have $u(x_0, t) \neq 0$ for any $t \in [0, T]$. If we let $z(t)$ denote the number of zeroes of $u(\cdot, t)$ in I ,

(a) $z(t)$ is finite for all $t > 0$.

(b) If (x_0, y_0) is a multiple zero of u , then for any $t_1 < t_0 < t_2$ we have $z(t_1) > z(t_2)$.

Remark 1.21. Theorem 1.20 has a powerful application to solutions of the curve shortening flow. Indeed, it yields several properties in dimension one that have no direct analogues in higher-dimensional mean curvature flow. For example, it can be shown that the number of critical points of the curvature is non-increasing. Another application is that the number of intersection points between two solutions is non-increasing. These results have analogues in the case of free boundary, but one has to impose some extra conditions to control it. The second mentioned property is a more refined version of the avoidance principle, which we will examine in the following section.

1.3.1 The Avoidance Principle

One extremely useful corollary of the maximum principle is the avoidance principle. Roughly speaking, this says that two free boundary curve shortening flows in Ω which do not intersect initially will never intersect. We shall make use of the ideas presented by Stahl in (Stahl, 1996b, Section 4).

Let $\gamma^+ : I \times [0, T) \rightarrow \mathbb{R}^2$ and $\gamma^- : I \times [0, T) \rightarrow \mathbb{R}^2$ be two one parameter families of curves inside Ω , such that $\partial\Gamma_t^\pm \subset \Sigma$ and which satisfy

$$\begin{cases} \left(\frac{\partial}{\partial t} \gamma^+ \right)^\perp \geq -\kappa^+ \nu^+ & \text{in } I \times [0, T) \\ \langle \nu^+, \nu_\Sigma \circ \gamma^+ \rangle \geq 0 & \text{on } \partial I \times [0, T), \end{cases}$$

and

$$\begin{cases} \left(\frac{\partial}{\partial t} \gamma^- \right)^\perp \leq -\kappa^- \nu^- & \text{in } I \times [0, T) \\ \langle \nu^-, \nu_\Sigma \circ \gamma^- \rangle \leq 0 & \text{on } \partial I \times [0, T), \end{cases}$$

where $(\cdot)^\perp$ denotes the projection in the normal direction. Let $\Omega^\pm(t) \subset \Omega$ be the domains in Ω bounded by $\gamma^\pm(I, t)$, respectively, which are such that $\nu^\pm$ coincides with their outward pointing unit normal.

Proposition 1.22. *With the notation as above, if $\Omega^+(0) \subset \Omega^-(0)$ and $\Gamma_0^+ \cap \Gamma_0^- = \emptyset$, then $\Omega^+(t) \subset \Omega^-(t)$, for all $t \in [0, T)$.*

Proof. Assume, on the contrary, that there exists a first time $t_0 \in (0, T)$ when $\Omega^+(t) \subseteq \Omega^-(t)$ and the intersection $\Gamma_{t_0}^+ \cap \Gamma_{t_0}^-$ is non-empty. We will use the maximum principle to conclude that this leads to a contradiction. First, notice that at any $p_0 \in \Gamma_{t_0}^+ \cap \Gamma_{t_0}^-$, we must have

$$T_{p_0}\Gamma_{t_0}^+ = T_{p_0}\Gamma_{t_0}^-,$$

even if $p_0 \in \Sigma$. Let $\mathcal{U}_\varepsilon$ be an ε tubular neighborhood of $\Gamma_{t_0}^+$, and as in Definition 1.5, introduce generalized Gaussian coordinates in $\mathcal{U}_\varepsilon$ centered at $\Gamma_{t_0}^+$. Then we obtain local graphical representations for $\Gamma^\pm(t)$, for t close to t_0 and around p_0 . That is, there exist $u^+, u^- : J \times [t_0 - \delta, t_0] \rightarrow \mathbb{R}$, where δ is sufficiently small, and where J is a subinterval of I containing p_0 . Recall also that we identify I with $\Gamma_{t_0}^+$. We furthermore choose J so that $\partial J \cap \partial I = \emptyset$ if p_0 is an interior point and $\partial J \cap \partial I = \{p_0\}$ otherwise. The evolution inequalities for $\gamma^\pm$ imply that the functions $u^\pm$ satisfy the following scalar boundary problems

$$\begin{cases} \frac{\partial}{\partial t} u^+ \geq -\frac{\kappa_{u^+}}{\langle \nu_{u^+}, \xi \rangle} & \text{in } J \times [t_0 - \delta, t_0] \\ \nu_\Sigma^1(u^+)_x \geq \nu_\Sigma^2 & \text{on } (\partial J \cap \partial I) \times [t_0 - \delta, t_0], \end{cases}$$

and

$$\begin{cases} \frac{\partial}{\partial t} u^- \leq -\frac{\kappa_{u^-}}{\langle \nu_{u^-}, \xi \rangle} & \text{in } J \times [t_0 - \delta, t_0] \\ \nu_\Sigma^1(u^-)_x \leq \nu_\Sigma^2 & \text{on } (\partial J \cap \partial I) \times [t_0 - \delta, t_0], \end{cases}$$

where ξ is as in Definition 1.5, and $\nu_\Sigma = (\nu_\Sigma^1, \nu_\Sigma^2)$ is written in generalized Gaussian coordinates, as per Remark 1.6. Moreover, $u^+ > u^-$ on $J \times [t_0 - \delta, t_0]$. We now define the function $\varphi := u^+ - u^-$ which satisfies $\varphi \geq 0$ and standard PDE methods give that it satisfies a linear equation of the form

$$\frac{\partial \varphi}{\partial t} \geq \Delta \varphi + b \cdot \nabla \varphi + c \varphi,$$

with $|b|$ and c bounded functions. This computation is in fact easy to deduce from the later Section 3.3, where we analyze in depth this equation.

Moreover if $p_0 \in \Sigma$, then at that point $(\nu_\Sigma^1, \nu_\Sigma^2) = (1, 0)$. Therefore, the maximum principle, together with the Hopf boundary lemma imply that φ has to remain strictly positive. $\blacksquare$

1.4 Long-Time Existence – Curvature Blow-Up

We now turn our attention to the long-time behavior of the free boundary curve shortening flow. We will show that the flow can only be interrupted by a blow up of the curvature. More precisely we will prove the following.

Theorem 1.23. *Let $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$ be a solution to the free boundary curve shortening flow in a closed convex domain Ω , defined on a maximal time-interval. Then*

$$\limsup_{t \rightarrow T} \sup_{\Gamma_t} |\kappa| \rightarrow \infty.$$

To prove Theorem 1.23, we shall once again follow the program set out in Stahl (Stahl, 1996b), whose methods closely resemble that of Ecker-Huisken in (Ecker; Huisken, 1991).

The idea is relatively straightforward. If the curvature remains bounded on a maximal time interval $[0, T)$, then we may extract a *smooth* limiting surface as $t \rightarrow T$. By subsequently applying the short-time existence result, Theorem 1.4, we can extend the solution to a time interval $[0, T + \delta)$ for some $\delta > 0$, contradicting the maximality of $[0, T)$.

To make this argument precise, we show that there exists a collection of domains covering the curves Γ_t , for all $t \in (T - \delta, T)$, for some $\delta > 0$, such that, in each domain, the family of curves can be represented as a graph of a $C^{1+\alpha, \frac{1+\alpha}{2}}$ function. Standard PDE estimates then allow us to extract a smooth limiting curve Γ_T as $t \rightarrow T$.

To this end, we assume that for some $\rho > 0$, we have the global curvature bound

$$\limsup_{t \rightarrow T} \sup_{\Gamma_t} |\kappa| < \frac{1}{2\rho}. \tag{1.6}$$

and show that, under this assumption, T cannot be a maximal time of existence. We first give two definitions that will be useful for our exposition.

Definition 1.24. For a point $p = \gamma(x, t)$, we define the space cylinder around p with axis $\nu(x, t)$ as

$$\mathcal{P}_r(p) := Q((-r, r) \times (-\infty, \infty)),$$

where Q is the orthogonal transformation that maps 0 to p and maps e_2 to $\nu(x, t)$.

Moreover, for $\Delta > 0$, we define

$$\Sigma_{t_0}^\Delta := \{\gamma(x, t) \mid x \in \partial I, |t - t_0| \leq \Delta\} \subset \Sigma$$

to be the portion of the boundary covered by the curves at times within Δ of t_0 .

Definition 1.25. Fix a point $p_0 := \gamma(x_0, t_0)$. By a local graphical representation of Γ_t around $\gamma(x, t)$, we shall always mean a local representation of Γ_t around $\gamma(x, t)$ as a graph over the tangent space of Γ_{t_0} at the point p_0 . More precisely, there is a function $y : (-r, r) \times \{t\} \rightarrow \mathbb{R}$, for some $r > 0$, such that $Q(\text{graph}(y)) \cap \mathcal{P}_r(p_0)$ is equal to the connected component of $\Gamma_t \cap \mathcal{P}_r(p_0)$ containing $\gamma(x, t)$, where Q here is the orthogonal transformation that maps 0 to p_0 and maps e_2 to $\nu(x_0, t_0)$. Furthermore, we define the parametric height function

$$u : I \times [0, T] \rightarrow \mathbb{R}; \quad u(x, t) := \langle \gamma(x, t), \omega \rangle.$$

where $\omega = \nu(x_0, t_0)$.

Remark 1.26. For the special case of the local graphical representation of Γ_{t_0} around p_0 , we have

1. $p_0 = Q(0, 0) = Q(0, y(0, t_0))$,
2. $y'(0, t_0) = 0$.

We remark also, that the free boundary condition allows us to conclude that 1. and 2. above are still satisfied when $p_0 \in \partial \Gamma_{t_0}$.

We begin with the standard result that a curve with bounded curvature may be locally written as a graph with bounded gradient.

Lemma 1.27. *Consider a point $p_0 = \gamma(x_0, t_0)$. The connected component of $\mathcal{P}_{\rho/2}(p_0) \cap \Gamma_{t_0}$ containing p_0 can be written as a graph over the tangent space of Γ_{t_0} at p_0 , where ρ is as in (1.6). Moreover, at all points $\gamma(x, t_0)$ in this component, the normal satisfies*

$$\langle \nu(x, t_0), \omega \rangle^{-1} \leq 1 + \frac{|\gamma(x, t_0)|^2 - u^2(x, t_0)}{4\rho^2}$$

and the height function satisfies

$$|u(x, t_0)| \leq \frac{|\gamma(x, t_0)|^2 - u^2(x, t_0)}{\rho} \leq \frac{\rho}{4},$$

where $\omega = \nu(x_0, t_0)$.

Proof. We first show that the connected component of $\mathcal{P}_{\rho/2}(p_0) \cap \Gamma_{t_0}$ containing p_0 can be written as a graph over the tangent space of Γ_{t_0} at p_0 . Suppose not, then there must be a point x_1 such that $\langle \nu(x_1, t_0), \omega \rangle = 1/2$ and moreover, for all $x \in I$ between x_0 and x_1 we have $\gamma(x, t_0) \subset \mathcal{P}_{\rho/2}(p_0)$ and $\langle \nu(x, t_0), \omega \rangle \geq 1/2$. Note that the portion of Γ_{t_0} between p_0 and $\gamma(x_1, t_0)$ is a graph over the tangent space of Γ_{t_0} at p_0 , and, by the fundamental theorem of calculus, its length ℓ satisfies $\ell \geq \rho$, since

$$\frac{1}{2} = |\langle \nu(x_1, t_0), \omega \rangle - 1| \leq \ell \kappa \leq \ell \frac{1}{2\rho}.$$

But then, its projection onto the tangent space of Γ_{t_0} at p_0 has length $d \geq \rho/2$, since

$$\ell = \int \langle \nu(x, t_0), \omega \rangle^{-1} dx \leq 2d,$$

which contradicts the fact that $\gamma(x_1, t_0)$ is in the cylinder $\mathcal{P}_{\rho/2}(p_0)$.

To prove the estimates on the normal and the height function, we will assume without loss of generality that p_0 is the origin and the tangent space of Γ_{t_0} at p_0 is the x -axis. Let's denote by $y(z)$ the local graphical parametrisation of the connected component of $\mathcal{P}_{\rho/2}(p_0) \cap \Gamma_{t_0}$ containing p_0 . We will first show the estimate on the normal, which is equivalent to showing that for any z in the domain of y , we have

$$\sqrt{1 + y'(z)^2} < 1 + \left(\frac{|z|}{2\rho} \right)^2. \quad (1.7)$$

To see this, fix such an z , and note that $|z| < \frac{\rho}{2}$. Then for any s with $|s| \leq |z|$, we have

$$\begin{aligned} |y'(s)| &= |y'(s) - y'(0)| = \left| \int_0^s y''(\sigma) d\sigma \right| \\ &\leq \int_0^s |y''(\sigma)| d\sigma \\ &\leq \int_0^s \frac{1}{2\rho} (1 + y'(\sigma)^2)^{3/2} d\sigma \\ &\leq \frac{1}{2\rho} \sqrt{|z|} \left(\int_0^s (1 + y'(\sigma)^2)^3 d\sigma \right)^{1/2}, \end{aligned}$$

where at the second inequality above we used the formula for the curvature of a graph and the assumption (1.6). Hence,

$$1 + y'(s)^2 \leq 1 + \frac{|z|}{4\rho^2} \int_0^s (1 + y'(\sigma)^2)^3 d\sigma.$$

We now claim that this implies (1.7). Indeed, let $Y(s) = 1 + \frac{|z|}{4\rho^2} \int_0^s (1 + y'(\sigma)^2)^3 d\sigma$ and notice that $Y(0) = 1$. Clearly,

$$Y'(s) = \frac{|z|}{4\rho^2} (1 + y'(s)^2)^3 \leq \frac{|z|}{4\rho^2} Y(s)^3,$$

and so by comparing with the ODE solution we see that

$$Y(s) \leq \left(1 - \frac{|z|^2}{2\rho^2}\right)^{-\frac{1}{2}},$$

from which (1.7) follows since

$$\sqrt{1 + y'(z)^2} \leq Y(s)^{\frac{1}{2}} \leq \left(1 - \frac{|z|^2}{2\rho^2}\right)^{-1/4} \leq 1 + \left(\frac{|z|}{2\rho}\right)^2.$$

Finally, to prove the height estimate, since $\frac{|z|}{2\rho} < 1$, we have

$$1 + y'(z)^2 \leq 1 + 2 \left(\frac{|z|}{2\rho}\right)^2 + \left(\frac{|z|}{2\rho}\right)^4 \implies y'(z)^2 \leq 4 \left(\frac{|z|}{2\rho}\right)^2,$$

so

$$|y'(z)| < \frac{|z|}{\rho}.$$

Thus,

$$\begin{aligned} |u(x, t_0)| &= |\langle \gamma(x, t_0) - \gamma(x_0, t_0), e_2 \rangle| \\ &= \left| \left\langle \int_0^s \frac{d}{d\sigma} (\sigma, y(\sigma)) d\sigma, e_2 \right\rangle \right| < \frac{|z|^2}{\rho}. \end{aligned} \quad \blacksquare$$

Observe that Lemma 1.27 is a general lemma on curves and is therefore true for any curve $\gamma(\cdot, t)$ in the evolution. However, a priori, even at points in $\mathcal{P}_r(p_0)$, the tangent space over which we can write the curve as a graph might vary a lot along the flow. The crucial aspect of this section is to show that this cannot happen for r sufficiently small.

Lemma 1.28. *With u as defined in Definition 1.25 (with respect to a fixed normal $\omega = \nu(x_0, t_0)$), we have that for any $t_1, t_2 \in [0, T]$:*

$$|\gamma(x, t_2) - \gamma(x, t_1)| \leq \frac{1}{2\rho} |t_2 - t_1|,$$

and

$$(|\gamma|^2 - u^2)(x, t_2) \leq \left(\frac{1}{2\rho} |t_2 - t_1| + \sqrt{|\gamma|^2 - u^2}(x, t_1) \right)^2.$$

Proof. We compute, using the evolution of γ and (1.6),

$$\begin{aligned} \left| \frac{d}{dt} |\gamma(x, t) - \gamma(x, t_1)|^2 \right| &= 2 | \langle \partial_t \gamma(x, t), \gamma(x, t) - \gamma(x, t_1) \rangle | \\ &\leq \frac{1}{\rho} |\gamma(x, t) - \gamma(x, t_1)|, \end{aligned}$$

which implies that

$$\left| \frac{d}{dt} |\gamma(x, t) - \gamma(x, t_1)| \right| \leq \frac{1}{2\rho}.$$

Integrating both from t_1 to t_2 and from t_2 to t_1 , we obtain

$$|\gamma(x, t_2) - \gamma(x, t_1)| \leq \frac{1}{2\rho} |t_2 - t_1|.$$

Next, again using the evolution of γ and (1.6), we compute

$$\begin{aligned} \left| \frac{d}{dt} (|\gamma|^2 - u^2) \right| &= 2 | \langle -\kappa \nu, \gamma - \langle \gamma, \omega \rangle \omega \rangle | \\ &\leq \frac{1}{\rho} |\gamma - \langle \gamma, \omega \rangle \omega| = \frac{1}{\rho} \sqrt{|\gamma|^2 - u^2}, \end{aligned}$$

which implies

$$\left| \frac{d}{dt} \sqrt{|\gamma|^2 - u^2} \right| \leq \frac{1}{2\rho}.$$

Integrating, as before, between t_1 and t_2 we obtain

$$\left| \sqrt{|\gamma|^2 - u^2}(x, t_2) - \sqrt{|\gamma|^2 - u^2}(x, t_1) \right| \leq \frac{1}{2\rho} |t_2 - t_1|,$$

which finishes the proof. ■

By combining Lemmas 1.27 and 1.28, we may now conclude that, for r small enough and times close to t_0 , we can control the behavior of $\Gamma_t \cap \mathcal{P}_r(p_0)$. It will be useful to introduce one more definition.

Definition 1.29. Given a time t_0 , let $r_0 > 0$ be such that the intersection of Γ_{t_0} with any ball of radius r_0 has at most one connected component

Lemma 1.30. *Let $\gamma : I \times [0, T] \rightarrow \mathbb{R}^2$ be a solution to the free boundary curve shortening flow in Ω and consider a point $p_0 = \gamma(x_0, t_0)$. Then for all $(x, t) \in I \times [0, T]$ such that $\gamma(x, t) \in \mathcal{P}_r(p_0) \cap B_{r_0/2}(p_0)$, with $r \leq \rho/4$ and $|t - t_0| \leq 2\rho \min\{r, \frac{r_0}{2}\}$,*

$$|u(x, t)| \leq \frac{|t - t_0|}{2\rho} + \frac{4r^2}{\rho}.$$

Proof. Let $\gamma(x, t) \in \mathcal{P}_r(p_0) \cap B_{r_0/2}(p_0)$ with $|t - t_0| \leq 2\rho \min\{r, \frac{r_0}{2}\}$. Then, by Lemma 1.28, we have

$$(|\gamma|^2 - u^2)(x, t_0) \leq \left(r + \sqrt{|\gamma|^2 - u^2}(x, t)\right)^2 \leq (2r)^2$$

and

$$|\gamma(x, t_0) - p_0| \leq r_0.$$

This means that $\gamma(x, t_0)$ is contained in $\mathcal{P}_{2r}(p_0) \cap B_{r_0}(p_0)$. Therefore, Lemma 1.27 implies

$$|u(x, t_0)| < \frac{4r^2}{\rho}.$$

Moreover, by the evolution of γ , and the bound on the speed (1.6), we have

$$|u(x, t) - u(x, t_0)| \leq \frac{1}{2\rho} |t - t_0|.$$

Therefore, we obtain

$$|u(x, t)| \leq |u(x, t) - u(x, t_0)| + |u(x, t_0)| \leq \frac{|t - t_0|}{2\rho} + \frac{4r^2}{\rho},$$

as required. ■

Finally, we get an estimate for the distance of a point $p \in \Gamma_{t_0}$ from the boundary portion $\Sigma_{t_0}^\Delta$, as in Definition 1.24.

Lemma 1.31. *Let $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$ be a solution to the free boundary curve shortening flow and let $t_0 \in [0, T)$ and $\Delta > 0$. Then, for any $p_0 \in \Gamma_{t_0} \setminus \Sigma$,*

$$\text{dist}(p_0, \Sigma_{t_0}^\Delta) \geq \text{dist}(p_0, \partial\Gamma_{t_0}) - \frac{1}{2\rho}\Delta.$$

where $\Sigma_{t_0}^\Delta$ is as in Definition 1.24.

Proof. Let $t \in [t_0 - \Delta, t_0 + \Delta]$ and let $x \in \partial I$. Then arguing as in Lemma 1.28, the evolution of γ together with equation (1.6) imply

$$\left| \frac{d}{dt} |\gamma(x, t) - p_0|^2 \right| \leq \frac{1}{\rho} |\gamma(x, t) - p_0|,$$

which in turn yields

$$\left| \frac{d}{dt} |\gamma(x, t) - p_0| \right| \leq \frac{1}{2\rho}.$$

Integrating between t_0 and t_1 with $t_1 \in [t_0 - \Delta, t_0 + \Delta]$ implies

$$||\gamma(x, t_0) - p_0| - |\gamma(x, t_1) - p_0|| \leq \frac{1}{2\rho}\Delta.$$

Therefore,

$$|\gamma(x, t_0) - p_0| \leq \frac{1}{2\rho}\Delta + |\gamma(x, t_1) - p_0|$$

and the claim follows. ■

1.4.1 The Interior Estimate

In this section, we still consider a solution to the free boundary curve shortening flow $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$, and our goal is to show local interior gradient bounds. We adapt Stahl's methods from (Stahl, 1996b) to obtain bounds for the function

$$v = \langle \nu(x, t), \omega \rangle^{-1},$$

given a fixed normal $\omega = \nu(x_0, t_0)$, which in turn are based off of the ideas presented in (Ecker; Huisken, 1991). To simplify the exposition, we will assume that $t_0 = 0$, $p_0 = \gamma(x_0, t_0)$ is the origin and $\omega = e_2$. First, notice that v satisfies

$$(\partial_t - \Delta)v = -\kappa v - 2\frac{v_s^2}{v},$$

which shows that a lower bound on v is preserved. The problem however is that v is only locally bounded, so we need to consider a localization of v . Because of Lemma 1.30, it makes sense to introduce a smooth cut-off function $\eta : I \times [0, T] \rightarrow \mathbb{R}$ which vanishes outside $(\mathcal{P}_r(0) \cap B_{r_0/2}(0)) \times [0, T]$, where $r \leq \rho/4$ and r_0 is as in Definition 1.29. Then $v\eta$ is bounded. We choose η to be of the form

$$\eta = e^{\lambda\varphi} - 1,$$

where the constant $\lambda > 0$ and the function φ will be chosen momentarily, and compute

$$\begin{aligned} (\partial_t - \Delta)(v\eta) &= \lambda v(\eta + 1) ((\partial_t - \Delta)\varphi - \lambda(\varphi_s)^2) \\ &\quad + \eta \left((\partial_t - \Delta)v + 2\frac{v_s^2}{v} \right) - 2\frac{(v_s)(v\eta)_s}{v}. \end{aligned}$$

Therefore, if (x, t_1) is a maximum of $v\eta$ in $I \times [0, t_1]$, with x an interior point of I , then at that point we must have

$$(\partial_t - \Delta)\varphi \geq \lambda(\varphi_s)^2. \tag{1.8}$$

Given $r \leq \rho/4$, we now choose $\varphi : I \times [0, T] \rightarrow \mathbb{R}_{\geq 0}$ to be

$$\varphi(x, t) := \begin{cases} (\frac{r}{2\beta}(u - M_{r,t_0}) + r^2 - (|\gamma|^2 - u^2))_+, & \gamma(x, t) \in \mathcal{P}_r(0) \cap B_{\frac{r_0}{2}}(0) \\ 0 & \text{otherwise.} \end{cases}$$

where

$$M_{r,t_0} := \sup\{|u(x, t)| : \gamma(x, t) \in \mathcal{P}_r(0) \cap B_{r_0/2}(0), 0 \leq t \leq t_0\},$$

for some $t_0 \in (0, T)$, and $\beta > 0$ is any constant. Observe that η is Lipschitz with compact support in $(\mathcal{P}_r(0) \cap B_{r_0/2}(0)) \times [0, t_0]$, and smooth when $\varphi > 0$. Moreover, when 0 is an interior point, by Lemma 1.31 and by taking r even smaller if necessary (depending on the distance of the origin to the boundary), we can ensure that the support of η is away from the boundary Σ . In particular, it suffices to take

$$r \leq \text{dist}(0, \partial\Sigma) - \frac{t_0}{2\rho}.$$

Therefore if (x, t_1) is a maximum of $v\eta$ in $I \times [0, t_1]$ with $0 < t_1 \leq t_0$, then x has to be an interior point. Then, applying (1.8) at that point, yields, after choosing $\lambda = 16\beta^2/r^2$,

$$v(x, t_1) \leq 2(1 + 8\beta).$$

Moreover, since $\varphi(x, t) \leq r^2$, by Lemma 1.27 we have

$$v(x, t_0) \leq 1 + \frac{r^2}{4\rho^2} \leq 1 + \frac{1}{64}.$$

Therefore, we conclude that for any (x, t) satisfying $\gamma(x, t) \in \mathcal{P}_r(0) \cap B_{r_0/2}(0)$ with $t \in [0, t_0]$,

$$(\eta v)(x, t) \leq (e^{16\beta^2} - 1)2(1 + 8\beta).$$

Finally, in order to get a bound for v , we need to bound η from below. We do this by considering points in a smaller cylinder; let (x, t) be such that $\gamma(x, t) \in \mathcal{P}_{r/2}(0) \cap B_{r_0/2}(0)$, and $t \leq 2\rho \min\{r, \frac{r_0}{2}\}$, then by Lemma 1.30,

$$\varphi(x, t) \geq -\frac{r}{2\beta} \left(\frac{|t|}{2\rho} + \frac{4r^2}{\rho} \right) + \frac{3}{4}r^2 \geq -\frac{r^2}{\beta} + \frac{3}{4}r^2.$$

Therefore, choosing $\beta = 4$ we obtain $\varphi(x, t) \geq \frac{1}{2}r^2$. The above exposition proves the following:

Theorem 1.32. *Let $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$ be a solution to the free boundary curve shortening flow. There exists a constant C such that for any interior point $p_0 = \gamma(x_0, t_0)$ where $(x_0, t_0) \in I \times [0, T)$, the following holds: Let $d_0 = \text{dist}(p_0, \partial\Sigma)$, $r \leq \min\{\frac{\rho}{4}, \frac{d_0}{2}\}$ and $\delta \leq 2\rho \min\{r, \frac{r_0}{2}, \frac{d_0}{2}\}$, then for any (x, t) , with $t_0 \leq t \leq t_0 + \delta$ and $\gamma(x, t) \in \mathcal{P}_{r/2}(p_0) \cap B_{r_0/2}(p_0)$,*

$$v(x, t) \leq C.$$

1.4.2 The Boundary Estimate

In this section we will complete the gradient estimates, by ruling out boundary maxima for the function

$$v = \langle \nu(x, t), \omega \rangle^{-1},$$

given a fixed normal $\omega = \nu(x_0, t_0)$. As we did in Section 1.4.1 we will consider a localization of this function ηv , and we will show that for sufficiently large values of ηv , the gradient at the boundary satisfies $\langle \nabla(\eta v), \nu^\Sigma \rangle < 0$. By applying the Hopf boundary lemma, we will be then able to prove that the gradient estimate holds up to the boundary. We introduce the following bound on the curvature of Σ , κ_Σ ,

$$\sup_{\Sigma} |\kappa_\Sigma| \leq \frac{1}{2\rho_\Sigma}. \quad (1.9)$$

We will assume that $t_0 = 0$ and that the fixed point $p_0 = \gamma(x_0, 0)$ is on the boundary Σ . Define the nearby point in Ω

$$p_m := p_0 - \frac{1}{K} r \nu^\Sigma(p_0),$$

where $K > 1$ and $r \leq \rho_\Sigma$. Now similar to the interior case, we consider $r \leq \rho/4$, and set

$$\eta = e^{\lambda\varphi} - 1$$

with φ defined by

$$\varphi(x, t) := \begin{cases} \left(\frac{r}{2\beta} (u - M_{r,t_0}) + r^2 - (|\gamma - p_m|^2 - u^2) \right)_+ & \gamma(x, t) \in \mathcal{P}_r(p_0) \cap B_{\frac{r_0}{2}}(p_0) \\ 0 & \text{otherwise,} \end{cases}$$

where

$$M_{r,t_0} := \sup\{|u(x, t)| : \gamma(x, t) \in \mathcal{P}_t(p_m), 0 \leq t \leq t_0\}.$$

At a boundary point we may compute

$$\langle \nabla(\eta v), \nu_\Sigma \rangle = v \langle \nabla \eta, \nu_\Sigma \rangle + \eta \langle \nabla v, \nu_\Sigma \rangle = \lambda v (\eta + 1) \langle \nabla \varphi, \nu_\Sigma \rangle + \eta \langle \nabla v, \nu_\Sigma \rangle.$$

By the curvature assumption, $\langle \nabla v, \nu_\Sigma \rangle \leq \frac{v^2}{2\rho}$. To estimate the $\nabla \varphi$ term we compute

$$\langle \nabla \varphi, \nu_\Sigma \rangle = \left(2u + \frac{r}{2\beta} \right) \langle \nu_\Sigma, \omega \rangle - 2 \langle \gamma - p_m, \nu_\Sigma \rangle.$$

Now note that by Lemma 1.30, as long as $t \leq 2\rho \min\{r, r_0/2\}$, we have $|u(x, t)| \leq 2r$. Therefore

$$|\langle \nu_\Sigma, \omega \rangle| \leq |\langle \nu_\Sigma, \nu_\Sigma(p_0) \rangle| \leq \frac{r}{\rho_\Sigma}$$

and

$$\begin{aligned} \langle \gamma - p_m, \nu_\Sigma \rangle &= \langle \gamma - p_0, \nu_\Sigma \rangle + \langle p_0 - p_m, \nu_\Sigma(p_0) \rangle + \langle p_0 - p_m, \nu_\Sigma - \nu_\Sigma(p_0) \rangle \\ &\geq \frac{r}{K} - \frac{r^2}{\rho_\Sigma}. \end{aligned}$$

So further taking, $\beta > 1$, $r < \frac{\rho_\Sigma}{12K}$ we obtain

$$\langle \nabla \varphi, \nu_\Sigma \rangle < -\frac{r}{K},$$

and, choosing $\lambda = \beta^2/r^2$ we have

$$\langle \nabla(\eta v), \nu_\Sigma \rangle < \frac{v}{2\rho}(\eta v - \frac{\beta^2}{K}).$$

Therefore, we note that we cannot have a boundary maximum that is less than β^2/K . Therefore, taking β large enough, so that it exceeds the initial value of ηv , we can argue exactly as in the interior case to bound the cutoff function from above and below to obtain:

Theorem 1.33. *Let $\gamma : I \times [0, T) \rightarrow \mathbb{R}^2$ be a solution to the free boundary curve shortening flow. There exists a constant C such that for any boundary point $p_0 = \gamma(x_0, t_0)$ with $(x_0, t_0) \in I \times [0, T)$, the following holds: Let $r \leq \min\{\frac{\rho}{4}, \frac{d_0}{2}, \rho_\Sigma\}$ and $\delta \leq 2\rho \min\{r, \frac{r_0}{2}\}$, then for any (x, t) , with $t_0 \leq t \leq t_0 + \delta$ and $\gamma(x, t) \in P_{r/2}(p_0) \cap B_{r_0/2}(p_0)$*

$$v(x, t) = \langle \nu(x, t), \nu(x_0, t_0) \rangle^{-1} \leq C.$$

Theorems 1.32 and 1.33 imply a gradient bound of the local graph representations in generalized Gaussian coordinates, $w : \mathcal{U}_r \times [t_0, t_0 + \delta] \rightarrow \mathbb{R}$, centred at a given point $\gamma(x_0, t_0)$ independent of such a point. Therefore, we have a C^1 -estimate for w on the domain $\mathcal{U}_r \times [t_0, t_0 + \delta]$, where r and δ have the correct scaling behaviour; $r \sim \rho$, $\delta \sim \rho^2$.

Proof of Theorem 1.23. Assume that we have a solution defined on a maximal time interval $[0, T_C)$. Then, by Lemma 1.30, we have the following height estimate

$$|\gamma(x, t_1) - \gamma(x, t_0)| \leq \frac{1}{2\rho} |t_1 - t_0|$$

for any $x \in I$ and any $t_0, t_1 \in [0, T_C)$. Therefore, by taking a sequence of times $t_j \rightarrow T_C$, then we can extract a limit curve

$$\gamma(x, T_C) := \lim_{t_j \rightarrow T_C} \gamma(x, t).$$

By the gradient estimates given in Theorems 1.32 and 1.33, we see that we may cover Γ_{T_C} by local graph representations, each of which satisfies a uniformly parabolic differential equation with a transversal boundary condition. Therefore, we conclude that each graph representation is C^∞ for all $t > 0$, and so Γ_{T_C} is actually smooth. Applying Theorem 1.4 now implies that we may extend T_C to some time $T_C + \delta$, contradicting the maximal time of existence. ■

2

Forwards in Time Convergence and Stability

The aim of this chapter is to better understand the long-time behavior of a convex free boundary curve shortening flow. By results of Stahl (Stahl, 1996a), it is known that such flows converge to a point. In this chapter, we present more recent results showing that the flow in fact converges to a round point (Langford; Zhu, 2023), and moreover, we establish an optimal rate of convergence (Bourni; Burns; Langford, 2026).

2.1 Forwards in Time Convergence

In this section, we prove convergence to a round point. The method we employ is based on distance comparison techniques developed by Langford and Zhu (Langford; Zhu, 2023). This approach parallels earlier work of Huisken (Huisken, 1998), who showed, using only the maximum principle, that the ratio of extrinsic to intrinsic distances for a periodic curve shortening flow remains controlled throughout the evolution. A blow-up argument can then be used to describe the long-time behavior of the flow. The case of Neumann boundary conditions is handled through the introduction of a particularly effective tool, the so-called *extended chord-arc* profile.

Let $\gamma : I \rightarrow \mathbb{R}^2$ be a curve inside a convex domain Ω with endpoints on $\Sigma = \partial\Omega$. We define the formal double

$$I = (I \sqcup I)/\partial I,$$

and write $\mathbf{x} = (x, \text{sign}(\mathbf{x}))$ for elements of I , where $\text{sign}(\mathbf{x})$ distinguishes to which copy of I it belongs. We also write

$$\gamma : I \rightarrow \mathbb{R}^2,$$

defined by $\gamma(\mathbf{x}) = \gamma(x)$. We will denote by $d(x, y)$ the Euclidean distance between the points $x, y \in \Omega$ (the *extrinsic* distance) and by $l(x, y)$ the arc-length along the curve Γ between $x, y \in \Gamma$ (the *intrinsic* distance). We can extend these notions to a *reflected distance* $\tilde{d}(x, y)$ between two points in Ω and the *reflected arc-length* $\tilde{l}(x, y)$ between two points $x, y \in \Gamma$ by

$$\tilde{d}(x, y) = \min_{z \in \Sigma} (|x - z| + |y - z|),$$

and

$$\tilde{l}(x, y) = \min_{s \in \partial\Gamma} (l(x, s) + l(y, s)).$$

Finally, we can define doubled versions on these quantities on $\mathbf{I} \times \mathbf{I}$ by

$$\mathbf{l}(\mathbf{x}, \mathbf{y}) := \begin{cases} l(x, y), & \text{if } \text{sign}(\mathbf{x}) = \text{sign}(\mathbf{y}) \\ \tilde{l}(x, y), & \text{if } \text{sign}(\mathbf{x}) \neq \text{sign}(\mathbf{y}), \end{cases}$$

and

$$\mathbf{d}(\mathbf{x}, \mathbf{y}) := \begin{cases} d(\gamma(x), \gamma(y)), & \text{if } \text{sign}(\mathbf{x}) = \text{sign}(\mathbf{y}) \\ \tilde{d}(\gamma(x), \gamma(y)), & \text{if } \text{sign}(\mathbf{x}) \neq \text{sign}(\mathbf{y}). \end{cases}$$

The *completed chord-arc profile* ψ of Γ is then defined by

$$\psi(\delta) := \inf\{\mathbf{d}(x, y) \mid x, y \in \Gamma, \mathbf{l}(\mathbf{x}, \mathbf{y}) = \delta\}.$$

The main result of (Langford; Zhu, 2023) is given below.

Theorem 2.1. *Let $\{\Gamma_t\}_{t \in [0, T]}$ be a compact free boundary curve shortening flow in Ω . Let $L(t)$ denote the arc-length of Γ_t . Suppose that $L(0)(1 + C) \leq \frac{\epsilon}{1000}$ for some $\epsilon \in (0, \frac{1}{10})$, where here $C = \sup_{\Sigma} \kappa_{\Sigma}$. Given any $c \in (0, \frac{1}{100})$, if the inequality*

$$\psi(\delta, t) \geq c\mathbf{L}(t) \left(\sin \left((\pi - \epsilon) \frac{\delta}{\mathbf{L}(t)} + \frac{\epsilon}{2} \right) - \sin \left(\frac{\epsilon}{2} \right) \right)$$

holds at $t = 0$, then it holds for all $t \in [0, T]$.

We note that if $L(t) \rightarrow 0$ as $t \rightarrow T$, then $(1 + C)L(t) \leq \frac{\epsilon}{1000}$ is eventually satisfied for any $\epsilon \in (0, \frac{1}{10})$.

This is a highly technical and elegant proof based on two-point maximum principles. We believe it is a particularly insightful argument, and we refer the reader to the original paper (Langford; Zhu, 2023) for the details. Including the full proof here would make the chapter unnecessarily lengthy, so we have chosen to omit it.

2.1.1 The Intrinsic Blow-Up

Precisely as in (Langford; Zhu, 2023, Section 6.3), we will use Theorem 2.1 to classify smooth free boundary blow-ups. We begin with the following classification result.

Lemma 2.2. *The only convex ancient free boundary curve shortening flows in the half-plane $\mathbb{R}_+^2$ are the shrinking round semi-circles, the stationary half-lines, the half-Grim reapers, and the half-Engenent ovals.*

Proof. Let $\{\Gamma_t\}_{t \in (-\infty, \omega)}$ be a convex ancient free boundary curve shortening flow in $\mathbb{R}_+^2$. By reflecting Γ_t across $\partial \mathbb{R}_+^2$ and noting that all odd-order derivatives of κ vanish at $\partial \mathbb{R}_+^2$ (which can be seen by repeatedly differentiating the boundary condition with respect to time) we obtain a convex ancient periodic curve shortening flow in the plane. So the claim follows by the classification done in (Bourni; Langford; Tinaglia, 2020; Daskalopoulos; Hamilton; Sesum, 2010). ■

Theorem 2.3. *Let $\Omega \subset \mathbb{R}^2$ be a convex domain with C^2 boundary. Let $\{\Gamma_t\}_{t \in [0, T)}$ be a maximal free boundary curve shortening flow starting from an embedded convex curve Γ_0 . Then, as $t \rightarrow T$, Γ_t converges to a point $z \in \Sigma$, and*

$$\tilde{\Gamma}_t := \frac{\Gamma_t - z}{\sqrt{2(T-t)}}$$

converges uniformly in the smooth topology as $t \rightarrow T$ to the unit semi-circle in $T_z\Omega$.

Proof. Since the curvature satisfies

$$\begin{cases} (\partial_t - \Delta)\kappa = \kappa^3 & \text{in } \Gamma_t \\ \langle \nabla \kappa, \nu_\Sigma \rangle = \kappa \kappa_\Sigma & \text{on } \partial \Gamma_t, \end{cases}$$

the maximum principle implies that

$$\max_{\Gamma_t} \kappa^2 \geq \frac{1}{2(T-t)}.$$

We claim that $\limsup_{t \rightarrow T} \max_{\Gamma_t} (T-t)\kappa^2 < \infty$. If this is not the case, then we proceed with a standard blow-up procedure; choose a sequence of times $t_j \in [0, T - \frac{1}{j})$ and a sequence of points $x_j \in I$ such that

$$\kappa^2(x_j, t_j) \left(T - \frac{1}{j} - t_j \right) = \max_{(x,t) \in I \times [0, T - \frac{1}{j}]} \kappa^2(x, t) \left(T - \frac{1}{j} - t \right),$$

and set $\sigma_j := \lambda_j^2 t_j$ and $T_j = \lambda_j^2 \left(T - \frac{1}{j} - t_j \right)$ where $\lambda_j := |\kappa(x_j, t_j)|$. Now we take a sequence of rescaled flows:

$$\gamma_j : I \times (-\sigma_j, T_j) \rightarrow \lambda_j(\Omega - \gamma(x_j, t_j)),$$

defined by

$$\gamma_j(x, t) = \lambda_j \left(\gamma(x, t_j + \lambda_j^{-2} t) - \gamma(x_j, t_j) \right).$$

Since we are assuming $\max_{\Gamma_t}(T - t)\kappa^2 \rightarrow \infty$ as $t \rightarrow T$, we can take a subsequential limit $t_j \rightarrow T$ such that $\lambda_j \rightarrow \infty$, $\sigma_j \rightarrow \infty$, $T_j \rightarrow \infty$ and $\lambda_j(\Omega - \gamma(x_j, t_j)) \rightarrow \Omega_\infty$, where Ω_∞ is either the plane or some half-plane. Since $\gamma_j(x_j, 0) = 0$ and

$$\kappa_j^2(x, t) = \lambda_j^{-2} \kappa^2(x, t_j + \lambda_j^{-2} t) \leq \frac{T - \frac{1}{j} - t_j}{T - \frac{1}{j} - (t_j + \lambda_j^{-2} t)} = \frac{T_j}{T_j - t},$$

the curvature is uniformly bounded on any compact time interval and (up to a change of orientation) takes unit value at the space-time origin. Therefore, arguing as in the proof of Theorem 1.23, the rescaled solutions γ_j converge in C^∞ after passing to a subsequence, to a smooth, proper limiting solution $\gamma_\infty : I \times (-\infty, \infty) \rightarrow \Omega_\infty$ uniformly on compact subsets of $I \times (-\infty, \infty)$. Since we have assumed convexity of our solution, and that curvature takes unit value at the space-time origin, we conclude that the limit flow is not a critical chord and the limiting flow is locally uniformly convex (by the strong maximum principle). The classification result for convex ancient periodic curve shortening flows implies that γ_∞ is the half-Grim reaper (since it is the only eternal solution), which is impossible due to Theorem 2.1.

Next, we will show that both

$$\limsup_{t \rightarrow T} \max_{\Gamma_t} |2(T - t)\kappa^2 - 1| = 0, \quad (2.1)$$

and

$$\limsup_{t \rightarrow T} \max_{\Gamma_t} (T - t)^{-(m+1)} |\nabla^m \kappa|^2 = 0, \quad (2.2)$$

for all m . To do this, as before take a sequence of times $t_j \rightarrow T$, choose $x_j \in I$ so that

$$\kappa^2(x_j, t_j) = \max_{x \in I} \kappa^2(x, t_j),$$

and set $\lambda_j = (T - t_j)^{\frac{1}{2}}$ and $\sigma_j = \lambda_j^2 t_j$. The sequence of rescaled solutions

$$\gamma_j : I \times (-\sigma_j, 1) \rightarrow \lambda_j(\Omega - \gamma(x_j, t_j))$$

defined by

$$\gamma_j(x, t) = \lambda_j \left(\gamma(x, t_j + \lambda_j^{-2} t) - \gamma(x_j, t_j) \right),$$

converges in the smooth topology, after perhaps passing to a subsequence, to a smooth, proper limiting solution

$$\gamma_\infty : I \times (-\infty, 1) \rightarrow \Omega_\infty,$$

where Ω_∞ is either the plane or the half-plane. Since

$$\max_{\Gamma_t} \kappa^2 \geq \frac{1}{2(T-t)}$$

and the fact the chord-arc estimate in Theorem 2.1 is scale invariant, we deduce that γ_∞ is the shrinking semi-circle, and so the claim follows. Convergence to a point on Σ follows by integrating the curve shortening flow equation (1.1) and applying (2.1). Smooth convergence to the unit semi-circle after a rescaling follows by converting (2.2) into estimates for the rescaled immersions. ■

2.1.2 A Brief Comment on the General Case

We comment here that much more may be deduced from the chord-arc estimate of Langford and Zhu, Theorem 2.1. By adapting a monotonicity formula of Altschuler (Altschuler, 1991, Theorem 5.14), we may conclude that a blow-up limit is locally uniformly convex, even when we replace the convexity condition of our solution to any solution with a finite time of existence. Furthermore, carefully estimating the curvature and length of a solution whose time of existence is infinite, and applying some regularity theory for parabolic PDE's with a transversal boundary condition establishes the long-time behaviour of such a solution. The full statement is as follows:

Theorem 2.4. (Langford; Zhu, 2023, Theorem 6.1) *Let $\{\Gamma_t\}_{t \in [0, T)}$ be a maximal free boundary curve shortening flow starting from a properly embedded curve $\Gamma_0 \subset \Omega$. If $T < \infty$, then $\Gamma_t \rightarrow z \in \Sigma$ and*

$$\tilde{\Gamma}_t := \frac{\Gamma_t - z}{\sqrt{2(T-t)}}$$

converges uniformly in the smooth topology as $t \rightarrow T$ to the unit semi-circle in $T_z\Omega$. Alternatively, if $T = \infty$, then Γ_t converges smoothly as $t \rightarrow \infty$ to a chord in Ω which meets Σ orthogonally.

2.2 Forwards in time stability

Now we refine even further the forward-in-time behavior of solutions to the free boundary curve shortening flow described in Section 2.1. There, we established in Theorem 2.3 that any convex solution shrinks to a round half-point. We will here describe the asymptotic profile of the flow near the extinction time.

The main result shows that, after an appropriate rescaling about the extinction point, solutions converge to a shrinking semicircle at an explicit and optimal rate in all C^k norms, as shown in (Bourni; Burns; Langford, 2026).

Theorem 2.5. *Let $\Omega \subset \mathbb{R}^2$ be a convex domain with C^2 boundary and let $\{\Gamma_t\}_{t \in [0, T)}$ be a maximal free boundary curve shortening flow starting from a properly embedded curve Γ_0 inside Ω . If $T < \infty$, then Γ_t converges uniformly to some $p \in \partial\Omega$, and, denoting by $\mathcal{R}$ the rotation of $\mathbb{R}^2$ which maps the outward normal to $\partial\Omega$ at p to $-e_2$, the rescaled curves*

$$\tilde{\Gamma}_t := \frac{\mathcal{R}(\Gamma_t - p)}{\sqrt{2(T - t)}}$$

converge as $t \rightarrow T$ to the unit semi-circle in $\mathbb{R}_+^2$ uniformly in the C^k -topology, for every $k \geq 0$, at a rate no slower than $(T - t)^{1-\delta}$, for any $\delta > 0$.

The proof is based on a stability analysis of the shrinking semicircle, viewed as a self-similar solution of the rescaled flow. The associated linearized operator exhibits unstable modes corresponding to time translations and horizontal translations, which obstruct direct convergence estimates. To overcome this difficulty, we introduce a normalization of the flow which dynamically compensates for these symmetries. This is achieved by fixing two geometric quantities: the enclosed area between the evolving curve and the support curve, and a suitably defined center of mass. With this normalization, the unstable modes can be eliminated, allowing the stability argument to close and yielding sharp convergence rates.

This strategy is inspired by the approach of Ben Andrews in his treatment of the affine normal flow for closed hypersurfaces ([Andrews *et al.*, 2020](#), §16.5.2). Here, however, the method must be adapted to the free-boundary setting and to the specific features of curve shortening flow. The presence of a boundary introduces additional complications. In particular, when working with the Gauss map parametrization, the domain of definition is no longer fixed in time, which makes the linearized stability analysis around the shrinking semicircle considerably more delicate.

Finally, we note that in the special case where the domain Ω is a hyperplane, the order of decay given in the main theorem can be improved to $2 - \delta$ (instead of $1 - \delta$). This follows from the proof of the theorem by taking the rescaled barrier support function and curvature to be zero throughout. (Of course, this case can also be derived from the GageHamilton theorem ([GaHa86](#)), after reflecting the solution.) On the other hand, if the contraction point lies on a positively curved region of the barrier, then the rescaled barrier support function, and hence the derivative of the support function of the evolving curve (according to the free boundary condition) is of exponential decay order 1, and we therefore cannot expect a better decay order than $1 - \delta$ in that case.

Consider a compact solution $\{\Gamma_t\}_{t \in [0, T)}$ to the free boundary curve shortening flow in a convex, locally uniformly convex domain $\Omega \subset \mathbb{R}^2$. Denoting the boundary of this domain by $\Sigma := \partial\Omega$, suppose that Γ_t converges to the round half-point $p \in \Sigma$ as $t \rightarrow T < \infty$; i.e.,

$$\frac{\Gamma_t - p}{\sqrt{2(T - t)}} \xrightarrow{t \rightarrow T} \mathcal{C}^+ \tag{2.3}$$

in the smooth topology, where $\mathcal{C}^+$ is a unit semi-circle in $\overline{H}$, the closed half-plane which supports $\Omega - p$ at the origin. Up to a rigid motion, a time-translation and a parabolic rescaling, we may arrange so that p is the origin, that $\overline{H}$ is the upper half-plane and that the initial curve Γ_0 is arbitrarily close, in the smooth topology, to the unit semi-circle in $\overline{H}$. In particular, Γ_t will be convex for $t \geq 0$ and therefore we may consider the boundary value problem for its support function $\sigma(\cdot, t)$. We will adopt a Gauss map parametrisation, that is we will parametrize by the angle $\theta \in [\underline{\theta}(t), \overline{\theta}(t)]$ the outward unit normal $\mathbf{N} = (\cos \theta, \sin \theta)$ makes with the positive x -axis. By outward pointing here, we mean with respect to the convex region that the

solution and part of Σ bound. Then, σ satisfies the following problem:

$$\begin{cases} \sigma_t(\theta, t) &= -(\sigma_{\theta\theta} + \sigma)^{-1}(\theta, t), \quad (\theta, t) \in (\underline{\theta}, \bar{\theta}) \times [0, T), \\ \sigma_\theta(\underline{\theta}, t) &= -\sigma_\Sigma(\bar{\theta}_\Sigma), \quad t \in [0, T), \\ \sigma_\theta(\bar{\theta}, t) &= \sigma_\Sigma(\underline{\theta}_\Sigma), \quad t \in [0, T), \end{cases} \quad (2.4)$$

where σ_Σ is the support function of Σ parametrised by the Gauss map so that

$$(\underline{\theta}_\Sigma(t), \bar{\theta}_\Sigma(t)) = (\frac{\pi}{2} + \bar{\theta}(t), \frac{3\pi}{2} + \underline{\theta}(t)). \quad (2.5)$$

Note that, for notational convenience, we will often omit the argument t in the endpoints of the angle intervals $\bar{\theta}, \underline{\theta}, \bar{\theta}_\Sigma, \underline{\theta}_\Sigma$, as we did for example in (2.4).

By the smooth convergence in (2.3), there exists a constant C , such that the curvature κ of the evolving curves satisfies

$$C^{-1} \leq \sqrt{2(T-t)}\kappa(\theta, t) \leq C, \quad \forall (\theta, t) \in [\underline{\theta}, \bar{\theta}] \times [0, T). \quad (2.6)$$

Since

$$\frac{d}{dt}\bar{\theta} = \kappa_\Sigma(\underline{\theta}_\Sigma)\kappa(\bar{\theta}, t) \quad \text{and} \quad \frac{d}{dt}\underline{\theta} = -\kappa_\Sigma(\bar{\theta}_\Sigma)\kappa(\underline{\theta}, t), \quad (2.7)$$

where κ_Σ is the curvature of the boundary Σ , we obtain

$$\frac{C^{-1}}{\sqrt{2(T-t)}} \leq \frac{d}{dt}(\bar{\theta} - \underline{\theta}) \leq \frac{C}{\sqrt{2(T-t)}}. \quad (2.8)$$

Integrating this estimate from time t to T yields,

$$C^{-1}\sqrt{2(T-t)} \leq \pi - (\bar{\theta} - \underline{\theta}) \leq C\sqrt{2(T-t)}. \quad (2.9)$$

Linear stability

The linearisation of (2.4) at the shrinking semi-circle is given by

$$\begin{cases} v_t(\theta, t) &= \frac{-1}{2t}(v_{\theta\theta}(\theta, t) + v(\theta, t)), \quad \theta \in (0, \pi), \\ v_\theta(0, t) &= 0, \\ v_\theta(\pi, t) &= 0, \end{cases}$$

where $t \in (-\infty, 0)$. Introducing the rescaled variables

$$\tilde{v}(\theta, \tilde{t}) := e^{\tilde{t}} v(\theta, t), \quad \tilde{t} = -\frac{1}{2} \log(-2t)$$

transforms this equation into

$$\begin{cases} \tilde{v}_{\tilde{t}}(\theta, \tilde{t}) &= \tilde{v}_{\theta\theta}(\theta, \tilde{t}) + 2\tilde{v}(\theta, \tilde{t}), \quad \theta \in (0, \pi), \\ \tilde{v}_{\theta}(0, \tilde{t}) &= 0, \\ \tilde{v}_{\theta}(\pi, \tilde{t}) &= 0, \end{cases}$$

where $\tilde{t} \in (-\infty, +\infty)$ here. Separation of variables then gives the representation

$$\tilde{v}(\theta, \tilde{t}) = \sum_{j=0}^{\infty} C_j e^{(2-j^2)\tilde{t}} \cos(j\theta).$$

Note that the unstable mode $j = 0$ corresponds to time-translations and the unstable mode $j = 1$ corresponds to horizontal translation. Indeed, denoting by $\gamma(\cdot, t)$ the Gauss map parametrization of Γ_t , if $\gamma_{\varepsilon}(\theta, t) := \gamma(\theta, t - \varepsilon)$, then $\frac{d}{d\varepsilon}\big|_{\varepsilon=0} \langle \gamma_{\varepsilon}, \mathbf{N} \rangle = 1$, and if $\gamma_{\varepsilon}(\theta, t) := \gamma(\theta, t) + \varepsilon e_1$, then $\frac{d}{d\varepsilon}\big|_{\varepsilon=0} \langle \gamma_{\varepsilon}, \mathbf{N} \rangle = \cos \theta$.

2.2.1 Geometric Functionals and Their Evolution Equations

The Enclosed Area Functional

Consider a fixed convex domain $U \subset \Omega$ bounded by a locally uniformly convex free boundary portion Γ and convex barrier portion of $\partial\Omega$, which, by abusing notation, we will still denote by Σ . By denoting with $\mathbf{N}_{\Sigma}$ the outward pointing unit normal to Σ , and X the position vector, the area $|U|$, enclosed by Γ and Σ , is given by

$$\begin{aligned} 2|U| &= 2 \int_U 1 dX = \int_U \operatorname{div} X dX \\ &= \int_{\Gamma} X \cdot \mathbf{N} ds + \int_{\Sigma} X \cdot \mathbf{N}_{\Sigma} ds \\ &= \int_{\underline{\theta}}^{\bar{\theta}} \frac{\sigma}{\kappa} d\theta + \int_{\Sigma} X \cdot \mathbf{N}_{\Sigma} ds_{\Sigma} \\ &= \int_{\underline{\theta}}^{\bar{\theta}} \sigma(\sigma_{\theta\theta} + \sigma) d\theta + \int_{\Sigma} X \cdot \mathbf{N}_{\Sigma} ds \\ &=: \mu(\sigma) + \mu_{\Sigma}. \end{aligned}$$

The first variation of μ at $\sigma = 1$ and along $v := \frac{d}{d\varepsilon}\big|_{\varepsilon=0}\sigma_\varepsilon$ is given by

$$D\mu|_1(v) = \frac{d}{d\varepsilon}\bigg|_{\varepsilon=0} \mu(\sigma_\varepsilon) = v_\theta \bigg|_{\underline{\theta}}^{\bar{\theta}} + 2 \int_{\underline{\theta}}^{\bar{\theta}} v d\theta.$$

Therefore,

$$\begin{aligned} 2|U| &= \mu_\Sigma + \mu(1) + D\mu|_1(\sigma - 1) + O(\|\sigma - 1\|_{C^2}^2) \\ &= \mu_\Sigma + \bar{\theta} - \underline{\theta} + (\sigma - 1)_\theta \bigg|_{\underline{\theta}}^{\bar{\theta}} + 2 \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) d\theta + O(\|\sigma - 1\|_{C^2}^2) \\ &= \mu_\Sigma + \bar{\theta} - \underline{\theta} + \sum_{X \in \partial\Sigma} X \cdot \mathbf{N}_\Sigma + 2 \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) d\theta + O(\|\sigma - 1\|_{C^2}^2), \end{aligned}$$

which yields

$$2 \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) d\theta = 2|U| - (\bar{\theta} - \underline{\theta}) - \mu_\Sigma - \sum_{X \in \partial\Sigma} X \cdot \mathbf{N}_\Sigma + O(\|\sigma - 1\|_{C^2}^2). \quad \mathbf{(2.10)}$$

The Center of Mass Functional

With the same setup as in the enclosed area functional, we define the center of mass to be

$$q(U) = \int_U X \cdot e_1 dX,$$

for a domain U bounded by a locally uniformly convex free boundary portion Γ and a locally uniformly convex barrier portion Σ . The center of mass is then determined by

$$\begin{aligned} 2q(U) &= 2 \int_U X \cdot e_1 dX = \int_U (X \cdot e_1) \operatorname{div} X dX \\ &= \int_U (\operatorname{div}((X \cdot e_1)X) - X \cdot e_1) dX, \end{aligned}$$

which gives

$$\begin{aligned}
3q(U) &= \int_U \operatorname{div}((X \cdot e_1)X) dX \\
&= \int_{\Gamma} (X \cdot e_1)X \cdot \mathbf{N} ds + \int_{\Sigma} (X \cdot e_1)X \cdot \mathbf{N}_{\Sigma} ds \\
&= \int_{\underline{\theta}}^{\bar{\theta}} (\sigma \cos \theta + \sigma_{\theta} \sin \theta) \sigma (\sigma_{\theta\theta} + \sigma) d\theta + \int_{\Sigma} (X \cdot e_1)X \cdot \mathbf{N}_{\Sigma} ds \\
&=: \nu(\sigma) + \nu_{\Sigma}.
\end{aligned}$$

The first variation of ν at $\sigma = 1$ along $v := \frac{d}{d\varepsilon} \big|_{\varepsilon=0} \sigma_{\varepsilon}$ is given by

$$D\nu|_1(v) = \frac{d}{d\varepsilon} \bigg|_{\varepsilon=0} \nu(\sigma_{\varepsilon}) = 2v \sin \theta \Big|_{\underline{\theta}}^{\bar{\theta}} + v_{\theta} \cos \theta \Big|_{\underline{\theta}}^{\bar{\theta}} + \int_{\underline{\theta}}^{\bar{\theta}} v \cos \theta d\theta.$$

So preceding as with the area functional above we have

$$\begin{aligned}
3q(U) &= \nu_{\Sigma} + \nu(1) + D\nu|_1(\sigma - 1) + O(\|\sigma - 1\|_{C^2}^2) \\
&= \nu_{\Sigma} + [\sigma \sin \theta + \sigma_{\theta} \cos \theta + (\sigma - 1) \sin \theta]_{\underline{\theta}}^{\bar{\theta}} \\
&\quad + \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \cos \theta d\theta + O(\|\sigma - 1\|_{C^2}^2) \\
&= \int_{\Sigma} (X \cdot e_1)X \cdot \mathbf{N}_{\Sigma} ds + [X \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} + \sum_{\partial\Gamma} (1 - X \cdot \mathbf{N}) \mathbf{N}_{\Sigma} \cdot e_1 \\
&\quad \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \cos \theta d\theta + O(\|\sigma - 1\|_{C^2}^2).
\end{aligned}$$

Here we have used the identity

$$X \cdot e_2 = (\sigma \mathbf{N} + \sigma_{\theta} \mathbf{T}) \cdot e_2 = \sigma \sin \theta + \sigma_{\theta} \cos \theta,$$

where $\mathbf{T}$ is the unit tangent to the curve (counterclockwise rotation on $\mathbf{N}$ by $\pi/2$). We also adopt the notation for the difference of a function evaluated at the boundary points: $[f]_{\underline{\theta}}^{\bar{\theta}} = f(\bar{\theta}) - f(\underline{\theta})$, which typically arises from integrating a derivative. Therefore, we obtain

$$\begin{aligned}
\int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \cos \theta d\theta &= 3q(U) - \int_{\Sigma} (X \cdot e_1) \sigma_{\Sigma} ds - [X \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} \\
&\quad + \sum_{\partial\Gamma} (1 - \sigma)(\mathbf{N}_{\Sigma} \cdot e_1) + O(\|\sigma - 1\|_{C^2}^2).
\end{aligned} \tag{2.11}$$

Evolution equations

We now want to calculate the evolution equations of the enclosed area $|U_{(t)}|$ and the center of mass $q(U_{(t)})$ associated with the evolving curve Γ_t . After an appropriate translation and rescaling, we expect that these quantities satisfy $|U_{(t)}| - (\bar{\theta} - \underline{\theta}) \sim 0$ and $q(U_{(t)}) \sim 0$. We therefore derive their evolution equations for the rescaled flow,

$$\tilde{\Gamma}_{\tilde{t}} = \lambda(t)(\Gamma_t - p(t)e_1), \quad \frac{d\tilde{t}}{dt} = \lambda^2. \quad (2.12)$$

We will use the Gauss map parametrization for both the evolving curves and Σ , which we denote by γ and γ_Σ , respectively. Recall that the support function of Σ , denoted by σ_Σ , is defined on the interval $[\underline{\theta}_\Sigma, \bar{\theta}_\Sigma]$, while the support function $\sigma(\cdot, t)$ of Γ_t is defined on $[\underline{\theta}, \bar{\theta}]$, with the two intervals related as described in (2.5).

To simplify notation, we suppress the explicit dependence on t in the domains, and we adopt the same convention for other quantities throughout. Moreover, we use the notation $\{f\}$ to denote the sum of a function evaluated at the boundary points of Γ_t ; that is,

$$\{f\} := \sum_{\partial\Gamma_t} f.$$

Note that $\partial\Gamma_t = \partial\Sigma$. Since we are using the Gauss map parametrization, the boundary points correspond to the angles $\bar{\theta}$ and $\underline{\theta}$ when referring to functions defined on Γ_t , and to $\underline{\theta}_\Sigma$ and $\bar{\theta}_\Sigma$ when referring to functions defined on Σ . Because all functions are expressed in terms of the angular parameter, the meaning of this notation will be clear from the context. For example,

$$\{\sigma\} = \sigma(\bar{\theta}) + \sigma(\underline{\theta}), \quad \{\sigma_\Sigma\} = \sigma_\Sigma(\bar{\theta}_\Sigma) + \sigma_\Sigma(\underline{\theta}_\Sigma),$$

and

$$\{\sigma\kappa_\Sigma\} = \sigma(\bar{\theta})\kappa_\Sigma(\underline{\theta}_\Sigma) + \sigma(\underline{\theta})\kappa_\Sigma(\bar{\theta}_\Sigma).$$

Note that this notation should not be confused with the notation we will use for the *difference* of a function evaluated at the boundary points, $[f]_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} = f(\bar{\theta}_\Sigma) - f(\underline{\theta}_\Sigma)$ or $[f]_{\underline{\theta}}^{\bar{\theta}} = (\bar{\theta}) - f(\underline{\theta})$, which typically arises from integrating a derivative.

We also introduce the translated position vector

$$X_p := X - p e_1,$$

and define the corresponding support functions by

$$\sigma^p := X_p \cdot \mathbf{N}, \quad \sigma_\Sigma^p := X_p \cdot \mathbf{N}_\Sigma.$$

Finally we introduce the difference of boundary angles function

$$\Theta(t) = \bar{\theta} - \underline{\theta}$$

as well as the scale function

$$\Lambda(t) = \frac{1}{\sqrt{2(T-t)}}.$$

In terms of asymptotics, we will use the standard $O - o$ notation for quantities defined on the flow $\{\Gamma_t\}$. In particular, we will use $O(\Lambda^k)$, $k \in \mathbb{Z}$, to mean that a quantity divided by Λ^k is bounded in absolute value everywhere and for all times in $[0, T)$ and $o(\Lambda)$ to mean that moreover the bound tends to zero as $t \rightarrow T$.

Evolution of the enclosed area functional

We define the following quantity which is an approximation of

$$\int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) d\theta$$

for the rescaled flow, as seen by (2.10):

$$U(t) := 2\lambda^2(t)|U_{(t)}| - \Theta(t) - \lambda^2(t) \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} \frac{\sigma_\Sigma}{\kappa_\Sigma} d\theta - \Lambda(t)\{\sigma_\Sigma\}, \quad (2.13)$$

where with $U_{(t)}$ we denote the enclosed area, as defined in section 2.2.1, of the rescaled flow at time t . Note that this does not depend on the translation by $p \cdot e_1$.

Lemma 2.6. *The evolution of U , defined in (2.13), is given by*

$$\begin{aligned} \frac{d}{dt}U(t) = & \frac{\lambda_t}{\lambda} (2U(t) + 2\Theta(t) + 2\Lambda(t)\{\sigma_\Sigma\}) \\ & - \lambda^2(2(\bar{\theta} - \underline{\theta}) - \{\sigma_\Sigma\kappa\}) \\ & + \Lambda(t)\{\sigma\kappa\kappa_\Sigma\} - \{\kappa_\Sigma\kappa\} - \Lambda^3\{\sigma_\Sigma\}. \end{aligned} \quad (2.14)$$

Proof. We compute the evolution of each of the terms in the definition of U separately. Recall first (2.7), which gives

$$\Theta'(t) = \kappa_\Sigma(\underline{\theta}_\Sigma)\kappa(\bar{\theta}, t) + \kappa_\Sigma(\bar{\theta}_\Sigma)\kappa(\underline{\theta}, t) = \{\kappa_\Sigma\kappa\}.$$

The terms involving Σ are affected only by the evolution of the endpoints, therefore

$$\frac{d}{dt} \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} \frac{\sigma_\Sigma}{\kappa_\Sigma} d\theta = -\{\sigma_\Sigma\kappa\},$$

and

$$\frac{d}{dt}\{\sigma_\Sigma\} = -\{\sigma\kappa\kappa_\Sigma\}.$$

For the enclosed area, we have, by Lemma 1.12,

$$\frac{d}{dt}|U(t)| = -(\bar{\theta} - \underline{\theta}) = -\Theta(t).$$

Therefore, putting everything together, we have

$$\begin{aligned} \frac{d}{dt}U(t) = & 4\lambda\lambda_t|U(t)| - 2\lambda\lambda_t \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} \frac{\sigma_\Sigma}{\kappa_\Sigma} d\theta - \Lambda_t\{\sigma_\Sigma\} \\ & - \lambda^2(2(\bar{\theta} - \underline{\theta}) - \{\sigma_\Sigma\kappa\}) + \Lambda\{\sigma\kappa\kappa_\Sigma\} - \{\kappa_\Sigma\kappa\} \\ = & \frac{\lambda_t}{\lambda} (2U(t) + 2\Theta + 2\Lambda\{\sigma_\Sigma\}) \\ & - \Lambda^3\{\sigma_\Sigma\} - \lambda^2(2(\bar{\theta} - \underline{\theta}) - \{\sigma_\Sigma\kappa\}) + \Lambda\{\sigma\kappa\kappa_\Sigma\} - \{\kappa_\Sigma\kappa\}. \quad \blacksquare \end{aligned}$$

Evolution of the center of mass functional

We define the following quantity which is an approximation of

$$\int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \cos \theta d\theta$$

for the rescaled flow, as seen by (2.11):

$$q(t) = \lambda^3 \left(3q(U_{(t)}^p) - \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} (X \cdot e_1) \frac{\sigma_\Sigma}{\kappa_\Sigma} d\theta - \frac{1}{\Lambda(t)^2} [\gamma \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} \right), \quad (2.15)$$

where $q(U_{(t)}^p)$ is the center of mass, as defined in (2.2.1), for the rescaled flow, at time t .

Lemma 2.7. *The evolution of q , defined in (2.15), is given by*

$$\begin{aligned} \frac{d}{dt} q(t) &= 3 \frac{\lambda_t}{\lambda} q(t) \\ &+ \lambda^3 \left[- \int_{\underline{\theta}}^{\bar{\theta}} (X \cdot e_1 - p) d\theta - p_t |U_{(t)}| \right. \\ &\quad \left. + \{ (X \cdot e_1) \sigma_\Sigma \kappa \} + \frac{1}{\Lambda^2} [\kappa \sin \Gamma]_{\underline{\theta}}^{\bar{\theta}} + 2 [\gamma \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} \right]. \end{aligned} \quad (2.16)$$

Proof. We compute the evolution of each term in the definition of q separately. We first compute the evolution of the terms that involve only Σ , which only depends on the motion of the endpoints.

$$\frac{d}{dt} \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} (X \cdot e_1) \frac{\sigma_\Sigma}{\kappa_\Sigma} d\theta = - \{ (X \cdot e_1) \sigma_\Sigma \kappa \},$$

and

$$\frac{d}{dt} [\gamma \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} = - [\kappa \sin]_{\underline{\theta}}^{\bar{\theta}}.$$

To compute the evolution of $q(U_{(t)}^p)$ we proceed as follows.

$$\begin{aligned} \frac{d}{dt} q(U_{(t)}^p) &= \frac{d}{dt} \int_{\underline{\theta}}^{\bar{\theta}} (X_p \cdot e_1) \frac{\sigma^p}{\kappa} d\theta + \frac{d}{dt} \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} (X_p \cdot e_1) \frac{\sigma_\Sigma^p}{\kappa_\Sigma} d\theta \\ &= \{ X_p \cdot e_1 \sigma^p \kappa_\Sigma \} + \{ X_p \cdot e_1 \sigma_\Sigma^p \kappa \} \\ &\quad - \int_{\underline{\theta}}^{\bar{\theta}} (X_p \cdot e_1) (1 + \sigma^p (\kappa_{\theta\theta} + \kappa)) + \frac{\sigma^p}{\kappa} (\kappa \cos \theta - \kappa_\theta \sin \theta) d\theta \\ &\quad - \int_{\underline{\theta}}^{\bar{\theta}} \frac{p_t}{\kappa} (\sigma^p + X_p \cdot e_1 \cos \theta) d\theta - \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} \frac{p_t}{\kappa_\Sigma} (\sigma_\Sigma^p + X_p \cdot e_1 \cos \theta) d\theta. \end{aligned}$$

Observe that

$$\begin{aligned}
& \int_{\underline{\theta}}^{\bar{\theta}} (X_p \cdot e_1)(1 + \sigma^p(\kappa_{\theta\theta} + \kappa)) d\theta \\
&= [(X_p \cdot e_1)\sigma^p\kappa_{\theta}]_{\underline{\theta}}^{\bar{\theta}} - [(X_p \cdot e_1)\sigma_{\theta}^p\kappa]_{\underline{\theta}}^{\bar{\theta}} \\
&\quad + \int_{\underline{\theta}}^{\bar{\theta}} (X_p \cdot e_1)(1 + (\sigma^p + \sigma_{\theta\theta}^p)\kappa) d\theta - \int_{\underline{\theta}}^{\bar{\theta}} (X_p \cdot e_1)_{\theta}(\sigma^p\kappa_{\theta} - \sigma_{\theta}^p\kappa) d\theta \\
&= [(X_p \cdot e_1)\sigma^p\kappa_{\theta}]_{\underline{\theta}}^{\bar{\theta}} - [(X_p \cdot e_1)\sigma_{\theta}^p\kappa]_{\underline{\theta}}^{\bar{\theta}} \\
&\quad + \int_{\underline{\theta}}^{\bar{\theta}} (X_p \cdot e_1)(1 + (\sigma^p + \sigma_{\theta\theta}^p)\kappa) d\theta + \int_{\underline{\theta}}^{\bar{\theta}} \frac{\sin \theta}{\kappa}(\sigma^p\kappa_{\theta} - \sigma_{\theta}^p\kappa) d\theta,
\end{aligned}$$

and since

$$\sigma^p + \sigma_{\theta\theta}^p = \sigma + \sigma_{\theta\theta} = \frac{1}{\kappa}$$

and

$$\sigma_{\theta}^p \sin \theta - \sigma^p \cos \theta = -X \cdot e_1 + p,$$

we obtain

$$\begin{aligned}
\frac{d}{dt}q(U_{(t)}^p) &= - \int_{\underline{\theta}}^{\bar{\theta}} X_p \cdot e_1 d\theta - \frac{1}{3}p_t \int_U \operatorname{div}(X_p + (X_p \cdot e_1)e_1) \\
&= - \int_{\underline{\theta}}^{\bar{\theta}} X_p \cdot e_1 d\theta - p_t|U_{(t)}|. \tag{2.17}
\end{aligned}$$

Putting everything together, and noting that $\Lambda_t = \Lambda^3$ we obtain (2.16). ■

2.2.2 Dynamic stability

The linear analysis (recall subsection 2.2) suggests that the shrinking semicircle is dynamically stable after modding out time translations and horizontal spatial translations. To this end we will show that for the rescaling of the flow, (2.12), and for appropriately chosen functions λ and p , the support function $\tilde{\sigma}$ of $\tilde{\Gamma}$ has projections onto the unstable modes that satisfy decay estimates sufficient to yield ‘fast’ convergence to the round point. Roughly speaking, this normalization implies that, for the rescaled flow, the quantities

$$\int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) d\theta, \quad \int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) \cos \theta d\theta$$

are both of order $e^{-2\tilde{t}} + O(|\sigma - 1|_{C^2}^2)$. We will do this in two steps, first obtaining the weaker bound $e^{-\tilde{t}} + O(|\sigma - 1|_{C^2}^2)$, and then using this to refine our estimates and obtain the optimal bound.

Lemma 2.8. *There exists a scaling function $\lambda : [0, T) \rightarrow \mathbb{R}$ such that $U(t) = 0$, for all $t \in [0, T)$, with U as defined in (2.13), Moreover this λ satisfies*

$$\lambda(t) - \Lambda(t) = O(1)$$

and

$$\frac{\lambda_t}{\lambda^3} = 1 + O(\Lambda^{-1}).$$

Proof. Recall the evolution of U , given in Lemma 2.6. In order to find λ such that $U(t)$ is identically zero, we solve

$$\frac{\lambda_t}{\lambda} (2\Theta + 2\Lambda\{\sigma_\Sigma\}) = \lambda^2(2\Theta - \{\sigma_\Sigma\kappa\}) - \Lambda\{\sigma\kappa\kappa_\Sigma\} + \{\kappa_\Sigma\kappa\} + \Lambda^3\{\sigma_\Sigma\}$$

Note that, by the convergence (2.3), we have that

$$\kappa(\theta) = \Lambda + E(\theta) \quad \text{and} \quad \sigma(\theta) = \frac{1}{\Lambda} + e(\theta), \quad (2.18)$$

where the functions E, e satisfy

$$E(\theta) = o(\Lambda), \quad e(\theta) = o(\Lambda^{-1}). \quad (2.19)$$

Therefore λ should solve the ODE

$$\frac{\lambda_t}{\lambda} (2\Theta + 2\Lambda\{\sigma_\Sigma\}) = \lambda^2(2\Theta - \Lambda\{\sigma_\Sigma\} + \{E\sigma_\Sigma\}) - \{\Lambda e\kappa\kappa_\Sigma\} + \Lambda^3\{\sigma_\Sigma\},$$

which, after dividing by λ^2 , becomes

$$\frac{\lambda_t}{\lambda^3} = 1 + \frac{\{E\sigma_\Sigma\} - 3\Lambda\{\sigma_\Sigma\}}{2\Theta + 2\Lambda\{\sigma_\Sigma\}} + \frac{1}{\lambda^2} \frac{\Lambda^3\{\sigma_\Sigma\} - \{\Lambda e\kappa\kappa_\Sigma\}}{2\Theta + 2\Lambda\{\sigma_\Sigma\}}.$$

Next, observe that $\sigma_\Sigma = O(\Lambda^{-2})$, because both σ_{Σ_t} and its derivative $(\sigma_{\Sigma_t})_\theta$ vanish at $\theta = \frac{3\pi}{2}$. Hence, by Taylor's theorem, together with (2.9) and (2.5), we have

$$\sigma_{\Sigma_t}(\theta) = \frac{1}{2}(\theta - \frac{3\pi}{2})^2 \kappa_{\Sigma_t}^{-1}(\frac{3\pi}{2}) + O((\theta - \frac{3\pi}{2})^3) = O(\Lambda^{-2}). \quad (2.20)$$

Using (2.19) and (2.20), the ODE can be simplified as

$$2\frac{\lambda_t}{\lambda^3} = 2 + O\left(\frac{1}{\Lambda}\right) + \frac{1}{\lambda^2}O(\Lambda).$$

Hence, any solution with $\lambda(t) \rightarrow \infty$ as $t \rightarrow T$, satisfies

$$\frac{1}{\lambda^2} = \frac{1}{\Lambda^2} + O(\Lambda^{-3}).$$

To see this, note that there exists a constant C such that,

$$2 - \frac{C}{\Lambda} - \frac{C\Lambda}{\lambda^2} \leq 2\frac{\lambda_t}{\lambda^3} \leq 2 + \frac{C}{\Lambda} + \frac{C\Lambda}{\lambda^2}.$$

This implies

$$\left(e^{\frac{C}{\Lambda}}\lambda^{-2}\right)_t \geq e^{\frac{C}{\Lambda}}\left(2 - \frac{C}{\Lambda}\right) \quad \text{and} \quad \left(e^{-\frac{C}{\Lambda}}\lambda^{-2}\right)_t \leq e^{-\frac{C}{\Lambda}}\left(2 + \frac{C}{\Lambda}\right).$$

Integrating these inequalities from t to T then gives the stated estimate. Rearranging yields the required estimates on λ .

Finally, we note that with λ as above, $U(t) = 0$. Indeed, this choice of λ , together with (2.14), implies that

$$U'(t) = \frac{2\lambda_t}{\lambda}U(t).$$

Therefore

$$\frac{d}{dt}(\lambda^{-2}U(t)) = 0,$$

and since $\lim_{t \rightarrow T} U(t) = 0$, the result follows. ■

Lemma 2.9. *Consider q be as defined in (2.15), where λ is such that λ^{-1} is bounded. Then, there exists a translation function $p : [0, T) \rightarrow \mathbb{R}$, independent of λ , such that $q(t) = 0$ for all $t \in [0, T)$. Moreover, p satisfies*

$$p = O(\Lambda^{-1}) \quad \text{and} \quad p_t = O(\Lambda).$$

Proof. Recalling the evolution equation of q , (2.16), we choose p so that

$$p_t|U(t)| - p\Theta = - \int_{\underline{\theta}}^{\bar{\theta}} X \cdot e_1 d\theta + \{(X \cdot e_1)\sigma_{\Sigma}\kappa\} + \frac{1}{\Lambda^2}[\kappa \sin]_{\underline{\theta}}^{\bar{\theta}} + 2[\gamma \cdot e_2]_{\underline{\theta}}^{\bar{\theta}}.$$

Notice that this can be written as

$$(p|U_{(t)}|)_t = -\Delta(t) + \delta(t), \quad (2.21)$$

where $\Delta(t) = -\int_{\underline{\theta}}^{\bar{\theta}} X \cdot e_1 d\theta$ and $\delta(t)$ is the rest of the right hand side, and observe that these functions satisfy

$$\Delta(t) = o(\Lambda^{-1}) \text{ and } \delta(t) = O(\Lambda^{-2}). \quad (2.22)$$

Indeed, this can be easily verified by expressing the position in terms of the support function,

$$X = \sigma(\cos \theta, \sin \theta) + \sigma_\theta(-\sin \theta, \cos \theta),$$

and using the asymptotics (2.18). The estimate for $\delta(t)$ is then straightforward after recalling (2.20) and (2.4), whereas for $\Delta(t)$ is evident after integration by parts

$$\begin{aligned} \Delta(t) &= [-\sigma \sin \theta]_{\underline{\theta}}^{\bar{\theta}} + 2 \int_{\underline{\theta}}^{\bar{\theta}} \sigma \cos \theta d\theta \\ &= [(\Lambda^{-1} - e(\theta)) \sin \theta]_{\underline{\theta}}^{\bar{\theta}} + 2 \int_{\underline{\theta}}^{\bar{\theta}} e(\theta) \cos \theta d\theta. \end{aligned}$$

Therefore, integrating (2.21) from t to T , we obtain

$$p|U_{(t)}| = O(\Lambda^{-3}).$$

Now, recall the expression for $|U_{(t)}|$ from Section 2.2.1. Together with (2.18) and (2.20), this implies that $|U_{(t)}| = O(\Lambda^{-2})$ and therefore we obtain

$$p = O(\Lambda^{-1}).$$

Note also that (2.21), together with the estimates (2.22), give the required p_t estimate.

Finally, with this choice of p , and by (2.16), the evolution for q becomes

$$\left(\frac{q}{\lambda^3}\right)_t = 0$$

and since $q(t) \rightarrow 0$ and $t \rightarrow T$, and λ^{-1} is assumed bounded, integrating implies that q has to be constantly zero. $\blacksquare$

We can now obtain estimates on the unstable modes. As it will be convenient to express them in terms of the new time variable $\tilde{t}$, we first note that

$$e^{-\tilde{t}} = O(\lambda^{-1}) = O(\Lambda^{-1}). \quad (2.23)$$

To verify this, recall that, as in defined (2.12), the new time variable satisfies

$$\begin{cases} \frac{d\tilde{t}}{dt} = \lambda^2 \\ \tilde{t}(0) = -\frac{1}{2} \log(2T). \end{cases}$$

Using the inequalities for λ in Lemma 2.8, we have

$$\frac{d\tilde{t}}{dt} = \lambda^2 = \Lambda^2 + O(\Lambda),$$

integration of which yields the estimate $e^{\tilde{t}} = O(\Lambda)$ and therefore (2.23).

Lemma 2.10. *With λ and p as in Lemmas 2.8 and 2.9, the rescaled flow (2.12) satisfies*

$$\int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) = O(e^{-\tilde{t}}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2)$$

and

$$\int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) \cos \theta = O(e^{-\tilde{t}}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2).$$

Proof. The expression (2.10), and the estimates on λ given in Lemma 2.8, give

$$\begin{aligned} 2 \int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) &= 2\lambda^2(t) |U_{(t)}| - (\bar{\theta} - \underline{\theta}) \\ &\quad - \lambda^2(t) \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} \frac{\sigma_\Sigma^p}{\kappa_\Sigma} d\theta - \lambda(t) \{\sigma_\Sigma^p\} + O(\|\tilde{\sigma} - 1\|_{C^2}^2) \\ &= U(t) + O(\Lambda^{-1}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2). \end{aligned}$$

The expression (2.11), Lemma 2.8, and the estimates on p given in Lemma 2.9, give

$$\begin{aligned} 2 \int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) \cos \theta &= 3\lambda^3(t)q(U_{(t)}^p) - \lambda^3(t) \int_{\underline{\theta}_{\Sigma}}^{\bar{\theta}_{\Sigma}} (X_p \cdot e_1) \frac{\sigma_{\Sigma}^p}{\kappa_{\Sigma}} d\theta - \lambda(t)[\gamma \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} \\ &\quad + \sum_{\partial\Gamma} (1 - \sigma)(\mathbf{N}_{\Sigma} \cdot e_1) + O(\|\tilde{\sigma} - 1\|_{C^2}^2) \\ &= q(t) + O(\Lambda^{-1}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2). \end{aligned}$$

The result now follows by Lemmas 2.8 and 2.9, and (2.23). $\blacksquare$

2.2.3 L^2 decay

We consider now the normalised flow as defined in (2.12) with λ and p as defined in Lemmas 2.8, 2.9. Then, parametrizing by the Gauss map, and skipping the tildes here, the support function satisfies:

$$\partial_t \sigma = L \sigma + B \sigma \cos \theta - (\sigma_{\theta\theta} + \sigma)^{-1} =: F(\sigma), \quad (2.24)$$

where

$$L := \frac{\lambda_t}{\lambda}, \quad B := -\lambda p_t, \quad (2.25)$$

and note also that t is the new time variable defined in $t \in (-\frac{1}{2} \log(2T), +\infty)$. Note that, by Lemmas 2.8, 2.9 and (2.23),

$$|L - 1| \leq C e^{-t}, \quad |B| \leq C e^{-t}. \quad (2.26)$$

Moreover, the angle domains for the support function are not altered, however with the new time variable their

$$\frac{d}{dt} \Theta = O(\lambda^{-1}) = O(e^{-t})$$

We will show that $\sigma - 1$ converges to 0 exponentially fast.

Theorem 2.11. *For any $k \in \mathbb{N}$ and $\alpha < 1$, there exists $C(k, \alpha)$ such that the rescaled flow, 2.12, with λ and p as defined in Lemmas 2.8, 2.9, satisfies*

$$\|\sigma - 1\|_{C^k} \leq C(k, \alpha) e^{-\alpha t}, \quad \forall t \in (-\frac{1}{2} \log(2T), +\infty).$$

Remark 2.12. To prove Theorem 2.11 we first prove decay of the L^2 norm, which is a bit stronger,

$$\|\sigma - 1\|_{L^2} \leq C e^{-2t}, \quad \forall t \in (-\tfrac{1}{2} \log(2T), +\infty).$$

On higher derivatives the decay factor has to drop because we use interpolation to achieve the estimates.

Proof. We will show decay of $\|\sigma - 1\|_{L^2(\underline{\theta}, \bar{\theta})}$ by computing and estimating its evolution. The higher order norm estimates will then follow via interpolation.

We consider L, B , as defined in (2.25), *solely* as functions of t , so that the linearisation of the speed F , as in (2.24), about $\sigma = 1$ and along $v := \frac{d}{d\varepsilon}|_{\varepsilon=0} \sigma_\varepsilon$ is given by

$$DF|_1 v = v_{\theta\theta} + (L + 1)v + Bv \cos \theta.$$

Therefore we obtain

$$\begin{aligned} \frac{d}{dt} \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1)^2 d\theta &\leq C e^{-t} \|\sigma - 1\|_{C^0}^2 \\ &\quad + 2 \int_{\underline{\theta}}^{\bar{\theta}} \left[(\sigma - 1) \left(F(1) + DF|_1(\sigma - 1) \right) \right. \\ &\quad \left. + C |\sigma - 1| \|\sigma - 1\|_{C^2}^2 \right] d\theta. \end{aligned} \tag{2.27}$$

Since

$$F(1) = L - 1 + B \cos \theta,$$

we have, by Lemma 2.10 and (2.26), and applying the Cauchy-Schwarz inequality,

$$2 \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) F(1) d\theta \leq C (e^{-2t} + \|\sigma - 1\|_{C^2}^4).$$

The last term on the right hand side of (2.27) is estimated, using Hölder's inequality and Cauchy-Schwarz, as

$$\begin{aligned} \int_{\underline{\theta}}^{\bar{\theta}} |\sigma - 1| \|\sigma - 1\|_{C^2}^2 d\theta &\leq C \|\sigma - 1\|_{C^2}^2 \|\sigma - 1\|_{L^2} \\ &\leq \frac{C}{\epsilon} \|\sigma - 1\|_{C^2}^4 + \epsilon \|\sigma - 1\|_{L^2}^2, \end{aligned}$$

for any $\epsilon > 0$. Finally, for the gradient term we have

$$2 \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) DF|_1(\sigma - 1) d\theta = 2 \int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \left((\sigma - 1)_{\theta\theta} + 2(\sigma - 1) \right) d\theta \\ + O(e^{-2t}) + O(\|\sigma - 1\|_{C^2}^4).$$

In order to estimate the first term on the right hand side we construct a C^1 (piecewise C^2) extension of $\sigma - 1$ in the interval $[0, \pi]$ with Neumann boundary condition at the endpoints. This construction is performed independently for each time slice; hence, we suppress the t -dependence in the notation. We define the extension by setting

$$\bar{\sigma}(\theta) - 1 = \begin{cases} \underline{a}\theta^2 + \underline{b}\theta + \underline{c} & \theta \in [0, \underline{\theta}] \\ \bar{a}\theta^2 + \bar{b}\theta + \bar{c} & \theta \in [\bar{\theta}, \pi], \end{cases} \quad (2.28)$$

where

$$\underline{a} = \frac{\sigma_{\theta}(\underline{\theta})}{2\underline{\theta}}, \quad \underline{b} = 0, \quad \underline{c} = \sigma(\underline{\theta}) - 1 - \underline{a}\underline{\theta}^2,$$

and

$$\bar{a} = -\frac{\sigma_{\theta}(\bar{\theta})}{2(\pi - \bar{\theta})}, \quad \bar{b} = -2\pi\bar{a}, \quad \bar{c} = \sigma(\bar{\theta}) - 1 - \bar{a}\bar{\theta}^2 - \bar{b}\bar{\theta}.$$

Recalling (2.4) and (2.20), we note that $\underline{a}$, $\bar{a}$ are bounded, $\bar{\sigma}$ is decreasing in $[0, \underline{\theta}]$ and increasing in $[\bar{\theta}, \pi]$, and also

$$\|\sigma - 1\|_{C^0} \leq \|\bar{\sigma} - 1\|_{C^0} \leq \|\sigma - 1\|_{C^0} + Ce^{-2t}. \quad (2.29)$$

The gradient term we want to estimate, can now be written as

$$\int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \left((\sigma - 1)_{\theta\theta} + 2(\sigma - 1) \right) d\theta = \int_0^{\pi} (\bar{\sigma} - 1) \left((\bar{\sigma} - 1)_{\theta\theta} + 2(\bar{\sigma} - 1) \right) d\theta \\ + Err(\bar{\sigma}, \sigma),$$

where the error term, $Err(\bar{\sigma}, \sigma)$, can be bounded as follows

$$Err(\bar{\sigma}, \sigma) = - \int_{[0, \pi] \setminus [\underline{\theta}, \bar{\theta}]} (\bar{\sigma} - 1) \left((\bar{\sigma} - 1)_{\theta\theta} + 2(\bar{\sigma} - 1) \right) d\theta \\ \leq - \int_{[0, \pi] \setminus [\underline{\theta}, \bar{\theta}]} (\bar{\sigma} - 1) (\bar{\sigma} - 1)_{\theta\theta} d\theta \quad (2.30) \\ \leq Ce^{-t} \|\bar{\sigma} - 1\|_{L^2} \leq \frac{C}{\epsilon} e^{-2t} + \epsilon \|\sigma - 1\|_{L^2}^2,$$

for any $\epsilon > 0$.

We finally estimate the term

$$\int_0^\pi (\bar{\sigma} - 1) \left((\bar{\sigma} - 1)_{\theta\theta} + 2(\bar{\sigma} - 1) \right) d\theta$$

by expressing $\bar{\sigma} - 1$ as $\bar{\sigma} - 1 = \sum_j \sigma_j \phi_j$ in $L^2([0, \pi])$, where $\phi_j = \cos(j\theta)$ and $\sigma_j = \langle \bar{\sigma} - 1, \phi_j \rangle$. Since $\bar{\sigma}$ it is a C^1 , and piecewise C^2 function, with Neumann boundary data, we have

$$\int_0^\pi \phi(\bar{\sigma}_{\theta\theta} + 2\bar{\sigma}) = \int_0^\pi \bar{\sigma}(\phi_{\theta\theta} + 2\phi),$$

for any ϕ also satisfying Neumann boundary data. The L^2 convergence, implies weak L^2 convergence and therefore

$$\begin{aligned} \int_0^\pi (\bar{\sigma} - 1)((\bar{\sigma} - 1)_{\theta\theta} + 2(\bar{\sigma} - 1)) &= \int_0^\pi \sum_j \sigma_j \phi_j ((\bar{\sigma} - 1)_{\theta\theta} + 2(\bar{\sigma} - 1)) \\ &= \lim_N \sum_j^N \sigma_j \int_0^\pi (\bar{\sigma} - 1)((\phi_j)_{\theta\theta} + 2\phi_j) \\ &= \lim_N \sum_j^N (2 - j^2) \sigma_j \int_0^\pi (\bar{\sigma} - 1) \phi_j \\ &= \lim_N \sum_j^N (2 - j^2) \sigma_j^2 \\ &\leq 4(\sigma_0^2 + \sigma_1^2) - 2 \lim_N \sum_{j=0}^N \sigma_j^2 \\ &= 4(\sigma_0^2 + \sigma_1^2) - 2\|\bar{\sigma} - 1\|_{L^2}^2. \end{aligned} \tag{2.31}$$

The two unstable nodes can be controlled, using (2.29) and Lemma 2.10,

$$\begin{aligned} \sigma_0^2 + \sigma_1^2 &\leq C \left[\left(\int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) d\theta \right)^2 + \left(\int_{\underline{\theta}}^{\bar{\theta}} (\sigma - 1) \cos \theta d\theta \right)^2 \right. \\ &\quad \left. + e^{-4t} + \|\sigma - 1\|_{C^2}^4 \right]. \tag{2.32} \\ &\leq C (e^{-2t} + \|\sigma - 1\|_{C^2}^4) \end{aligned}$$

Therefore, replacing all the above estimates in (2.27), we obtain, for any $\epsilon > 0$,

$$\frac{d}{dt} \|\sigma - 1\|_{L^2}^2 \leq \frac{C}{\epsilon} (e^{-2t} + \|\sigma - 1\|_{C^2}^4) - (4 - \epsilon) \|\sigma - 1\|_{L^2}^2.$$

The Gagliardo-Nirenberg interpolation inequality in bounded domains yields

$$\|\sigma - 1\|_{C^2} \leq C \|\sigma - 1\|_{C^7}^{\frac{1}{3}} \|\sigma - 1\|_{L^2}^{\frac{2}{3}} + C \|\sigma - 1\|_{L^2}.$$

Since, by (2.3), $\|\sigma - 1\|_{C^k} = o(1)$, for all $k \in \mathbb{N}$, we can absorb the $\|\sigma - 1\|_{C^2}^4$ term and obtain that for any $\epsilon > 0$, there exists $t_\epsilon \geq -\frac{1}{2} \log(2T)$ such that

$$\frac{d}{dt} \|\sigma - 1\|_{L^2}^2 \leq C e^{-2t} - (4 - \epsilon) \|\sigma - 1\|_{L^2}^2, \quad \forall t \geq t_\epsilon.$$

This then yields that

$$\|\sigma - 1\|_{L^2} \leq C e^{-t}, \quad \forall t \geq \frac{1}{2} \log(2T), \quad (2.33)$$

with C depending on $\|\sigma(\cdot, \frac{1}{2} \log(2T)) - 1\|_{C^0}$ and T , since we consider the estimate in the whole time interval. Interpolation, via for instance the Gagliardo-Nirenberg inequality as above, yields the required estimates. ■

We can now use this to get estimates for the original (unscaled) solution.

Theorem 2.13. *The support function of the free boundary curve shortening flow solution $\{\Gamma_t\}_{t \in [0, T]}$ satisfies, for any $\alpha < 1$,*

$$\sigma = \sqrt{2(T - t)} + O((T - t)^{\frac{1+\alpha}{2}})$$

and

$$\kappa = \frac{1}{\sqrt{2(T - t)}} + O((T - t)^{\frac{\alpha-1}{2}}).$$

Proof. By Theorem 2.11, and (2.23), for any $\alpha < 1$,

$$|\lambda(\sigma - p) - 1| = |\tilde{\sigma} - 1| = O(\Lambda^{-\alpha})$$

and

$$|\lambda^{-1}\kappa - 1| = |\tilde{\kappa} - 1| = O(\Lambda^{-\alpha}),$$

where here we reintroduce tildes for the rescaled flow (2.12). The estimate for κ is then immediate recalling the estimates for λ from Lemma 2.8. To prove the estimate for the support function, we will first show that we can improve the estimates on p given in in Lemma 2.9. Recall that

$$p_t = -\frac{1}{|U(t)|} \int_{\underline{\theta}}^{\bar{\theta}} X_p \cdot e_1 d\theta + \frac{\delta(t)}{|U(t)|} = -\frac{1}{|U(t)|\lambda} \int_{\underline{\theta}}^{\bar{\theta}} \tilde{X} \cdot e_1 d\theta + \frac{\delta(t)}{|U(t)|},$$

where $\delta(t)$ is as defined in (2.22). Since $\tilde{X} \cdot e_1 = \tilde{\sigma} \cos \theta - \tilde{\sigma}_\theta \sin \theta$, we have

$$\begin{aligned} \int_{\underline{\theta}}^{\bar{\theta}} \tilde{X} \cdot e_1 &= \int_{\underline{\theta}}^{\bar{\theta}} \tilde{\sigma} \cos \theta - \tilde{\sigma}_\theta \sin \theta d\theta = 2 \int_{\underline{\theta}}^{\bar{\theta}} \tilde{\sigma} \cos \theta d\theta - [\tilde{\sigma} \sin \theta]_{\underline{\theta}}^{\bar{\theta}} \\ &= 2 \int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) \cos \theta d\theta + [(2 - \tilde{\sigma}) \sin \theta]_{\underline{\theta}}^{\bar{\theta}}. \end{aligned}$$

Therefore, by Lemma 2.10, the above quantity is of the order $O(\Lambda^{-1})$ and, together with (2.22), we obtain the improved estimate.

$$p_t = O(1).$$

Since $p(T) = 0$, integrating from t to T we find

$$p(t) = O(\Lambda^{-2}). \quad (2.34)$$

We can now get the required estimate by scaling back and recalling the estimates for λ from Lemma 2.8

$$\sigma = \frac{1}{\Lambda} + O(\Lambda^{-1-\alpha}). \quad \blacksquare$$

Note that in the proof of Theorem 2.13, we showed that after we know a certain decay on σ (Lemma 2.10 is sufficient), we can go back to Lemma 2.9 and improve on the estimate for p . We claim that we can also improve the estimate for λ in Lemma 2.8. To do this, note that the quantities E and e , as defined in 2.18, due to Theorem 2.13, satisfy the improved estimates

$$E(\theta) = O(\Lambda^{1-\alpha}), \quad e(\theta) = O(\Lambda^{-1+\alpha}), \quad \forall \alpha < 1. \quad (2.35)$$

Therefore, following the proof of Lemma 2.8, the ODE that λ solves can be better estimated as

$$2 \frac{\lambda_t}{\lambda^3} = 2 + O\left(\frac{1}{\Lambda^{1+\alpha}}\right) + \frac{1}{\lambda^2} O(\Lambda^{1-\alpha}).$$

Hence, a solution with $\lambda(t) \rightarrow \infty$ as $t \rightarrow T$, satisfies

$$\frac{1}{\lambda^2} = \frac{1}{\Lambda^2} + O(\Lambda^{-3-\alpha}),$$

which by rearranging gives

$$\lambda = \Lambda + O(\Lambda^{-\alpha}), \quad \forall \alpha < 1. \quad (2.36)$$

We can now improve Lemma 2.10 to the following.

Corollary 2.14. *With λ and p as in Lemmas 2.8 and 2.9, the rescaled flow (2.12) satisfies, for any $\alpha < 1$,*

$$\int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) = O(e^{-(1+\alpha)\tilde{t}}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2)$$

and

$$\int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) \cos \theta = O(e^{-(1+\alpha)\tilde{t}}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2).$$

Proof. Note that by (2.10) and (2.11), together with the new estimates on p and λ given in (2.34), (2.36), we have, for any $\alpha < 1$,

$$\begin{aligned} 2 \int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) &= 2\lambda^2(t)|U(t)| - (\bar{\theta} - \underline{\theta}) \\ &\quad - \lambda^2(t) \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} \frac{\sigma_\Sigma^p}{\kappa_\Sigma} d\theta - \lambda(t)\{\sigma_\Sigma^p\} + O(\|\tilde{\sigma} - 1\|_{C^2}^2) \\ &= U(t) + O(\Lambda^{-1-\alpha}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2), \end{aligned}$$

and

$$\begin{aligned} 2 \int_{\underline{\theta}}^{\bar{\theta}} (\tilde{\sigma} - 1) \cos \theta &= 3\lambda^3(t)q(U(t)^p) - \lambda^3(t) \int_{\underline{\theta}_\Sigma}^{\bar{\theta}_\Sigma} (X_p \cdot e_1) \frac{\sigma_\Sigma^p}{\kappa_\Sigma} d\theta - \lambda(t)[\gamma \cdot e_2]_{\underline{\theta}}^{\bar{\theta}} \\ &\quad + \sum_{\partial\Gamma} (1 - \sigma)(\mathbf{N}_\Sigma \cdot e_1) + O(\|\tilde{\sigma} - 1\|_{C^2}^2) \\ &= q(t) + O(\Lambda^{-1-\alpha}) + O(\|\tilde{\sigma} - 1\|_{C^2}^2). \end{aligned}$$

The result now follows by Lemmas 2.8 and 2.9, and (2.23). ■

Remark 2.15. Even though we can improve on the decay for the unstable modes, we cannot expect any better decay for the full L^2 norm, which is due to the boundary values. In particular, on the boundary we have $|\tilde{\sigma}_\theta| = \tilde{\sigma}_\Sigma$ which is of the order e^{-t} . Recalling (2.7) this cannot be improved.

3

Ancient solutions

3.1 Introduction

Ancient solutions to geometric flows are solutions defined on backwards-infinite time intervals and play a central role in the analysis of singularity formation ([Hamilton, 1995](#)). They also appear in quantum field theory, modeling the ultraviolet regime in certain Dirichlet sigma models ([Bakas; Sourdis, 2007](#)). From a geometric perspective, ancient solutions are particularly interesting because they often exhibit strong rigidity and symmetry properties, reminiscent of phenomena observed for entire minimal surfaces.

A major theme in the study of geometric flows is the classification of ancient solutions under suitable geometric assumptions. In many settings convexity plays a decisive role, leading to strong rigidity results even in higher dimensions. This chapter focuses on the one dimensional case of curve shortening flow, where the classification theory is particularly well developed. It provides a useful model for understanding higher dimensional problems and more complicated geometric flows.

Ancient solutions for curve shortening flow.

For curve shortening flow in the plane, convex ancient solutions are completely classified. This classification was obtained by Daskalopoulos–Hamilton–Sesum ([Daskalopoulos; Hamilton; Sesum, 2010](#)) and later re-proved by using more geometric methods by Bourni–Langford–Tinaglia ([Bourni; Langford; Tinaglia, 2020](#)), whose approach also treats the non-compact case. The only convex ancient solutions to curve shortening flow in $\mathbb{R}^2$ are: stationary lines, shrinking circles, Grim Reaper translators and Angenent ovals. The shrinking circle is given by the family $\{S^1_{\sqrt{-2t}}\}_{t \in (-\infty, 0)}$. The other two non-trivial examples admit explicit formulas. The Grim

Reaper $\{G_t\}_{t \in (-\infty, \infty)}$ and the Angenent oval $\{A_t\}_{t \in (-\infty, 0)}$ are described by

$$G_t := \{(x, y) : \cos x = e^{t-y}\}, \quad A_t := \{(x, y) : \cos x = e^t \cosh y\}.$$

Of course, applying rigid motions, time translations or parabolic rescalings to any of these solutions produces further examples.

On the sphere, the situation is even more rigid. Bryan and Louie (Bryan; Louie, 2016) proved that the shrinking parallel is the only convex ancient solution to curve shortening flow on $\mathbb{S}^2$, while Choi and Mantoulidis (Choi; Mantoulidis, 2022) showed that it is also the only embedded ancient solution with uniformly bounded length.

Higher-dimensional rigidity results.

Although our focus is on one dimension, it is instructive to recall some related rigidity results for mean curvature flow in higher dimensions. When $n \geq 2$, strong classification results hold under additional geometric assumptions such as uniform convexity, bounded eccentricity, Type-I curvature decay, or bounded isoperimetric ratio. Under such conditions, the only compact convex ancient solutions are shrinking spheres (Huisken; Sinestrari, 2015) (see also (Daskalopoulos; Hamilton; Sesum, 2010; Haslhofer; Hershkovits, 2016; Langford, 2017)). An extension of this result to the non-compact setting was obtained by Lynch (Lynch, 2023). In dimension $n = 1$, Wang proved an analogous rigidity statement for the shrinking circle, showing that it is the unique solution among entire convex ancient solutions (Wang, 2011). In fact, he established a striking dichotomy theorem: any convex ancient solution in Euclidean space either sweeps out the entire space or remains confined to a slab. Related connections between non-collapsed and entire solutions appear in (Bourni; Langford; Lynch, 2024; Brendle; Naff, 2023).

Convexity alone, however, does not force spherical symmetry in higher dimensions. There exist families of compact convex ancient solutions that shrink to round points as $t \rightarrow 0$ but become increasingly elongated as $t \rightarrow -\infty$, resembling a shrinking cylinder in the central region and a translating soliton near the tips (Angenent, 2013; Haslhofer; Hershkovits, 2016; White, 2003). A major classification result of Angenent–Daskalopoulos–Sesum (Angenent; Daskalopoulos; Sesum, 2019, 2020) and Brendle–Choi (Brendle; Choi, 2019) shows that the only uniformly two-convex, non-

collapsing ancient solutions are the shrinking sphere, shrinking cylinder, ancient ovaloid, and the bowl soliton.

There also exist examples with bounded convex time-slices that sweep out slab regions, known as ancient pancakes (Bourni; Langford; Tinaglia, 2019; Wang, 2011). Bourni–Langford–Tinaglia (Bourni; Langford; Tinaglia, 2019) proved that there is a unique rotationally symmetric ancient pancake and obtained a precise asymptotic description of it. In contrast, the collapsed regime is much less rigid; indeed, large families of solutions were constructed in (Bourni; Langford; Tinaglia, 2022).

Without convexity assumptions the landscape becomes far richer. Choi and Mantoulidis (Choi; Mantoulidis, 2022) proved that if a minimal surface has Morse index I , then there exists an I -parameter family of ancient mean curvature flows emanating from it. Moreover, solutions with sufficiently fast exponential decay toward the minimal surface must lie in this family. These results were later extended to the non-compact setting in (Choi; Huang; Lee, 2025) (see also (Han, 2026)).

Free boundary curve shortening flow.

In this chapter we are interested not in closed curves but rather in the free boundary problem for curve shortening flow. In this setting, recall that the endpoints of the curve are constrained to lie on a fixed barrier curve and meet it orthogonally. A natural question is whether ancient solutions in this setting can also be classified.

Until recently, very little was known about ancient solutions to free boundary curve shortening flow. The only explicit examples were obtained by restricting solutions from the closed or complete setting (for instance, a shrinking circle restricted to a half-plane). A first classification result was obtained by Bourni–Langford (Bourni; Langford, 2023), who classified convex ancient free boundary solutions in the unit disc. Later, Bourni–Burns–Catron (Bourni; Burns; Catron, 2025) extended this result to arbitrary convex domains in $\mathbb{R}^2$, overcoming the lack of symmetry by combining barrier arguments with careful boundary expansions. The goal of this chapter is to present these results.

Theorem 3.1 (Classification in the Disc). *Modulo rotation about the origin and translation in time, there exists a unique convex, locally uniformly convex ancient solution to free boundary curve shortening flow in the disc. It converges to the point $(0, 1)$ as $t \rightarrow 0$ and smoothly to the segment*

$[-1, 1] \times \{0\}$ as $t \rightarrow -\infty$. The solution is invariant under reflection across the y -axis, and as a graph over the x -axis satisfies

$$e^{-\lambda^2 t} y(x, t) \rightarrow A \cosh(\lambda x) \quad \text{uniformly in } x \text{ as } t \rightarrow -\infty$$

for some $A > 0$, where λ solves $\lambda \tanh \lambda = 1$.

More generally, a similar classification holds in arbitrary convex domains.

Theorem 3.2 (Classification in a Convex Domain). *Let $\Omega \subset \mathbb{R}^2$ be a smooth convex domain. Then, modulo time translation, for each diameter of Ω (a segment intersecting $\partial\Omega$ orthogonally) there exist precisely two convex, locally uniformly convex ancient solutions to free boundary curve shortening flow in Ω , one on each side of the diameter.*

We note that the exponential convergence of the solution to the segment arises from an analysis of the Jacobi operator on the segment. In the case of the circle, the constant λ and the limiting function $\phi = \cosh(\lambda x)$ correspond to the first eigenvalue and eigenfunction of

$$\begin{cases} \Delta u = -\lambda u \text{ in } (-1, 1) \\ \frac{\partial u}{\partial \eta} = u \text{ on } \{-1, 1\}, \end{cases}$$

where η denotes the outward pointing unit normal, so that $\frac{\partial u}{\partial \eta}(\pm 1) = \pm u_x(\pm 1)$. In a general domain, one obtains the same type of exponential convergence, with the rate determined by the first eigenvalue of the corresponding Jacobi problem and the solution converging to the associated eigenfunction. In this setting, the Jacobi equation is again given by the Laplacian, but the Robin boundary condition must be modified so that the right hand side equals the curvature of $\partial\Omega$ multiplied u . Both the eigenvalue and the eigenfunction can then be computed explicitly in terms of the boundary curvatures.

In higher dimensions the situation becomes more delicate. In (Bourni; Langford, 2025), it is shown that uniqueness still holds in the symmetric case of a ball, but for general domains the classification problem becomes considerably more complicated. A very general existence theory for ancient free boundary flows was recently developed by Bourni–Franz (Bourni; Franz, 2026). They showed that if a free boundary minimal hypersurface,

in a Riemannian manifold with convex boundary, has Morse index I , then there exists an I -parameter family of ancient free boundary mean curvature flows emanating from it. Moreover, any ancient solution that converges sufficiently rapidly to the minimal hypersurface must belong to this family. This result will play an important role in this chapter. We will give two proofs of the classification theorems 3.2 and 3.1, one of which follows ideas from (Bourni; Franz, 2026).

The first proof follows the original approach of (Bourni; Langford, 2023; Bourni; Burns; Catron, 2025). The general idea is to construct solutions as limits of ‘old but not ancient’ flows that start increasingly close to the diameter.

Maximum principles, together with curvature-to-height estimates, are then used to obtain precise asymptotics and ultimately uniqueness. We present this argument first in the case of the circle. The case of a general convex domain requires a more delicate analysis, but the underlying principles remain the same. In particular, the absence of symmetry forces one to combine barrier arguments with Taylor expansions near the boundary to obtain suitable curvature and height estimates. Although the ideas are similar, the general case is technically more involved. For this reason, and because the method is of independent interest, it is useful to develop the argument first in the symmetric setting.

The second proof, that applies to general convex domains, uses a different approach. In (Bourni; Franz, 2026), it is shown that a convex free-boundary minimal hypersurface can be deformed along its first eigenfunction while preserving the orthogonality condition at the boundary. We use this observation to construct a foliation of a neighborhood of the diameter by convex curves. This foliation is, in an appropriate sense, “almost” a solution to the free-boundary curve shortening flow and can therefore serve as a family of barriers. These barriers yield both the existence and uniqueness of the desired ancient solution. We present this argument in the setting of a general convex domain.

3.2 Classification in the case of S^1

3.2.1 Existence

Our first goal is the explicit construction of a non-trivial ancient free boundary curve shortening flow in the disc. It will be clear from the con-

struction that the solution is reflection symmetric about the vertical axis, emerges at time negative infinity from the horizontal diameter, and converges at time zero to the point $(0, 1)$. We shall also prove that the height of the constructed solution converges to the diameter exponentially as time goes to $-\infty$, and this estimate will be key in proving its uniqueness.

Barriers

The first curves one might consider as barriers are circular arcs. Indeed, these do serve as barriers as we show below. Given $\theta \in (0, \frac{\pi}{2})$, denote by C_θ the circle centered on the y -axis which meets ∂B^2 orthogonally at $(\cos \theta, \sin \theta)$. That is,

$$C_\theta \doteq \left\{ (x, y) \in \mathbb{R}^2 : x^2 + (\csc \theta - y)^2 = \cot^2 \theta \right\}. \quad (3.1)$$

If we set

$$\theta^-(t) \doteq \arcsin e^t \quad \text{and} \quad \theta^+(t) \doteq \arcsin e^{2t},$$

then we find that the inward normal speed of $C_{\theta^-(t)}$ is no greater than its curvature κ^- and the inward normal speed of $C_{\theta^+(t)}$ is no less than its curvature κ^+ . The maximum principle and the Hopf boundary point lemma then imply that

Proposition 3.3. *A solution to the free boundary curve shortening flow in B^2 which lies below (resp. above) the circle C_{θ_0} at time t_0 lies below $C_{\theta^+(t_0^+ + t - t_0)}$ (resp. above $C_{\theta^-(t_0^- + t - t_0)}$) for all $t > t_0$, where $2t_0^+ = \log \sin \theta_0$ (resp. $t_0^- = \log \sin \theta_0$).*

Although circular arcs do serve as barriers, one can construct more precise ones, closer to an actual solution, by using segments of Angenent ovals, as we show below.

Consider the shifted scaled Angenent oval $\{A_t^\lambda\}_{t \in (-\infty, 0)}$, where

$$A_t^\lambda \doteq \left\{ (x, y) \in \mathbb{R} \times (0, \frac{\pi}{2\lambda}) : \sin(\lambda y) = e^{\lambda^2 t} \cosh(\lambda x) \right\}.$$

For each λ , $\{A_t^\lambda\}_{t \in (-\infty, 0)}$ evolves by curve shortening flow and there is a unique $t_\lambda \in (-\infty, 0)$ such that $(\cos \theta, \sin \theta) \in A_{t_\lambda}^\lambda$, satisfying

$$e^{\lambda^2 t_\lambda} = \frac{\sin(\lambda \sin \theta)}{\cosh(\lambda \cos \theta)}.$$

The outward pointing unit normal ν_λ of A_t^λ at $(\cos \theta, \sin \theta)$ satisfies

$$\nu_\lambda(\cos \theta, \sin \theta) \cdot (\cos \theta, \sin \theta) = \frac{\cos \theta \tanh(\lambda \cos \theta) - \sin \theta \cot(\lambda \sin \theta)}{\sqrt{\tanh^2(\lambda \cos \theta) + \cot^2(\lambda \sin \theta)}}.$$

Lemma 3.4. *For each $\theta \in (0, \frac{\pi}{2})$, there is a unique $\lambda(\theta) \in (0, \frac{\pi}{2 \sin \theta})$ such that*

$$\nu_{\lambda(\theta)}(\cos \theta, \sin \theta) \cdot (\cos \theta, \sin \theta) = 0.$$

Given $\theta, \theta_0 \in (0, \frac{\pi}{2})$ with $\theta > \theta_0$,

$$\nu_{\lambda(\theta_0)}(\cos \theta, \sin \theta) \cdot (\cos \theta, \sin \theta) < 0.$$

Proof. Define

$$f(\lambda, \theta) \doteq \cos \theta \tanh(\lambda \cos \theta) - \sin \theta \cot(\lambda \sin \theta).$$

Observe that

$$\lim_{\lambda \searrow 0} f(\lambda, \theta) = -\infty, \quad \lim_{\lambda \nearrow \frac{\pi}{2 \sin \theta}} f(\lambda, \theta) = \cos \theta \tanh\left(\frac{\pi}{2} \cot \theta\right) > 0,$$

and

$$\frac{\partial f}{\partial \lambda} = \cos^2 \theta (1 - \tanh^2(\lambda \cos \theta)) + \sin^2 \theta (1 + \cot^2(\lambda \sin \theta)) > 0. \quad (3.2)$$

The first claim follows. Next observe that

$$\begin{aligned} \frac{\partial f}{\partial \theta} &= -\sin \theta \tanh(\lambda \cos \theta) - \lambda \cos \theta \sin \theta \operatorname{sech}^2(\lambda \cos \theta) \\ &\quad - \cos \theta \cot(\lambda \sin \theta) + \lambda \sin \theta \cos \theta \csc^2(\lambda \sin \theta). \end{aligned}$$

Given $\theta \in (0, \frac{\pi}{2})$, we obtain, at the unique zero $\lambda \in (0, \frac{\pi}{2 \sin \theta})$ of $f(\cdot, \theta)$,

$$\begin{aligned} \frac{\partial f}{\partial \theta} &= -\sin \theta \tan \theta \cot(\lambda \sin \theta) - \lambda \cos \theta \sin \theta (1 - \tan^2 \theta \cot^2(\lambda \sin \theta)) \\ &\quad - \cos \theta \cot(\lambda \sin \theta) + \lambda \sin \theta \cos \theta \csc^2(\lambda \sin \theta) \\ &= -\sec \theta \cot(\lambda \sin \theta) (1 - \lambda \sin \theta \cot(\lambda \sin \theta)). \end{aligned}$$

Since $Y \cot Y < 1$ for $Y \in (0, \frac{\pi}{2})$, we obtain $\frac{\partial f}{\partial \theta} < 0$. The second claim follows. ■

The maximum principle and the Hopf boundary point lemma now imply the following.

Proposition 3.5. *Let $\{\Gamma_t\}_{t \in [\alpha, \omega]}$ be a solution to free boundary curve shortening flow in B^2 . Suppose that $\lambda \leq \lambda(\theta_\alpha)$, where θ_α denotes the smaller, in absolute value, of the two turning angles to Γ_α at its boundary. If Γ_α lies above A_s^λ , then Γ_t lies above $A_{s+t-\alpha}^\lambda$ for all $t \in (\alpha, \omega) \cap (-\infty, \alpha - s)$.*

Proof. By the avoidance principle, the two families of curves can never develop contact at an interior point. Since the families are monotonic, they cannot develop boundary contact at a boundary point $(\cos \theta, \sin \theta)$ with $|\theta| \leq \theta_\alpha$. On the other hand, following the notation of the proof of Lemma 3.4, since $\lambda \leq \lambda(\theta_\alpha)$, (3.2) implies that

$$f(\lambda, \theta_\alpha) \leq f(\lambda_\alpha, \theta_\alpha) = 0,$$

and hence, by the argument of Lemma 3.4,

$$f(\lambda, \theta) \leq 0 \text{ for } \theta \geq \theta_\alpha.$$

So the Hopf boundary point lemma implies that no boundary contact can develop for $\theta \geq \theta_\alpha$ either. ■

Remark 3.6. Since $s \cot s \rightarrow 1$ as $s \rightarrow 0$, $f(\lambda, \theta)$ is non-negative at $\theta = 0$ so long as $\lambda \geq \lambda_0$, where $\lambda_0 \tanh \lambda_0 = 1$. This also yields that as $\theta \rightarrow 0$, the value $\lambda(\theta)$ from Lemma 3.4 decreases to λ_0 .

Old-but-not-ancient solutions

For each $\rho > 0$, choose a curve Γ^ρ in $\overline{B^2}$ with the following properties:

- Γ^ρ meets ∂B^2 orthogonally at $(\cos \rho, \sin \rho)$,
- Γ^ρ is reflection symmetric about the y -axis,
- $\Gamma^\rho \cap B^2$ is the relative boundary of a convex region $\Omega^\rho \subset B^2$, and
- $\kappa_s^\rho > 0$ in $B^2 \cap \{x > 0\}$.

For example, we could take $\Gamma^\rho \doteq A_{t_\rho}^{\lambda_\rho} \cap B^2$, where $\lambda_\rho > \lambda_0$ and t_ρ are (uniquely) chosen so that

$$\cos \rho \tanh(\lambda_\rho \cos \rho) - \sin \rho \cot(\lambda_\rho \sin \rho) = 0$$

and

$$-t_\rho = \lambda_\rho^{-2} \log \left(\frac{\cosh(\lambda_\rho \cos \rho)}{\sin(\lambda_\rho \sin \rho)} \right).$$

Work of Stahl, see in particular Theorem 2.3, and maximum principle arguments, yield the following *old-but-not-ancient solutions*.

Lemma 3.7. *For each $\rho \in (0, \frac{\pi}{2})$, there exists a smooth solution¹ $\{\Gamma_t^\rho\}_{t \in [\alpha_\rho, 0]}$ to curve shortening flow with $\Gamma_{\alpha_\rho}^\rho = \Gamma^\rho$ which satisfies the following properties:*

- Γ_t^ρ meets ∂B^2 orthogonally for each $t \in (\alpha_\rho, 0)$,
- Γ_t^ρ is convex and locally uniformly convex for each $t \in (\alpha_\rho, 0)$,
- Γ_t^ρ is reflection symmetric about the y -axis for each $t \in (\alpha_\rho, 0)$,
- $\Gamma_t^\rho \rightarrow (0, 1)$ uniformly as $t \rightarrow 0$,
- $\kappa_s^\rho > 0$ in $B^2 \cap \{x > 0\}$, and
- $\alpha_\rho < \frac{1}{2} \log \left(\frac{2 \sin \rho}{1 + \sin^2 \rho} \right) \rightarrow -\infty$ as $\rho \rightarrow 0$.

Remark 3.8. We refer to the solutions of Lemma 3.7, as “old-but-not-ancient” solutions.

Proof. Existence of a maximal solution to curve shortening flow out of Γ^ρ which meets ∂B^2 orthogonally follows from Theorem 2.3, and further that the solutions converges to a (half-round) point on the boundary of B^2 at the final time. We obtain $\{\Gamma_t^\rho\}_{t \in [\alpha_\rho, 0]}$ by time-translating this solution. By uniqueness of solutions, Γ_t^ρ remains reflection symmetric about the y -axis for $t \in (\alpha_\rho, 0)$, so the final point is $(0, 1)$. The reflection symmetry also implies that $\kappa_s^\rho = 0$ at the point $p_t \doteq \Gamma_t^\rho \cap \{x = 0\}$ for all $t \in [\alpha_\rho, 0)$. By Lemma 1.11, $|\kappa_s^\rho| = \kappa^\rho > 0$ at the boundary point $q_t \doteq \partial \Gamma_t^\rho \cap \{x > 0\}$ for all $t \in (\alpha_\rho, 0)$. Applying Sturm’s theorem (Theorem 1.20) to κ_s^ρ , we thus find that $\kappa_s^\rho > 0$ on $\Gamma_t^\rho \cap B^2 \cap \{x > 0\}$ for all $t \in (\alpha_\rho, 0)$.

Finally, observe that the circle C_{θ_ρ} defined by

$$\sin \theta_\rho = \frac{2 \sin \rho}{1 + \sin^2 \rho}$$

¹ Given by a one parameter family of immersions $\gamma : [-1, 1] \times [\alpha_\rho, 0] \rightarrow \overline{B^2}$ satisfying $\gamma \in C^\infty([-1, 1] \times (\alpha_\rho, 0)) \cap C^{2+\beta, 1+\frac{\beta}{2}}([-1, 1] \times [\alpha_\rho, 0])$ for some $\beta \in (0, 1)$.

is tangent to the line $y = \sin \rho$, and hence lies above Γ^ρ , that is $C_{\theta_\rho} \subset \Omega^\rho$. Therefore, the final property follows from Proposition 3.3. ■

We now fix $\rho > 0$ and consider a solution $\{\Gamma_t\}_{t \in (\alpha, 0)} = \{\Gamma_t^\rho\}_{t \in (\alpha_\rho, 0)}$, as in Lemma 3.7. We will suppress the ρ superscripts and subscripts often from the notation, when no confusion should arise. We introduce the following notation

$$\underline{\kappa}(t) \doteq \min_{\Gamma_t} \kappa = \kappa(p_t) \quad \text{and} \quad \bar{\kappa}(t) \doteq \max_{\Gamma_t} \kappa = \kappa(q_t),$$

and define $\underline{y}(t)$, $\bar{y}(t)$ and $\bar{\theta}(t)$ by

$$p_t = (0, \underline{y}(t)), \quad q_t = (\cos \bar{\theta}(t), \sin \bar{\theta}(t)), \quad \text{and} \quad \bar{y}(t) = \sin \bar{\theta}(t).$$

Often, and when there is no confusion, we will also suppress the t dependence. We will also use θ to denote the angle between the tangent vector τ and the x -axis, so that $\tau = (\cos \theta, \sin \theta)$, and since the curves are convex we can parametrize them by θ . Note that, with the above notation we have $\theta \in [-\bar{\theta}(t), \bar{\theta}(t)]$.

Lemma 3.9. *Each old-but-not-ancient solution, as in Lemma 3.7, satisfies*

$$\underline{\kappa} \leq \tan \bar{\theta} \leq \bar{\kappa}, \tag{3.3}$$

$$\sin \bar{\theta} \leq e^t, \tag{3.4}$$

and

$$\frac{\sin \bar{\theta}}{1 + \cos \bar{\theta}} \leq \underline{y} \leq \sin \bar{\theta}. \tag{3.5}$$

Proof. If we parametrize by turning angle $\theta \in [-\bar{\theta}, \bar{\theta}]$, then

$$\cos \bar{\theta} = x(\bar{\theta}) = \int_0^{\bar{\theta}} \frac{\cos u}{\kappa(u)} du,$$

and by the monotocity of κ we have

$$\cos \bar{\theta} \geq \frac{1}{\bar{\kappa}} \int_0^{\bar{\theta}} \cos u \, du = \frac{\sin \bar{\theta}}{\bar{\kappa}},$$

and similarly $\underline{\kappa} \leq \tan \bar{\theta}$, proving (3.3). The estimate (3.4) now follows by integrating the inequality

$$\frac{d}{dt} \sin \bar{\theta} = \cos \bar{\theta} \bar{\kappa} \geq \sin \bar{\theta}$$

between any initial time t and the final time 0 (at which $\bar{\theta} = \frac{\pi}{2}$ since the solution contracts to the point $(0, 1)$).

The upper bound for $\underline{y}$ follows from convexity and the boundary condition $\bar{y} = \sin \bar{\theta}$. To prove the lower bound, we will show that the circle $C_{\bar{\theta}(t)}$ lies nowhere above Γ_t . Suppose that this is not the case. Then, since $C_{\bar{\theta}(t)}$ lies locally below Γ_t near q_t , we can move $C_{\bar{\theta}(t)}$ downwards until it is tangent from below to a point p'_t on $\Gamma_t \cap \{x \geq 0\}$, at which we must have $\kappa \geq \tan \bar{\theta}(t)$. But then, since $\kappa_s \geq 0$ in $\{x > 0\}$, we find that $\kappa \geq \tan \bar{\theta}(t)$ for all points between p'_t and q_t . But this implies that this whole arc (including p'_t) lies above $C_{\bar{\theta}(t)}$, a contradiction. ■

Taking the limit

Proposition 3.10. *There exists a convex, locally uniformly convex ancient curve shortening flow in the disc with free boundary on the circle.*

Proof. For each $\rho > 0$, consider the old-but-not-ancient solution $\{\Gamma_t^\rho\}_{t \in [\alpha_\rho, 0]}$ constructed in Lemma 3.7. If we represent Γ_t^ρ as a graph $x \mapsto y^\rho(x, t)$ over the x -axis, then convexity, the boundary condition and (3.4) imply that

$$|y_x^\rho(x, t)| \leq \tan \bar{\theta}(t) \leq \frac{e^t}{\sqrt{1 - e^{2t}}}.$$

Since the bound is independent of ρ , the estimates in Theorems 1.32 and 1.33 imply uniform-in- ρ bounds for the curvature and its derivatives. So the limit

$$\{\Gamma_t^\rho\}_{t \in [\alpha_\rho, 0]} \rightarrow \{\Gamma_t\}_{t \in (-\infty, 0)}$$

exists in C^∞ (globally in space on compact subsets of time) and the limit $\{\Gamma_t\}_{t \in (-\infty, 0)}$ satisfies curve shortening flow with free boundary in B^2 . Moreover, to see that the limit is not trivial (the static diameter) we integrate $\underline{y}_t = \underline{\kappa}$ and use (3.3) to obtain

$$1 - \underline{y} = \int_t^0 \underline{\kappa} dt \leq \int_t^0 \tan \bar{\theta} dt \leq \int_t^0 \frac{e^t}{\sqrt{1 - e^{2t}}} dt = \frac{\pi}{2} - \arcsin e^t,$$

which gives a uniform (in ρ) positive lower bound for y for t not too negative, in particular for $t > \log(\sin(\frac{\pi}{2} - 1))$. Finally, the solution inherits the reflection symmetry with respect to the y -axis and therefore converges to $(0, 1)$ as $t \rightarrow 0$. ■

3.2.2 Asymptotics for the Height of a Constructed Solution

For the purposes of this section, we fix an ancient solution $\{\Gamma_t\}_{(-\infty, 0)}$ obtained as in Proposition 3.10 by taking a sublimit as $\lambda \searrow \lambda_0$ of the specific old-but-not ancient solutions $\{\Gamma_t^\lambda\}_{t \in [\alpha_\lambda, 0)}$ corresponding to $\Gamma_{\alpha_\lambda}^\lambda = A_{t_\lambda}^\lambda \cap B^2$, t_λ being the time at which $\{A_t^\lambda\}_{t \in (-\infty, 0)}$ meets ∂B^2 orthogonally. In this section we will show that the solution converges exponentially fast to the diameter, which will be essential to prove its uniqueness. We will do this by showing bounds on the ratio of the curvature over the height (the distance to the diameter). We first show a lower bound.

Lemma 3.11. *The ancient solution $\{\Gamma_t\}_{(-\infty, 0)}$ satisfies*

$$\frac{\kappa}{\langle \gamma, e_2 \rangle} \geq \lambda_0^2, \quad (3.6)$$

where γ is a parametrization (or the position).

Proof. Consider the old-but-not-ancient solution $\{\Gamma_t^\lambda\}_{t \in [\alpha_\lambda, 0)}$ and for simplicity let's use the notation $y := \langle \gamma, e_2 \rangle$. Since

$$(\partial_t - \Delta) \frac{\kappa}{y} = \kappa^2 \frac{\kappa}{y} + 2 \left\langle \nabla \frac{\kappa}{y}, \frac{\nabla y}{y} \right\rangle,$$

and at a boundary point,

$$\left(\frac{\kappa}{y} \right)_s = \frac{\kappa_s}{y} - \frac{\kappa}{y} \frac{y_s}{y} = 0,$$

the Hopf boundary point lemma and the maximum principle imply then that the minimum of $\frac{\kappa}{y}$ can not decrease under the flow. Therefore it is bounded by its value on the initial timeslice $\Gamma_{\alpha_\lambda}^\lambda = A_{t_\lambda}^\lambda$, where

$$\kappa = \lambda \cos \theta \tan(\lambda y) \geq \lambda^2 y \cos \bar{\theta}.$$

The claim then follows by letting $\lambda \rightarrow \lambda_0$, since then $\bar{\theta} \rightarrow 0$. ■

We next show that the curvature decays exponentially fast.

Lemma 3.12. *There exist $T > -\infty$ and $C < \infty$ such that*

$$\bar{\kappa} \leq Ce^t \text{ for all } t < T. \quad (3.7)$$

Proof. We will prove the estimate for each old-but-not-ancient solution $\{\Gamma_t^\lambda\}_{t \in [\alpha_\lambda, 0)}$. We first prove the estimate.

$$|\kappa_s| \leq 2\kappa \quad (3.8)$$

for t sufficiently negative. It will suffice to prove that

$$|\kappa_s| - \kappa + \langle \gamma, \nu \rangle \leq 0. \quad (3.9)$$

Indeed, since $\langle \gamma, \nu \rangle_s = \kappa \langle \gamma, \tau \rangle$ has the same sign as the x -coordinate, we may estimate, as in (3.6),

$$|\langle \gamma, \nu \rangle| \leq |\langle \gamma, \nu \rangle|_{x=0} \leq \lambda_0^{-2} \kappa|_{x=0} = \lambda_0^{-2} \min_{\Gamma_t} \kappa \leq \kappa. \quad (3.10)$$

For λ sufficiently close to λ_0 , we have $\kappa|_{t=\alpha_\lambda} < 1/2$. Denote by T^λ the first time at which κ reaches $1/2$. Since κ is continuous up to the initial time α_λ , we have $T^\lambda > \alpha_\lambda$. We claim that (3.9) holds for $t < T^\lambda$. Indeed, it is satisfied on the initial timeslice $\Gamma_{\alpha_\lambda}^\lambda = A_{t_\lambda}^\lambda$ since

$$\kappa_s^2 - \kappa^2 = \lambda^2 (\cos^2 \theta \sin^2 \theta - \sin^2 \theta - a_\lambda^2) = -\lambda^2 (\sin^4 \theta + a_\lambda^2) < 0,$$

whereas $\langle \gamma, \nu \rangle \leq 0$. We will show that

$$f_\varepsilon \doteq |\kappa_s| - \kappa + \langle \gamma, \nu \rangle - \varepsilon e^{t-\alpha_\lambda}$$

remains negative up to time T^λ . Suppose, to the contrary, that f_ε reaches zero at some time $t < T^\lambda$ at some point $p \in \bar{\Gamma}_t$. Since $|\kappa_s| - \kappa + \langle \gamma, \nu \rangle$ vanishes at the boundary, p must be an interior point. Since κ_s vanishes at the y -axis, and the curve is symmetric, we may assume that $x(p) > 0$. At such a point,

$$\begin{aligned} 0 &\leq (\partial_t - \Delta) f_\varepsilon = \kappa^2 (4\kappa_s - \kappa + \langle \gamma, \nu \rangle) - 2\kappa - \varepsilon e^{t-\alpha_\lambda} \\ &= \kappa^2 (3[\kappa - \langle \gamma, \nu \rangle] + 4\varepsilon e^{t-\alpha_\lambda}) - 2\kappa - \varepsilon e^{t-\alpha_\lambda}. \end{aligned}$$

Recalling (3.10) and estimating $\kappa \leq \frac{1}{2}$ yields

$$0 \leq 6\kappa^3 - 2\kappa + (4\kappa^2 - 1)\varepsilon e^{t-\alpha_\lambda} < 0,$$

which is absurd. So f_ε does indeed remain negative, and taking $\varepsilon \rightarrow 0$ yields (3.8) for $t < T^\lambda$. Since $\text{Length}(\Gamma_t \cap \{x \geq 0\}) \leq 1$, integrating (3.8) yields

$$\bar{\kappa} \leq e^2 \underline{\kappa} \text{ for } t < T^\lambda.$$

Recalling (3.3) and (3.4), this implies that

$$\bar{\kappa} \leq e^2 \frac{e^t}{\sqrt{1-e^{2t}}} \text{ for } t < T^\lambda.$$

This also shows, that $T^\lambda \geq T$, for some T that is independent of λ , so we conclude that

$$\bar{\kappa} \leq C e^t \text{ for } t < T,$$

where C and T do not depend on λ , which finishes the proof. $\blacksquare$

We next prove an upper bound on the ratio of the curvature over the height.

Lemma 3.13. *There exist $C < \infty$ and $T > -\infty$ such that*

$$\frac{\kappa}{\langle \gamma, e_2 \rangle} \leq \lambda_0^2 + C e^{2t} \text{ for } t < T,$$

where γ is a parametrization (or the position).

Proof. Consider the old-but-not-ancient solution $\{\Gamma_t^\lambda\}_{t \in [a_\lambda, 0)}$ and once more we use the notation $y = \langle \gamma, e_2 \rangle$. By (3.7), we can find $C < \infty$ and $T > -\infty$ such that

$$\begin{aligned} (\partial_t - \Delta) \frac{\kappa}{y} &= \kappa^2 \frac{\kappa}{y} + 2 \left\langle \nabla \frac{\kappa}{y}, \frac{\nabla y}{y} \right\rangle \\ &\leq C e^{2t} \frac{\kappa}{y} + 2 \left\langle \nabla \frac{\kappa}{y}, \frac{\nabla y}{y} \right\rangle \text{ for } t < T. \end{aligned}$$

Since, at a boundary point,

$$\left(\frac{\kappa}{y} \right)_s = \frac{\kappa_s}{y} - \frac{\kappa}{y} \frac{y_s}{y} = 0,$$

the Hopf boundary point lemma and the ODE comparison principle yield

$$\max_{\Gamma_t^\lambda} \frac{\kappa}{y} \leq C \max_{\Gamma_{\alpha_\lambda}^\lambda} \frac{\kappa}{y} \text{ for } t \in (\alpha_\lambda, T).$$

But now

$$(\partial_t - \Delta) \frac{\kappa}{y} \leq C e^{2t} \max_{\Gamma_{\alpha_\lambda}^\lambda} \frac{\kappa}{y} + 2 \left\langle \nabla \frac{\kappa}{y}, \frac{\nabla y}{y} \right\rangle \text{ for } t < T,$$

and hence, by ODE comparison,

$$\max_{\Gamma_t^\lambda} \frac{\kappa}{y} \leq \max_{\Gamma_{\alpha_\lambda}^\lambda} \frac{\kappa}{y} (1 + C e^{2t}) \text{ for } t \in (\alpha_\lambda, T).$$

Since, on the initial timeslice $\Gamma_{\alpha_\lambda}^\lambda = A_{t_\lambda}^\lambda$,

$$\frac{\kappa}{y} = \frac{\lambda \tan(\lambda y)}{y} \cos \theta,$$

the claim follows upon taking $\lambda \rightarrow \lambda_0$. ■

Lemma 3.14. *The limits*

$$\lim_{t \rightarrow -\infty} e^{-\lambda_0^2 t} \underline{y}(t), \quad \lim_{t \rightarrow -\infty} e^{-\lambda_0^2 t} \bar{y}(t)$$

exist in $(0, \infty)$ on the particular ancient solution $\{\Gamma_t\}_{(-\infty, 0)}$.

Proof. Integrating the estimate from Lemma 3.12 between $-\infty$ and $t < T$ we obtain

$$\bar{\theta}(t) \leq C e^t,$$

and therefore

$$\bar{y} = \sin \bar{\theta} \leq C e^t.$$

The estimates from Lemmas 3.11, 3.13 at the point lying on the y -axis read as

$$\lambda_0^2 \leq \frac{\underline{y}_t}{\underline{y}} \leq \lambda_0^2 + C e^{2t}, \forall t < T,$$

which when integrated between t and T yields

$$\lambda_0^2(T - t) \leq \log \frac{\underline{y}(T)}{\underline{y}(t)} \leq \lambda_0^2(T - t) + \frac{C}{2}(e^{2T} - e^{2t})$$

and therefore

$$C^{-1} \leq \underline{y} e^{-\lambda_0^2 t} \leq C,$$

for some constant C . The result now follows, because the same estimate (with modified constants) holds for $\bar{y}$, since by (3.5), the two are comparable. ■

3.2.3 Uniqueness

Now let $\{\Gamma_t\}_{t \in (-\infty, 0)}$, $\Gamma_t = \partial_{\text{rel}} \Omega_t$, be *any* convex, locally uniformly convex ancient free boundary curve shortening flow in the disc. By Theorem 2.3, we may assume that Γ_t contracts to a point on the boundary as $t \rightarrow 0$.

Backwards Convergence

We first show that $\bar{\Gamma}_t$ converges to a diameter as $t \rightarrow -\infty$.

Lemma 3.15. *Up to a rotation of the plane,*

$$\bar{\Gamma}_t \xrightarrow{C^\infty} [-1, 1] \times \{0\} \text{ as } t \rightarrow -\infty.$$

Proof. Set $A(t) \doteq \text{area}(\Omega_t)$. Integrating the variational formula for area yields

$$A(t) = \int_t^0 \int_{\Gamma_t} d\theta,$$

where θ is the turning angle. Since convexity ensures that the total turning angle $\int_{\Gamma_t} d\theta$ is increasing and $A(t) \leq \pi$ for all t , we find that

$$\int_{\Gamma_t} d\theta \rightarrow 0 \text{ as } t \rightarrow -\infty.$$

Monotonicity of the flow, the free boundary condition and convexity now imply that the enclosed regions Ω_t satisfy, after a rotation,

$$\bar{\Omega}_t \rightarrow B^2 \cap \{y \geq 0\} \text{ as } t \rightarrow -\infty$$

in the Hausdorff topology.

If we now represent Γ_t graphically over the x -axis, then convexity and the boundary condition ensure that the height and gradient are bounded by the height at the boundary. The estimates in Theorems 1.32 and 1.33 now

give bounds for κ and its derivatives up to the boundary depending only on the height at the boundary. We then get smooth subsequential convergence along any sequence of times $t_j \rightarrow -\infty$. The claim follows since any sublimit is the horizontal segment. ■

We henceforth assume, without loss of generality, that the backwards limit is the diameter contained in the x -axis.

Reflection Symmetry

We prove that the solution is reflection symmetric using Alexandrov reflection across lines through the origin (see Chow and Gulliver ([Chow; Gulliver, 2001](#))).

Lemma 3.16. *Γ_t is reflection symmetric about the y -axis for all t .*

Proof. Given any $\omega \in (0, \pi)$, we define the halfspace

$$H_\omega = \{(x, y) : (x, y) \cdot (\sin \omega, -\cos \omega) > 0\}$$

and denote by R_ω the reflection about ∂H_ω . We first claim that, for every ω , there exists $t = t_\omega$ such that

$$(R_\omega \cdot \Gamma_t) \cap (\Gamma_t \cap H_\omega) = \emptyset \text{ for all } t < t_\omega.$$

Assume that the claim is not true. Then there exists $\omega \in (0, \pi)$, a sequence of times $t_i \rightarrow -\infty$, and a sequence of pairs of points $p_i, q_i \in \Gamma_{t_i}$ such that $R_\omega(p_i) = q_i$. This implies that the line passing through p_i and q_i is parallel to the vector $(\sin \omega, -\cos \omega)$, so the mean value theorem yields for each i a point r_i on Γ_{t_i} where the normal is parallel to $(\cos \omega, \sin \omega)$. This contradicts Lemma [3.15](#). ■

Asymptotics for the Height

We begin with a lemma that it is key in showing that any solution satisfies the same asymptotic estimates as the constructed solution, as given in Lemma [3.14](#).

Lemma 3.17. *For all $t < 0$,*

$$\kappa_s > 0 \text{ in } \{x > 0\} \cap \Gamma_t$$

and hence

$$\frac{\sin \bar{\theta}}{1 + \cos \bar{\theta}} \leq \underline{y}. \quad (3.11)$$

Proof. Choose $T > -\infty$ so that $\kappa < \frac{2}{7}$ for $t < T$ and, given $\varepsilon > 0$, set

$$v_\varepsilon \doteq \kappa_s + \varepsilon(1 - \langle \gamma, \nu \rangle).$$

We claim that $v_\varepsilon > 0$ in $\{x \geq 0\} \cap (-\infty, T)$. Suppose that this is not the case. Since at the spatial boundary $v_\varepsilon > \varepsilon$, and $v_\varepsilon \rightarrow \varepsilon$ as $t \rightarrow -\infty$, there must exist a first time in $(-\infty, T)$ and an interior point at which $v_\varepsilon = 0$. But, at such a point,

$$\begin{aligned} 0 &\geq (\partial_t - \Delta) v_\varepsilon = \kappa^2(\kappa_s - \varepsilon \langle \gamma, \nu \rangle) + 3\kappa^2 \kappa_s + 2\varepsilon \kappa \\ &= -\varepsilon \kappa^2 - 3\varepsilon \kappa^2(1 - \langle \gamma, \nu \rangle) + 2\varepsilon \kappa \\ &\geq \varepsilon(2 - 7\kappa)\kappa \\ &> 0, \end{aligned}$$

which is absurd. Now take $\varepsilon \rightarrow 0$ to obtain $\kappa_s \geq 0$ in $\{x \geq 0\} \cap \Gamma_t$ for $t \in (-\infty, T]$. Since $\kappa_s = 0$ at the y -axis and $\kappa_s = \kappa > 0$ at the right boundary point, the strong maximum principle and the Hopf boundary point lemma imply that $\kappa_s > 0$ in $\{x > 0\} \cap \Gamma_t$ for $t \in (-\infty, T]$. But then Sturm's theorem, Theorem 1.20, implies that κ_s does not develop additional zeroes up to time 0. Having established the first claim, the second follows as in Lemma 3.9. ■

Proposition 3.18. *Let $x \mapsto y(x, t)$ be the graphical parametrization of Γ_t as a graph over the x -axis. There exists a constant $A \in (0, \infty)$ such that*

$$e^{\lambda_0^2 t} y(x, t) \rightarrow A \cosh(\lambda_0 x) \text{ uniformly as } t \rightarrow -\infty.$$

Proof. Given $\tau < 0$, consider the rescaled height function

$$y^\tau(x, t) \doteq e^{-\lambda_0^2 \tau} y(x, t + \tau),$$

which is defined on the time-translated flow $\{\Gamma_t^\tau\}_{t \in (-\infty, -\tau)}$, where $\Gamma_t^\tau \doteq \Gamma_{t+\tau}$. Note that

$$\begin{cases} (\partial_t - \Delta^\tau) y^\tau = 0 & \text{in } \{\Gamma_t^\tau\}_{t \in (-\infty, -\tau)} \\ \langle \nabla^\tau y^\tau, N \rangle = y & \text{on } \{\partial \Gamma_t^\tau\}_{t \in (-\infty, -\tau)}, \end{cases} \quad (3.12)$$

where ∇^τ and Δ^τ are the gradient and Laplacian on $\{\Gamma_t^\tau\}_{t \in (-\infty, -\tau)}$, respectively, and N is the outward unit normal to ∂B^2 .

Since $\{\Gamma_t^\tau\}_{t \in (-\infty, 0)}$ reaches the origin at time zero, it must intersect the constructed solution for all $t < 0$. In particular, the value of $\underline{y}$ on the former can at no time exceed the value of $\bar{y}$ on the latter. But then Lemma 3.14 and (3.11) yield

$$\limsup_{t \rightarrow -\infty} e^{-\lambda_0^2 t} \bar{y} < \infty. \quad (3.13)$$

This implies a uniform bound for y^τ on $\{\Gamma_t^\tau\}_{t \in (-\infty, T]}$ for any $T \in \mathbb{R}$. So Alaoglu's theorem yields a sequence of times $\tau_j \rightarrow -\infty$ such that y^{τ_j} converges in the weak* topology as $j \rightarrow \infty$ to some $y^\infty \in L_{\text{loc}}^2([-1, 1] \times (-\infty, \infty))$. Since convexity and the boundary condition imply a uniform bound for $\nabla^\tau y^\tau$ on any time interval of the form $(-\infty, T]$, we may also arrange that the convergence is uniform in space at time zero, say. Weak* convergence ensures that y^∞ satisfies the problem

$$\begin{cases} y_t = & y_{xx} \text{ in } [-1, 1] \times (-\infty, \infty) \\ y_x(\pm 1) = & \pm y(\pm 1). \end{cases} \quad (3.14)$$

Indeed, a smooth function y^τ satisfies the boundary value problem (3.12) (and analogously for (3.14)) if and only if

$$\int_{-\infty}^{-\tau} \int_{\Gamma_t^\tau} y^\tau (\partial_t - \Delta^\tau)^* \eta = 0$$

for all smooth η which are compactly supported in time and satisfy

$$\nabla^\tau \eta \cdot N = \eta \text{ on } \partial \Gamma_t^\tau,$$

where $(\partial_t - \Delta^\tau)^* \doteq -(\partial_t + \Delta^\tau)$ is the formal L^2 -adjoint of the heat operator. Since $\{\Gamma_t^\tau\}_{t \in (-\infty, -\tau)}$ converges uniformly in the smooth topology to the stationary interval $\{[-1, 1] \times \{0\}\}_{t \in (-\infty, \infty)}$ as $\tau \rightarrow -\infty$, we conclude that the limit y^∞ must satisfy (3.14) in the L^2 sense (and hence in the classical sense due to the L^2 theory for the heat equation). Indeed, by the definition of smooth convergence, we may (after possibly applying a diffeomorphism) parametrize each flow $\{\bar{\Gamma}_t^{\tau_j}\}_{t \in (-\infty, -\tau_j)}$ over $I \doteq [-1, 1]$ by a family of embeddings $\gamma_t^j : I \times (-\infty, -\tau_j) \rightarrow \bar{B}^2$ which converge in $C_{\text{loc}}^\infty(I \times (-\infty, \infty))$ as $j \rightarrow \infty$ to the stationary embedding $(x, t) \mapsto xe_1$.

Given $\eta \in C_0^\infty(I \times (-\infty, \infty))$ satisfying $\eta_\zeta(\pm 1) = \pm \eta$, set $\eta^j \doteq \varphi^j \eta$, where $\varphi^j : [-1, 1] \times (-\infty, -\tau^j) \rightarrow \mathbb{R}$ is defined by

$$\varphi_\zeta^j + (1 - |\gamma_\zeta^j|)\varphi^j = 0, \quad \varphi^j(0, t) = 1.$$

That is, $\varphi^j(\zeta, t) = e^{s^j(\zeta, t) - \zeta}$, where $s^j(\zeta, t) \doteq \int_0^\zeta |\gamma_\xi^j(\xi, t)| d\xi$. This ensures that $\nabla^{\tau^j} \eta^j \cdot N = \eta^j$ at the boundary, and hence

$$0 = \int_{-\infty}^{\infty} \int_I y^{\tau_j} (\partial_t - \Delta^{\tau_j})^* \eta^j ds^j dt.$$

Since $\varphi^j \rightarrow 1$ in $C_{\text{loc}}^\infty(I \times (-\infty, \infty))$, a short computation reveals that

$$0 = \int_{-\infty}^{\infty} \int_I y^\infty (\partial_t - \Delta)^* \eta d\zeta dt.$$

Finally, we characterize the limit (uniqueness of which implies full convergence, completing the proof). Separation of variables leads us to consider the problem

$$\begin{cases} -\phi_{xx} = \mu \phi & \text{in } [-1, 1] \\ \phi_x(\pm 1) = \pm \phi(\pm 1). \end{cases}$$

There is only one negative eigenspace, and its frequency turns out to be $k_{-1} = \lambda_0$, with the corresponding mode given by

$$\phi_{-1}(x) \doteq \cosh(k_{-1}x).$$

Thus, recalling (3.13), we are able to conclude that

$$y^\infty(x, t) = Ae^{\lambda_0^2 t} \cosh(\lambda_0 x)$$

for some $A \geq 0$. In particular,

$$e^{-\lambda_0^2 \tau_j} y(x, \tau_j) = y^{\tau_j}(x, 0) \rightarrow A \cosh(\lambda_0 x) \quad \text{uniformly as } j \rightarrow \infty. \quad \blacksquare$$

Uniqueness

Uniqueness of the constructed ancient solution now follows directly from the avoidance principle.

Proposition 3.19. *Modulo time translation and rotation about the origin, there is only one convex, locally uniformly convex ancient solution to free boundary curve shortening flow in the disc.*

Proof. Denote by $\{\Gamma_t\}_{t \in (-\infty, 0)}$ the constructed ancient solution and let $\{\Gamma'_t\}_{t \in (-\infty, 0)}$ be a second ancient solution which, without loss of generality, contracts to the point $(0, 1)$ at time 0. By Proposition 3.18 we have that the graphical (over the x -axis) parametrizations of the solutions satisfy

$$e^{\lambda_0 t} y(x, t) \rightarrow A \cosh(\lambda_0 x) \quad \text{and} \quad e^{\lambda_0 t} y'(x, t) \rightarrow A' \cosh(\lambda_0 x)$$

respectively, with $A > 0$ by Lemma 3.14. We claim that $A' = A$. To see this note that if A' is not equal to A , then one of the two solutions must lie above the other at some time $t \ll 1$. But this violates the avoidance principle.

Given any $\tau > 0$, consider the time-translated solution $\{\Gamma_t^\tau\}_{t \in (-\infty, -\tau)}$ defined by $\Gamma_t^\tau = \Gamma'_{t+\tau}$. By Proposition 3.18,

$$e^{-\lambda_0^2 t} y^\tau(x, t) \rightarrow A e^{\lambda_0^2 \tau} \cosh(\lambda_0 x) \quad \text{as } t \rightarrow -\infty$$

uniformly in x . So Γ_t^τ lies above Γ_t for $-t$ sufficiently large. The avoidance principle then ensures that Γ_t^τ lies above Γ_t for all $t \in (-\infty, 0)$. Taking $\tau \rightarrow 0$, we find that Γ'_t lies above Γ_t for all $t < 0$. Since the two curves reach the point $(0, 1)$ at time zero, they intersect for all $t < 0$ by the avoidance principle. The strong maximum principle then implies that the two solutions coincide for all t . ■

3.3 The General Case; A Second Proof

In this section we will describe a second method to obtain the classification of convex ancient solutions in compact domains. In this second method, we construct the ancient solutions in a new way by perturbing a diameter along the first eigenfunction of its Jacobi operator. Uniqueness will be similar to that presented in the first method.

From now on, we let D be a diameter and by shifting, rotating and dilating we can assume that the diameter is given by $D = [-1, 1] \times \{0\}$. We first note that the Jacobi operator for D is the Laplacian. In particular, the second variation of the length functional along a variation generated by the vector field ue_2 is given by

$$\int_D |\nabla u|^2 d\mathcal{L}^1 - \sum_{\partial D} \kappa_\Sigma u^2, \quad (3.15)$$

where κ_Σ denotes the curvature of Σ . We consider $\lambda_0 < \lambda_1 \leq \lambda_2 \leq \dots$ to be the spectrum of

$$\begin{cases} -\phi_{xx} &= \lambda\phi \text{ in } [-1, 1] \\ \frac{\partial\phi}{\partial\eta} &= \phi\kappa_\Sigma \text{ on } \partial[-1, 1] = \{-1, 1\}, \end{cases} \quad (3.16)$$

where η is the outward pointing unit normal on Σ . Note that the boundary condition reads as

$$\phi_x(\pm 1) = \pm \phi(\pm 1) \kappa_\Sigma(\pm 1).$$

The eigenfunctions ϕ_k corresponding to λ_k satisfy

$$\int_D |\nabla \phi_k|^2 d\mathcal{L}^1 - \sum_{\partial D} \kappa_\Sigma \phi_k^2 = \lambda_k.$$

Lemma 3.20. *The spectrum of (3.16) satisfies the following. There is always a negative eigenvalue and zero is an eigenvalue if and only if $\kappa_\Sigma^{-1}(1) + \kappa_\Sigma^{-1}(-1) = 2$.*

Proof. The associated quadratic form is

$$Q(u) = \int_{-1}^1 |u'(x)|^2 dx - \kappa_\Sigma(1)|u(1)|^2 - \kappa_\Sigma(-1)|u(-1)|^2,$$

defined on $H^1(-1, 1)$. By the Rayleigh–Ritz characterization, the first eigenvalue is given by

$$\lambda_0 = \inf_{u \in H^1(-1, 1) \setminus \{0\}} \frac{Q(u)}{\int_{-1}^1 |u(x)|^2 dx}.$$

By considering the test function $u \equiv 1$, and since κ_Σ is strictly positive, we obtain that $\lambda_1 < 0$.

Suppose that $\lambda = 0$ is an eigenvalue. Then the eigenvalue problem becomes

$$-u'' = 0,$$

and every solution is of the form

$$u(x) = Ax + B,$$

for constants A, B . The Robin boundary conditions then gives that

$$A = \kappa_{\Sigma}(1)(A + B) = -\kappa_{\Sigma}(-1)(-A + B).$$

And now it is straightforward to deduce that a nontrivial solution $(A, B) \neq (0, 0)$ of this system exists if and only if the determinant of the coefficient matrix vanishes:

$$\begin{vmatrix} 1 - \kappa_{\Sigma}(1) & -\kappa_{\Sigma}(1) \\ 1 - \kappa_{\Sigma}(-1) & \kappa_{\Sigma}(-1) \end{vmatrix} = 0.$$

or equivalently,

$$\kappa^{-1}(1) + \kappa^{-1}(-1) = 2.$$

■

Remark 3.21. In fact one can check that there are at most two negative eigenvalues and if $\kappa_{\Sigma}^{-1}(1) + \kappa_{\Sigma}^{-1}(-1) > 2$, then there is exactly one negative eigenvalue.

From now on we denote by $\lambda_0 < 0$ the first eigenvalue, ϕ_0 the corresponding eigenfunction, normalized so that $\|\phi_0\|_{L^2(D)} = 1$, and $\lambda_I \geq \lambda_0$ the largest negative eigenvalue. We will present an alternative construction of a convex ancient solution to the free boundary curve shortening flow emerging from the diameter D , distinct from the method developed in Section 3.2. The construction proceeds by first constructing sub- and super-solutions obtained by perturbing the diameter in the direction of the first eigenfunction. A key difficulty is that such perturbations do not automatically preserve the orthogonality condition at the boundary. We overcome this by a careful application of the implicit function theorem, which allows us to foliate a neighborhood of D by free boundary convex curves. This construction extends to higher dimensions and, more generally, to mean-convex solutions, as established in (Bourni; Franz, 2026).

3.3.1 Graphs Using a Local Parametrization

Let $\bar{\nu}$ be a smooth vector field in Ω , which is equal to e_2 on D , and such that $\bar{\nu}$ is tangent to Σ . Moreover, we let ψ be the flow of $\bar{\nu}$ passing

through D at time zero. Note that

$$\psi: D \times (-r_0, r_0) \rightarrow \Omega$$

is a diffeomorphism between $D \times (-r_0, r_0)$, for some $r_0 > 0$, and a tubular neighborhood of D in Ω , so it provides a foliation of this neighborhood (à la Definition 1.5). Using the variable notation $\psi(x, s)$, we have that $\frac{d}{ds}\psi = \bar{\nu}$. We remark that $\bar{\nu}$ is not necessarily the normal to the leaves $\psi(D, s)$. However, since it is for the leaf $s = 0$, we have that

$$\begin{aligned} \langle \psi_x, \bar{\nu} \rangle &= \int_0^s \frac{d}{d\bar{s}} \langle \psi_x, \bar{\nu} \rangle d\bar{s} = \int_0^s \frac{d}{d\bar{s}} \langle \psi_x, \bar{\nu} \rangle \frac{d}{d\bar{s}} (\bar{s} - s) d\bar{s} \\ &= s \frac{d}{d\bar{s}} \langle \psi_x, \bar{\nu} \rangle |_{\bar{s}=0} - \int_0^s \frac{d^2}{d\bar{s}^2} \langle \psi_x, \bar{\nu} \rangle (\bar{s} - s) d\bar{s} \leq Cs. \end{aligned} \quad (3.17)$$

Similarly, since $\psi_x = e_1$ at $s = 0$,

$$\begin{aligned} |\psi_x|^2 - 1 &= \int_0^s \frac{d}{d\bar{s}} \langle \psi_x, \psi_x \rangle d\bar{s} = \int_0^s \frac{d}{d\bar{s}} \langle \psi_x, \psi_x \rangle \frac{d}{d\bar{s}} (\bar{s} - s) d\bar{s} \\ &= - \int_0^s \frac{d^2}{d\bar{s}^2} \langle \psi_x, \psi_x \rangle (\bar{s} - s) d\bar{s} \leq Cs^2, \end{aligned} \quad (3.18)$$

where in both the bounds above, $C > 0$ is a constant that depends on $\|\bar{\nu}\|_{C^2(D \times (-r_0, r_0))}$ and $\|\psi\|_{C^1(D \times (-r_0, r_0))}$.

Consider now a smooth function $u: D \rightarrow \mathbb{R}$ with $\|u\|_{C^0(D)} < r_0$, and the curve $\Gamma_u \subset \Omega$ defined as the graph of u via ψ . In other words, Γ_u is defined as the image of the function $F: D \rightarrow \Omega$, given by

$$F(x) := \psi(x, u(x)).$$

Then, a tangent vector to Γ_u is given by

$$F_x := \psi_x + u_x \bar{\nu}, \text{ with } |F_x|^2 = |\psi_x|^2 + 2u_x \langle \psi_x, \bar{\nu} \rangle + u_x^2. \quad (3.19)$$

The unit normal ν_u is given by

$$\nu_u = \frac{\bar{\nu} - \langle F_x, \bar{\nu} \rangle |F_x|^{-2} F_x}{\sqrt{1 - |F_x|^{-2} \langle F_x, \bar{\nu} \rangle^2}},$$

therefore

$$\langle \nu_u, \bar{\nu} \rangle = \sqrt{1 - |F_x|^{-2} \langle F_x, \bar{\nu} \rangle^2}. \quad (3.20)$$

Observe that, by (3.17), (3.18) and (3.19),

$$|\langle F_x, \bar{\nu} \rangle - u_x| \leq C|u|, \quad (3.21)$$

$$||F_x|^2 - 1| \leq C\|u\|_{C^1(D)}^2, \quad (3.22)$$

and

$$|\nu_u - \bar{\nu}| \leq C\|u\|_{C^1(D)}^2. \quad (3.23)$$

It is known that the linearization of the curvature $-\kappa_u := |F_x|^{-2}\langle F_{xx}, \nu_u \rangle$ of Γ_u (the curvature computed with respect to ν_u) is given by Δu and the linearization of the contact angle $\langle \nu_u, \eta \rangle$ is given by $-\frac{\partial u}{\partial \eta} + \kappa_\Sigma u$. These, in general, are derived by considering the first variation of the quantities (the curvature and the contact angle respectively). In dimension one it is also not that difficult to obtain the linearization ‘by hand’, by expanding the terms in the definition of the quantities.

In what follows (see (3.25)), we will need the linearization of $-\frac{\kappa_u}{\langle \nu_u, \bar{\nu} \rangle}$, which is also Δ since the linearization of $\langle \nu_u, \bar{\nu} \rangle$ is 1, as we show below. This motivates the following definition.

Definition 3.22. Given $u: D \rightarrow \mathbb{R}$ with $\|u\|_{C^0(D)} < r_0$, we define the functions $E(u): D \rightarrow \mathbb{R}$ and $\varepsilon(u): \partial D \rightarrow \mathbb{R}$ as

$$\begin{cases} -\frac{\kappa_u}{\langle \nu_u, \bar{\nu} \rangle} = \Delta u + E(u) & \text{on } D \\ \langle \nu_u, \eta \rangle = -\frac{\partial u}{\partial \eta} + \kappa_\Sigma u + \varepsilon(u) & \text{on } \partial D. \end{cases} \quad (3.24)$$

The functions E and ε satisfy

$$\|E(u)\|_{C^0(D)} \leq C\|u\|_{C^2(D)}\|u\|_{C^1(D)} \quad \text{and} \quad |\varepsilon(u)(\pm 1)| \leq C\|u\|_{C^1(D)}^2,$$

where C depends only on Σ .

We can verify the bounds on E and ε by computing them explicitly. For E we observe that

$$F_{xx} = \psi_{xx} + 2\psi_{xs}u_x + \psi_s u_{xx} + \psi_{ss}u_x^2.$$

and therefore

$$\begin{aligned} \langle F_{xx}, \nu_u \rangle &= u_{xx} \langle \bar{\nu}, \nu_u \rangle + \langle F_{xx} - \psi_s u_{xx}, \bar{\nu} \rangle + \langle F_{xx} - \psi_s u_{xx}, \nu_u - \bar{\nu} \rangle \\ &= u_{xx} \langle \bar{\nu}, \nu_u \rangle + \langle \psi_{xx}, \bar{\nu} \rangle + \langle F_{xx} - \psi_s u_{xx}, \nu_u - \bar{\nu} \rangle. \end{aligned}$$

Computing as in (3.17) with ψ_x replaced by ψ_{xx} (which is zero at $s = 0$) we obtain $\langle \psi_{xx}, \bar{\nu} \rangle \leq Cs^2$, with the constant C now depending on the C^2 norm of ψ . This together with (3.23) give

$$|\langle F_{xx}, \nu_u \rangle - u_{xx} \langle \bar{\nu}, \nu_u \rangle| \leq C \|u\|_{C^1(D)}^2.$$

Finally, using (3.22) and (3.23) again, we obtain the estimate for E , since

$$\left| -\frac{\kappa_u}{\langle \nu_u, \bar{\nu} \rangle} - u_{xx} \right| \leq C \|u\|_{C^1(D)}^2 (1 + \|u\|_{C^2(D)}^2).$$

For the ε estimate, we first observe that (3.17) at a boundary point becomes

$$\langle \psi_x, \bar{\nu} \rangle \leq s\kappa_\Sigma + Cs^2.$$

Therefore, we obtain

$$-\langle F_x, \bar{\nu} \rangle = -u_x - \langle \psi_x, \bar{\nu} \rangle = -u_x + \kappa_\Omega u + O(u^2),$$

and since $\langle \nu_u, \eta \rangle = -\langle \frac{F_x}{|F_x|}, \bar{\nu} \rangle$, using (3.22) we get the bound for ε .

3.3.2 Graphical Curve Shortening Flow

Let $u: D \times (T_0, T) \rightarrow \mathbb{R}$ be a time-dependent function over D , and consider the associated family of curves Γ_t defined as the graph of $u(\cdot, t)$ over ψ . In other words, Γ_t is the image of the function $F(\cdot, t): D \rightarrow M$ given by

$$F(x, t) := \psi(x, u(x, t)).$$

Then the family of curves Γ_t moves with velocity

$$\frac{d}{dt} F = u_t \bar{\nu}.$$

In particular, the normal speed is given by $u_t \langle \nu_u, \bar{\nu} \rangle$. Consequently (cf. proof of Theorem 1.4), the family evolves by graphical free boundary curve shortening flow if and only if

$$\begin{cases} u_t = -\frac{\kappa_u}{\langle \nu_u, \bar{\nu} \rangle} & \text{on } D \\ \langle \nu_u, \eta \rangle = 0 & \text{on } \partial D. \end{cases} \quad (3.25)$$

Using the linearization of the curvature and the Neumann condition in (3.24), we infer that graphical free boundary curve shortening flow is equivalent to a solution of the equation

$$\begin{cases} u_t - \Delta u = E(u) & \text{on } D \\ \frac{\partial u}{\partial \eta} - \kappa_\Sigma u = \varepsilon(u) & \text{on } \partial D. \end{cases} \quad (3.26)$$

First, let us state here, as we need it later, the Schauder estimates for linear parabolic equations as (3.26), see for example (Simon, 1997), (Baderko, 1998, Theorem 1), and (Gilbarg; Trudinger, 2001, Theorem 6.30) for the elliptic case.

Theorem 3.23 (Parabolic Schauder estimates). *Let $w: D \times (T_0, T) \rightarrow \mathbb{R}$ be a solution to the equation*

$$\begin{cases} w_t - \Delta w = f(x, t) & \text{on } D \\ \frac{\partial w}{\partial \eta} - \kappa_\Sigma w = g(x, t) & \text{on } \partial D, \end{cases}$$

in an interval $t \in (T_0, T)$. Then, for all $t \in (T_0 + 2, T)$,

$$\|w\|_{C^{2,\alpha}(D \times (t-1, t))} \leq C \left(\|w\|_{L^2(D \times (t-2, t))} + \|f\|_{C^{0,\alpha}(D \times (t-2, t))} + \|g\|_{C^0(\partial D \times (t-2, t))} \right),$$

where $C > 0$ is a constant depending only on Σ .

In the case of a convex solution, which recall is a property that is preserved under the flow, Stahl's (global in space, interior in time) Ecker–Huisken type estimates (Theorems 1.32 and 1.33) imply that the curvature and all its derivatives admit bounds that depend only on the height. In particular, by convexity, the quantity $\langle \nu, e_2 \rangle$ (i.e., the vertical component of the normal) is bounded from below by its minimum along the boundary. Consequently, the gradient is controlled in terms of the height, and therefore all curvature estimates ultimately depend only on the height. To summarize,

Theorem 3.24 ((Stahl, 1996b)). *If the solution $\{\Gamma_t\}_{t \in (T_0, T)}$, as defined above, is convex, then $\|u\|_{C^k(D)} \leq C(k, C_0)$, where C_0 is such that $\|u\|_{C^0(D)} \leq C_0$.*

3.3.3 Constructing an Ancient Solution

Theorem 3.25. *For every $\rho \in (\lambda_I, 0)$ and $\delta > 0$, there exist a $T \in \mathbb{R}$ and a $C^{2,\alpha}$ -function $u: D \times (-\infty, T) \rightarrow \mathbb{R}$ with $\|u(\cdot, t)\|_{C^{2,\alpha}(D)} < \delta$ for all $t \in (-\infty, T)$, satisfying the following property: Consider the function*

$$w(x, t) = \varphi_0(x)e^{-\lambda_0 t} + u(x, t).$$

Then, the curve

$$G_t = G_{w(\cdot, t)} := \{\psi(x, w(x, t)), x \in D\} \quad (3.27)$$

has free boundary on Σ with curvature κ_w (with respect to a choice of normal ν_w) satisfying

$$-\frac{\kappa_w}{\langle \nu_w, \bar{\nu} \rangle} = -\lambda_0 \varphi_0 e^{-\lambda_0 t} - \rho u. \quad (3.28)$$

Moreover,

$$|u(x, t)| \leq Ce^{-2\lambda_0 t}, \quad |u_t(x, t)| \leq Ce^{-2\lambda_0 t}. \quad (3.29)$$

In particular, G_t converges exponentially fast to D as $t \rightarrow -\infty$.

Remark 3.26. The point of the proposition is that we expect the existence of an ancient curve shortening flow whose graph over D , at first order, is given by $\varphi_0 e^{-\lambda_0 t}$ for $t \in (-\infty, T)$, with curvature, at first order, being $-\lambda_0 \varphi_0 e^{-\lambda_0 t}$. Our proposition does not directly give this ancient solution, but it adds a correction u to the first order approximation to satisfy the orthogonality condition at the boundary. Indeed, notice that a priori the graph of $\varphi_0 e^{-\lambda_0 t}$ over D is not necessarily orthogonal to Σ .

Proof of Theorem 3.25. Observe that since $\rho \in (\lambda_I, 0)$ is not an eigenvalue, the problem (3.16) with λ replaced by ρ does not admit nontrivial solutions. It is convenient to adopt the change of variable $\tau = e^{-\lambda_0 t}$ and look for τ defined in $(-\delta, \delta)$ for some $\delta > 0$. Let us define

$$h_{\varphi_0 \tau + u}(x, \tau) := -\lambda_0 \varphi_0 \tau \langle \nu_{\varphi_0 \tau + u}, \bar{\nu} \rangle,$$

where $\nu_{\varphi_0 \tau + u}$ is the upward pointing unit normal, and the map $\Phi: (-\delta, \delta) \times (B(0, \delta) \subset C^{2,\alpha}(D)) \rightarrow C^{0,\alpha}(D) \times \mathbb{R}^2$ by

$$\Phi(\tau, u) = \left(-\kappa_{\varphi_0 \tau + u} - h_{\varphi_0 \tau + u} + \rho u \langle \nu_{\varphi_0 \tau + u}, \bar{\nu} \rangle, \sum_{\partial D} \langle \nu_{\varphi_0 \tau + u}, \eta_{\varphi_0 \tau + u} \rangle \right).$$

Recall that $\nu_{\varphi_0\tau+sv}$ coincides with $\bar{\nu}$ for $\tau = 0$ and $s = 0$, and $\frac{d}{ds}|_{s=0}\langle\nu_{sv}, \bar{\nu}\rangle = 0$ (see (3.20) and (3.23)). Therefore, by the linearization operators (see Definition 3.24), we have

$$D\Phi_{(0,0)}(0, v) = \frac{d}{ds}|_{s=0}\Phi(0, sv) = \left(\Delta v + \rho v, \sum_{\partial D} -\frac{\partial v}{\partial x} + \kappa_{\Sigma} v \right).$$

Then $D\Phi_{(0,0)}(0, \cdot): C^{2,\alpha}(D) \rightarrow C^{0,\alpha}(D) \times \mathbb{R}^2$ is invertible. Indeed, it is injective by the choice of ρ , and it is also surjective (see for instance (Ladyzhenskaya; Ural'tseva, 1968, Theorem 3.2, pp. 137)).

As a result, by the implicit function theorem (possibly taking $\delta > 0$ smaller), there exists a function $u: (-\delta, \delta) \rightarrow B(0, \delta) \subset C^{2,\alpha}(\Sigma)$ such that $u(0) = 0$ and $\Phi(\tau, u(\tau)) = 0$ for all $\tau \in (-\delta, \delta)$. Since Φ is smooth, u is smooth as well. Let us define $w(x, \tau) = \varphi_0(x)\tau + u(\tau)(x)$, then the curve $G_\tau = G_{w(\cdot, \tau)}$ is a free boundary curve with curvature κ_w satisfying (3.28). Let us now compute the derivative of

$$0 = \Phi(\tau, u(\tau)) = \left(-\kappa_w - h_w + \rho u \langle \nu_w, \bar{\nu} \rangle, \sum_{\partial D} \langle \nu_w, \eta_w \rangle \right)$$

with respect to τ at $\tau = 0$. At $\tau = 0$,

$$h'_w = -\lambda_0 \varphi_0.$$

Thus, defining $w' = \frac{\partial w}{\partial \tau}|_{\tau=0} = \varphi_0 + u'$ we get

$$\begin{aligned} 0 &= \left(\Delta w' - (h_w)' + \rho u', \sum_{\partial D} -\frac{\partial w'}{\partial \eta} + \kappa_{\Sigma} w' \right) \\ &= \left(\Delta u' + \rho u', \sum_{\partial D} -\frac{\partial u'}{\partial \eta} + \kappa_{\Sigma} u' \right), \end{aligned}$$

where in the second equality we used that φ_0 is an eigenfunction of the Jacobi operator with respect to eigenvalue λ_0 . This implies that

$$\begin{cases} -\Delta u' = \rho u' & \text{on } D \\ \frac{\partial u'}{\partial \nu} = \kappa_{\Sigma} u' & \text{on } \partial D, \end{cases}$$

namely u' is an eigenfunction of the Jacobi operator relative to eigenvalue ρ and it is therefore equal to 0. As a result, since $u(0) = u'(0) = 0$ and u

is smooth (C^2 would be sufficient), there exists $C > 0$ such that $|u(\tau)| \leq C\tau^2$ and $|u'(\tau)| \leq C\tau$. By changing variable back to t , we get the desired estimates (3.29) on $|u|$ and $|u_t|$, concluding the proof. ■

We now construct sub- and super-solutions to the free boundary curve shortening flow that are close to the family $\{w(\cdot, t)\}_t$ constructed in Theorem 3.25. Note that this w satisfies

$$\begin{cases} w_t = -\lambda_0 \phi_0 e^{-\lambda_0 t} + u_t = -\frac{\kappa_w}{\langle \nu_w, \bar{\nu} \rangle} + \rho u + u_t & \text{on } D \\ \langle \nu_w, \bar{\nu} \rangle = 0 & \text{on } \partial D, \end{cases} \quad (3.30)$$

and therefore, comparing with (3.25), the evolving hypersurfaces $\Sigma_{w(\cdot, t)}$ “almost” satisfy curve shortening flow. Let us now construct sub- and super-solutions by slightly reparametrizing the time variable.

Lemma 3.27. *Given $\lambda \in (\lambda_0, 0)$, we define the families*

$$w^\pm(x, t) = w(x, t \pm e^{-\lambda t}),$$

where $\{w(\cdot, t)\}_t$ is the function constructed in Theorem 3.25 associated to $\phi_0 > 0$. Then there exists $t_\lambda \in \mathbb{R}$ such that w^+ is a supersolution and w^- is a subsolution of the free boundary curve shortening flow for all $t < t_\lambda$, that is,

$$\frac{dw^+}{dt} \geq -\frac{\kappa_{w^+}}{\langle \nu_{w^+}, \bar{\nu} \rangle} \quad \text{and} \quad \frac{dw^-}{dt} \leq -\frac{\kappa_{w^-}}{\langle \nu_{w^-}, \bar{\nu} \rangle}.$$

Remark 3.28. Note that Lemma 3.27 is also true for $\phi_0 < 0$, reversing the roles of w^+ and w^- .

Proof. Let $w_f(x, t) = w(x, t + f(t))$. Then, using (3.30),

$$\frac{dw_f}{dt} = \frac{\partial w}{\partial t}(1 + f'(t)) = \left(-\frac{\kappa_{w_f}}{\langle \nu_{w_f}, \bar{\nu} \rangle} + \rho u + \frac{\partial u}{\partial t} \right) (1 + f'(t)),$$

which implies

$$\frac{dw_f}{dt} + \frac{\kappa_{w_f}}{\langle \nu_{w_f}, \bar{\nu} \rangle} = -\lambda_0 \phi_0 e^{-\lambda_0(t+f(t))} f'(t) + \left(\rho u + \frac{\partial u}{\partial t} \right) (1 + f'(t)). \quad (3.31)$$

Recall that, by Theorem 3.25, $|u| + |u_t| \leq C e^{-2\lambda_0 t}$ and $\phi_0 > 0$. Therefore, for any $\lambda \in (\lambda_0, 0)$, if we choose $f(t) = \pm e^{-\lambda t}$, we obtain that for all t sufficiently small, the right hand side of (3.31) is positive and negative respectively. Hence, the result follows for $w^\pm = w_{\pm e^{-\lambda t}}$. ■

Theorem 3.29. *There exists a graphical ancient solution $\{\Gamma_t\}_{t \in (-\infty, T)}$, given by the graph of a function $v: D \times (-\infty, T) \rightarrow \mathbb{R}$ satisfying*

$$\|v(\cdot, t)\|_{C^0(D)} \leq Ce^{-\lambda_0 t},$$

for a constant $C > 0$ depending only on Σ , and

$$\lim_{t \rightarrow -\infty} v(x, t)e^{\lambda_0 t} = \phi_0(x).$$

Furthermore, this solution is convex.

Proof. Let $w(x, t) = \phi_0 e^{-\lambda_0 t} + u(x, t)$ be as in Theorem 3.25, and $w^\pm$ the corresponding barriers constructed in Lemma 3.27 for some $\lambda \in (\lambda_0, 0)$. Without loss of generality assume that $\phi_0 > 0$ (the case $\phi_0 < 0$ being similar, see Remark 3.28). We claim that we can choose t_λ , possibly smaller than the one required in Lemma 3.27, so that we additionally have

$$w^+(x, t) \geq w(x, t) \geq w^-(x, t) \quad \forall t < t_\lambda. \quad (3.32)$$

Indeed, using $|u_t(x, t)| \leq Ce^{-2\lambda_0 t}$, by Theorem 3.25 we obtain

$$\begin{aligned} w^+(x, t) - w(x, t) &= \phi_0 e^{-\lambda_0 t} (e^{-\lambda_0 e^{-\lambda t}} - 1) + u(x, t + e^{-\lambda t}) - u(x, t) \\ &\geq -\lambda_0 \phi_0 e^{-(\lambda_0 + \lambda)t} - Ce^{-(2\lambda_0 + \lambda)t}, \end{aligned}$$

which is clearly positive for t small enough. Similarly, we can argue that w^- lies below w , obtaining (3.32).

Now, consider a decreasing sequence of times $t_\lambda > t_0 > t_1 > \dots \rightarrow -\infty$ and, for every $i \in \mathbb{N}$, consider the free boundary curve shortening flow $\{G_t^i\}_{t \in [t_i, T_i]}$, starting from $G_{t_i} = G_{w(\cdot, t_i)}$. Note that the curves are convex since the initial curve is, and by Theorem 2.3, the flow is smooth and converges to a point.

Next, we claim that the curves remain graphical for all $t < t_\lambda$. This is true for a short time by smoothness, so each flow can be written as the graph of a function $v^i(x, t)$ over D . Moreover, v^i satisfies the graphical curve shortening flow (3.26), and by Lemma 3.27 it has to stay graphical with

$$w^-(x, t) \leq v^i(x, t) \leq w^+(x, t) \quad \forall t \in [t_i, t_\lambda].$$

This ensures that the flow remains graphical, and also, by Theorem 3.24, we have uniform-in- i bounds on the curvature and all its derivatives. Consequently, we can pass to a limit, as $i \rightarrow \infty$, and obtain an ancient solution with the desired properties as stated in the theorem. $\blacksquare$

3.3.4 Uniqueness

In this section, we show that for each diameter, there exist exactly two ancient convex solutions emanating from it, namely, the ones constructed in Theorem 3.29 for $\phi_0 > 0$ and $\phi_0 < 0$ respectively. Uniqueness is understood modulo time translations.

These solutions also exhaust all convex ancient solutions to the free boundary curve shortening flow, since any such solution, backward in time, converges to a diameter. The proof of this fact is analogous to Lemma 3.15. In that lemma, the statement is proved for the circle, but the argument applies verbatim for any bounded convex domain, with the only change being that the bound on the total enclosed area is replaced by a constant instead of π .

Therefore, to prove our classification theorem, Theorem 3.2, it suffices to establish uniqueness for solutions converging to a given diameter from a specified side. We accomplish this by showing that any such solution converges exponentially fast to the diameter, and hence shares the same asymptotic expansion as the solutions constructed in Theorem 3.29. The avoidance principle then yields uniqueness, as in Proposition 3.19.

From now on we assume that $\{\Gamma_t\}_{t \in (-\infty, 0)}$ is any maximal convex ancient free boundary curve shortening flow. Moreover, we let D be the backward limit diameter, which, after a rotation, we may assume lies along the x -axis. We first show exponential decay of the solution to D .

Lemma 3.30. *Γ_t converges backwards in time to the diameter D exponentially fast.*

Proof. Let $\gamma(\cdot, t)$ denote a parametrization of the time slices by turning angle, so that at $\gamma(\theta, t)$ the tangent (in counterclockwise orientation) is given by $(\cos \theta, \sin \theta)$. Using this parametrization, define $u = \langle \gamma, e_2 \rangle$, the height function, and let κ denote the curvature with respect to $(-\sin \theta, \cos \theta)$, the downward pointing unit normal, so that it is positive. We first show the following claim.

Claim. There exists a constant $C > 0$ such that

$$u \leq C\kappa. \tag{3.33}$$

Proof of Claim. To prove (3.33), note that at the boundary

$$\lim_{t \rightarrow -\infty} \frac{u}{\kappa} = \frac{1}{(\kappa^\Omega)^2}, \tag{3.34}$$

which follows from an application of L'Hôpital's rule. To see this, first, recall that at the boundary $\kappa_\theta = \kappa^\Omega$. And second, to compute u_θ , we let Φ be a parametrization of Σ by turning angle, and write

$$u(\theta) = \langle \Phi(\theta + \frac{\pi}{2}), e_2 \rangle = \int_{\frac{\pi}{2}}^{\frac{\pi}{2} + \theta} \frac{\sin u}{\kappa^\Omega(u)} du. \quad (3.35)$$

(3.34) implies that we can find a constant $C > 0$ such that (3.33) holds at the boundary. In the interior, we compute

$$(\partial_t - \Delta)(u - C\kappa) = -C\kappa^3 < 0.$$

The maximum principle, together with the fact that $\lim_{t \rightarrow -\infty} (u - C\kappa) = 0$, then establishes (3.33). $\blacklozenge$

The claim now implies that

$$\frac{u_t}{u} = -\frac{\kappa}{u} \langle \nu, e_2 \rangle \geq \frac{1}{2C},$$

where, as before, ν denotes the downward pointing unit normal, and with the last inequality being true for all $t \leq t_0$, for t_0 small enough, so that $-\langle \nu, e_2 \rangle \leq \frac{1}{2}$. Integration then yields the desired estimate

$$\log \frac{u(t_0)}{u(t)} \geq \frac{1}{2C}(t_0 - t)$$

implying

$$u(t) \leq u(t_0)e^{(2C)^{-1}(t-t_0)}.$$

■

Next we show the following Harnack type estimate.

Lemma 3.31. *There is a constant C so that for all times sufficiently negative*

$$|\kappa_s| \leq C\kappa,$$

where κ is regarded with respect to the downward pointing unit normal.

Proof. Let $\gamma(\cdot, t)$ denote a parametrization of Γ_t and ν the downward unit normal. We first show the following claim.

Claim. There exists Λ , depending only on Σ , such that

$$|\langle \gamma, \nu \rangle| \leq \Lambda\kappa.$$

Proof of Claim. Choose Λ so that

$$\frac{|X|}{\kappa_\Sigma(X)} < \Lambda, \quad \forall X \in \Sigma.$$

and consider the quantity $\langle \gamma, \nu \rangle - \Lambda \kappa$. Note that, since $\Gamma_t \xrightarrow{t \rightarrow -\infty} D$, we have that $\lim_{t \rightarrow -\infty} \langle \gamma, \nu \rangle - \Lambda \kappa = 0$. We claim that the quantity remains non positive. To see this, assume on the contrary that there is a first space-time point for which the quantity is equal to some positive number $\delta < 1$. By the Hopf boundary lemma, that point cannot occur at the boundary, since at any boundary point the derivative with respect to the outward pointing conormal η satisfies

$$\frac{\partial}{\partial \eta}(\langle \gamma, \nu \rangle - \Lambda \kappa) \leq \kappa(|\gamma| - \Lambda \kappa_\Sigma) < 0.$$

It also cannot occur in the interior, for times negative enough, when $\kappa < 1$, because then we would have

$$\begin{aligned} 0 &\leq (\partial_t - \Delta)(\langle \gamma, \nu \rangle - \Lambda \kappa) \\ &= -2\kappa + \kappa^2(\langle \gamma, \nu \rangle - \Lambda \kappa) \\ &= \kappa(-2 + \delta \kappa) < 0, \end{aligned}$$

which is absurd. Taking $\delta \rightarrow 0$ finishes the proof of the claim. $\blacklozenge$

Having this claim in our disposal, in order to prove the lemma, it suffices to show that

$$|\kappa_s| - C\kappa + \langle \gamma, \nu \rangle \leq 0,$$

for some constant C . Let $C = \Lambda + \max \kappa_\Sigma$ and for any $0 < \varepsilon \leq 1$, define the function

$$f_\varepsilon := |\kappa_s| - C\kappa + \langle \gamma, \nu \rangle - \varepsilon(e^t + 1).$$

Clearly, $\lim_{t \rightarrow -\infty} f_\varepsilon = -\varepsilon < 0$ and, at the boundary points,

$$f_\varepsilon \leq \kappa \kappa_\Sigma - C\kappa + \langle \gamma, \nu \rangle \leq \kappa(\kappa_\Sigma - C + \Lambda) < 0.$$

We claim that $f_\varepsilon < 0$ also at the interior for all sufficiently negative times. Indeed, if this is not the case then there exists a first time $t < T$ for which

$f_\varepsilon = 0$ at an interior point. Then, at such a point we have

$$\begin{aligned}
 0 &\leq (\partial_t - \Delta)f_\varepsilon \\
 &= 4\kappa^2\kappa_s - C\kappa^3 + \kappa^2 \langle \gamma, \nu \rangle - 2\kappa + \varepsilon e^t \\
 &\leq 4\kappa^2(C\kappa - \langle \gamma, \nu \rangle + \varepsilon(e^t + 1)) - C\kappa^3 + \kappa^2 \langle \gamma, \nu \rangle - 2\kappa - \varepsilon e^t \\
 &= 3C\kappa^3 + 4\kappa^2\varepsilon + (4\kappa^2 - 1)\varepsilon e^t - 3\kappa^3 \langle \gamma, \nu \rangle - 2\kappa,
 \end{aligned}$$

and assuming that κ is small enough, the term -2κ dominates and we obtain a contradiction. ■

We can now use Lemma 3.31 to refine the estimate of Lemma 3.30. We will use a graphical parametrization of the solution using the flow lines ψ of the vector field $\bar{\nu}$ (an extension of e_2), as defined in Section 3.3.1. By convexity and the backwards convergence to D we know that, at least for negative enough times, we can express the flow via a function $u : D \times (-\infty, T) \rightarrow \mathbb{R}_+$, so that

$$\Gamma_t = \{\psi(x, u(x, t)) : x \in D\}. \quad (3.36)$$

We prove the following estimate for u .

Lemma 3.32. *Let u be a graphical parametrization of Γ_t via ψ as in (3.36). Then*

$$\limsup_{t \rightarrow -\infty} e^{\lambda_0 t} u(x, t) < \infty.$$

Remark 3.33. Since u satisfies the graphical curve shortening flow 3.26. By the estimates for E, ε (see Definition 3.22) and the Schauder estimates (see 3.23), one also obtains $e^{\lambda_0 t} \|u(\cdot, t)\|_{C^{2,\alpha}} < \infty$. Higher derivative estimates then follow by elliptic theory.

Proof of Lemma 3.32. We first note that by Lemma 3.31 we have $(\log \kappa)_s \leq C$. We consider the curvature with respect to the downward pointing unit normal so that it is positive. By integrating from the point of minimum curvature, $\underline{\kappa}$, to the point of maximum curvature, $\bar{\kappa}$, along with the fact that $\text{Length}(\Gamma_t) \leq 2$, we have that for sufficiently negative times

$$2C \geq \log \frac{\bar{\kappa}}{\underline{\kappa}}$$

which of course implies

$$\bar{\kappa} \leq e^{2C} \underline{\kappa}. \quad (3.37)$$

Denote by $\{\Gamma_t^v\}_{t \in (-\infty, 0)}$ the solution constructed in Lemma 3.29, given as the graph of a function v via ψ , which we time translate so that it is maximal in the time interval $(-\infty, 0)$. Then for all sufficiently negative times, $\Gamma_t \cap \Gamma_t^v \neq \emptyset$. Indeed, if this were not true, then they would be disjoint for all times, which, by the avoidance principle contradicts the fact that they contract to the same point on the boundary at $t = 0$. Therefore, for any t sufficiently negative, there exists an $x_0 \in (-1, 1)$ such that $u(x_0, t) < v(x_0, t)$, and therefore, by construction of v ,

$$e^{\lambda_0 t} u(x_0, t) \leq e^{\lambda_0 t} v(x_0, t) < C.$$

Consider now t negative enough so that $\langle \nu, \bar{\nu} \rangle > \frac{1}{2}$. Then for any x , and such t , we have, using 3.37,

$$\begin{aligned} u(x, t) &= \int_{-\infty}^t u_t(x, t) dt = \int_{-\infty}^t \frac{\kappa(x, t)}{\langle \nu(x, t), \bar{\nu} \rangle} dt \leq 2e^{2C} \int_{-\infty}^t \frac{\kappa(x_0, t)}{\langle \nu(x_0, t), \bar{\nu} \rangle} dt \\ &= 2e^{2C} u(x_0, t). \end{aligned}$$

■

We can now prove the uniqueness theorem, Theorem 3.2, which is a direct corollary of the following theorem (see Remark 3.35).

Theorem 3.34. *Let u be a graphical parametrization of Γ_t via ψ as in (3.36). Then there exists a constant $A \in \mathbb{R}$ such that*

$$e^{\lambda_0 t} u(x, t) \rightarrow A \phi_0,$$

uniformly as $t \rightarrow -\infty$.

Remark 3.35. We remark that Theorem 3.2 follows directly from Theorem 3.34. Arguing as in Proposition 3.19, we observe that a different constant $\tilde{A}$ in Theorem 3.34 with the same sign corresponds to a time translation of the same solution. Moreover, two solutions where the constants A have different sign correspond to solutions on different sides of the diameter.

Proof of Theorem 3.34. For each $\tau < 0$, let $u^\tau(x, t) := e^{\lambda_0 \tau} u(x, t + \tau)$. Since u satisfies (3.26), u^τ satisfy

$$\begin{cases} \frac{d}{dt} u^\tau - \Delta u^\tau = e^{\lambda_0 \tau} E(e^{-\lambda_0 \tau} u^\tau) & \text{on } D \\ \frac{\partial u^\tau}{\partial \eta} - \kappa_\Sigma u^\tau = e^{\lambda_0 \tau} \varepsilon(e^{-\lambda_0 \tau} u^\tau) & \text{on } \partial D. \end{cases} \quad (3.38)$$

By Lemma 3.32 (see also Remark 3.33), the functions u^τ have uniform (in τ) bounds on their C^∞ norm. Moreover, by these estimates, together with the estimates of E and ε given in Definition 3.22, the right hand sides of both the equations and their boundary conditions in the problem (3.38) converge to zero as $\tau \rightarrow -\infty$. Therefore, for any sequence $\tau_j \rightarrow 0$, after passing to a subsequence, u^{τ_j} converge in the C^∞ norm to a function $u^\infty : D \times (-\infty, \infty) \rightarrow \mathbb{R}_+$ that satisfies

$$\begin{cases} \frac{d}{dt} u^\infty - \Delta u^\infty = 0 & \text{on } D \\ \frac{\partial u^\infty}{\partial \eta} - \kappa_\Sigma u^\infty = 0 & \text{on } \partial D. \end{cases}$$

We can therefore express u^∞ via the eigenfunctions ϕ_k of (3.16). Since, we additionally have that u^∞ does not change sign, we obtain that $u^\infty = A e^{-\lambda_0 t} \phi_0$ for a constant A . ■

Remark 3.36. We remark that one can compute the eigenfunction ϕ_0 and indeed it is given by

$$\phi_0 := \cosh \sqrt{-\lambda_0} x + \frac{\kappa_1 - \kappa_2}{2\sqrt{-\lambda_0} - (\kappa_1 + \kappa_2) \tanh \sqrt{-\lambda_0}} \sinh \sqrt{-\lambda_0} x,$$

where $\kappa_1 := \kappa^\Omega(1, 0)$, $\kappa_2 := \kappa^\Omega(-1, 0)$.

References

ALTSCHULER, S. J. Singularities of the curve shrinking flow for space curves. **Journal of Differential Geometry**, v. 34, n. 2, 1991.

ANDREWS, B. *et al.* **Extrinsic geometric flows**. American Mathematical Society, Providence, RI, 2020. v. 206. (Graduate Studies in Mathematics).

ANGENENT, S. The zero set of a solution of a parabolic equation. **J. Reine Angew. Math.**, v. 390, 1988.

ANGENENT, S.; DASKALOPOULOS, P.; SESUM, N. Unique asymptotics of ancient convex mean curvature flow solutions. **J. Differential Geom.**, v. 111, n. 3, 2019.

ANGENENT, S.; DASKALOPOULOS, P.; SESUM, N. Uniqueness of two-convex closed ancient solutions to the mean curvature flow. **Ann. of Math. (2)**, v. 192, n. 2, 2020.

ANGENENT, S. B. Formal asymptotic expansions for symmetric ancient ovals in mean curvature flow. **Networks and Heterogeneous Media**, v. 8, n. 1, 2013.

BADERKO, E. A. Schauder estimates for oblique derivative problems. **C. R. Acad. Sci. Paris Sér. I Math.**, v. 326, n. 12, 1998.

BAKAS, I.; SOURDIS, C. Dirichlet sigma models and mean curvature flow. **Journal of High Energy Physics**, v. 2007, n. 06, 2007.

BOURNI, T.; BURNS, N.; CATRON, S. Classification of convex ancient solutions to free boundary curve shortening flow in convex domains. **Journal of Geometric Analysis**, v. 35, n. 7, 2025.

BOURNI, T.; BURNS, N.; LANGFORD, M. Stability of the Shrinking Semi-Circle Under the Free Boundary Curve Shortening Flow. Preprint, [arXiv:2603.06949](#). [*S. I.*], 2026. arXiv: [2603.06949](#) [[math.DG](#)].

BOURNI, T.; FRANZ, G. Ancient solutions to free boundary mean curvature flow. **arXiv:2602.08850**, 2026. Preprint, available at <https://arxiv.org/abs/2602.08850>. arXiv: [2602.08850](#) [[math.DG](#)].

BOURNI, T.; LANGFORD, M. Classification of convex ancient free boundary mean curvature flows in the ball. **Annali della Scuola Normale Superiore di Pisa, Classe di Scienze** (5), v. 26, n. 2, 2025.

BOURNI, T.; LANGFORD, M. Classification of convex ancient free-boundary curve-shortening flows in the disc. **Analysis & PDE**, v. 16, n. 9, 2023.

BOURNI, T.; LANGFORD, M.; LYNCH, S. Collapsing and noncollapsing in convex ancient mean curvature flow. **Transactions of the American Mathematical Society**, v. 377, n. 5, 2024.

BOURNI, T.; LANGFORD, M.; TINAGLIA, G. Ancient mean curvature flows out of polytopes. **Geometry & Topology**, v. 26, n. 4, 2022. arXiv: [2006.16338](#) [[math.DG](#)].

BOURNI, T.; LANGFORD, M.; TINAGLIA, G. Collapsing ancient solutions of mean curvature flow. **Journal of Differential Geometry**, v. 113, n. 1, 2019. arXiv: [1705.06981](#) [[math.DG](#)].

BOURNI, T.; LANGFORD, M.; TINAGLIA, G. Convex ancient solutions to curve shortening flow. **Calculus of Variations and Partial Differential Equations**, v. 59, n. 4, 2020.

BRENDLE, S.; CHOI, K. Uniqueness of convex ancient solutions to mean curvature flow in $\mathbb{R}^3$. **Invent. Math.**, v. 217, n. 1, 2019.

BRENDLE, S.; NAFF, C. A local noncollapsing theorem for mean curvature flow. **Journal of Differential Geometry**, 2023.

BRYAN, P.; LOUIE, J. Classification of convex ancient solutions to curve shortening flow on the sphere. English. **J. Geom. Anal.**, Springer US, New York, NY; Mathematica Josephina, St. Louis, MO, v. 26, n. 2, 2016.

CHOI, K.; HUANG, J.; LEE, T. Ancient mean curvature flows with finite total curvature. **Transactions of the American Mathematical Society**, v. 378, n. 9, 2025. arXiv: [2405.01062](#) [[math.DG](#)].

- CHOI, K.; MANTOULIDIS, C. Ancient gradient flows of elliptic functionals and Morse index. **American Journal of Mathematics**, v. 144, n. 2, 2022.
- CHOW, B.; GULLIVER, R. Aleksandrov reflection and geometric evolution of hypersurfaces. **Comm. Anal. Geom.**, v. 9, n. 2, 2001.
- DASKALOPOULOS, P.; HAMILTON, R.; SESUM, N. Classification of compact ancient solutions to the curve shortening flow. **J. Differential Geom.**, v. 84, n. 3, 2010.
- ECKER, K.; HUISKEN, G. Interior estimates for hypersurfaces moving by mean curvature. English. **Invent. Math.**, Springer, Berlin/Heidelberg, v. 105, n. 3, 1991.
- GAGE, M.; HAMILTON, R. The heat equation shrinking convex plane curves. English. **J. Differ. Geom.**, International Press of Boston, Somerville, MA, v. 23, 1986.
- GILBARG, D.; TRUDINGER, N. S. **Elliptic partial differential equations of second order. Reprint of the 1998 ed.** Reprint of the 1998 ed. Berlin: Springer, 2001.
- HAMILTON, R. S. **Harmonic maps of manifolds with boundary.** Springer, 1975. v. 471. (Lecture Notes in Mathematics).
- HAMILTON, R. S. The formation of singularities in the Ricci flow. *In: SURVEYS in differential geometry*, Vol. II (Cambridge, MA, 1993). Int. Press, Cambridge, MA, 1995.
- HAN, Y. Ancient mean curvature flows from minimal hypersurfaces. **Geometriae Dedicata**, v. 220, n. 2, 2026. arXiv: [2311.15278](#) [[math.DG](#)].
- HASLHOFER, R.; HERSHKOVITS, O. Ancient solutions of the mean curvature flow. **Comm. Anal. Geom.**, v. 24, n. 3, 2016.
- HUISKEN, G. A distance comparison principle for evolving curves. **Asian J. Math.**, v. 2, n. 1, 1998.
- HUISKEN, G.; SINISTRARI, C. Convex ancient solutions of the mean curvature flow. **J. Differential Geom.**, v. 101, n. 2, 2015.
- LADYZHENSKAYA, O. A.; SOLONNIKOV, V. A.; URAL'TSEVA, N. N. **Linear and Quasi-linear Equations of Parabolic Type.** American Mathematical Society, 1968. v. 23. (Translations of Mathematical Monographs). Translated from the Russian by S. Smith.

LADYZHENSKAYA, O. A.; URAL'TSEVA, N. N. **Linear and quasilinear elliptic equations**. Academic Press, New York-London, 1968. Translated from the Russian by Scripta Technica, Inc, Translation editor: Leon Ehrenpreis.

LANGFORD, M. A general pinching principle for mean curvature flow and applications. **Calculus of Variations and Partial Differential Equations**, v. 56, n. 4, July 2017.

LANGFORD, M.; ZHU, J. A distance comparison principle for curve shortening flow with free boundary. Preprint, [arXiv:2302.14258](https://arxiv.org/abs/2302.14258). [S. l.], 2023.

LIEBERMAN, G. M. **Second order parabolic differential equations**. World Scientific Publishing Co., Inc., River Edge, NJ, 1996.

LYNCH, S. Uniqueness of convex ancient solutions to hypersurface flows. **Journal of Differential Geometry**, v. 125, n. 2, 2023.

SIMON, L. Schauder estimates by scaling. **Calc. Var. Partial Differential Equations**, v. 5, n. 5, 1997.

STAHL, A. Convergence of solutions to the mean curvature flow with a Neumann boundary condition. **Calc. Var. Partial Differential Equations**, v. 4, n. 5, 1996.

STAHL, A. Regularity estimates for solutions to the mean curvature flow with a Neumann boundary condition. **Calculus of Variations and Partial Differential Equations**, v. 4, n. 4, 1996.

WANG, X.-J. Convex solutions to the mean curvature flow. **Ann. of Math. (2)**, v. 173, n. 3, 2011.

WHITE, B. The nature of singularities in mean curvature flow of mean-convex sets. **Journal of the American Mathematical Society**, v. 16, n. 1, 2003.



impa



Instituto de
Matemática
Pura e Aplicada